



FDS_WP_20012

Production ticket

WHOLESALE

Purpose of this document	The functional design specification is a document developed before the programming of a solution is initiated.	
Audience	The A. Business Responsible knows the business, the processes and the requirements in the business. This person is responsible for understanding the designed solution and responsible for the approval of the design. The B. AX Solution Responsible is typically the author of the document. The AX solution responsible knows how Dynamics AX works and knows the principles of AX. The AX solution responsible is responsible for the design of the solution, the details in this and that the design is feasible and will be able to work. The C. Solution Architect is reviewer of the document, responsible for the coherence of the solution and that the design is in agreement with Dynamics AX and the architecture principles in WDH. The D. Technical Architect will review the functional design from a technical perspective, and review that the technical design matches the functional design. The E. Developer will use the document as a part of the development process and decide who will find layouts, labels, security and testing information in the Functional design document. The F. Test Manager will use the document to outline test scripts to be included in the different test phases. The G. Change Management responsible will use the document for training material, user manuals and communication in the organization.	
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Terminology	Terminology Used	Description
	Route	A route contains the set of operations that are required to produce a product or an item. This includes the sequence of operations, the site where the production occurs, the resource requirements that are involved, and the standard times that are required to set up and run the operations
	Operation	Operations represent the tasks that are required to produce items in production
	Production order	A Production order is usually a task for a customer that can be scheduled or assigned to someone

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2 Purpose and Short Description

[Outline the purpose with this development, and write a short description of the solution and reference to the context it relates to. An illustration is mandatory as well as indications of the types of roles involved. If there are different systems involved, an illustration of this is required too. Are you basing the description below on some assumptions?]

In present solution for local production BOM journal is being used to consume Kits and to have the finished item into stock. This process works fine for simple scenarios but it has its own complications when we think about a larger setup where we need a better control on production, quality checkpoint and material movement. With these points in view it become evident to have a separate work area (form) for production related activities. For this purpose Production control module in standard AX will be used.

3 Evaluation of Standard AX/ISV

[A short description of why we cannot live with what is already provided in Standard AX, and if we have evaluated any ISV solutions which failed]

The **Production control** module in Microsoft Dynamics AX is linked to other modules. These include **Inventory management**, **General ledger**, and **Organization administration**. This integration supports the information flow that is required to complete the manufacturing of a finished item. AX creates one production order per sales line which means for a binaural order AX will create two production orders. Since for a binaural order both the items move together in production, we need a form- Production ticket, which will hold both production order together. The purpose of this documentation is to design Production ticket form needed to support business requirements related to local production.

4 Requirements

[This contains references to the requirements from the RTM met with this modification]

Requirement Id	Requirement text
34580	Electronic production ticket
34061	Handle standard activity per step in the production
34349	Track order inside production
34465	Shift order backwards in production process
34474	Improved access to production data
34363	Easy access to BOM on all stages
34484	Binaural orders on production ticket
35092	Site ownership of orders
33843	View and handle quality check list information for each order
34321	MDSAP control points with state machine
34336	Validate engravement with ordered
34388	Compare order and actual model
35145	Make sure all steps have been scanned
34292	Capture listening test results
34293	Capture KIT programming test results

5 BPD Reference

[Reference to the process step in the BPD where this development will be triggered].

6 Detailed Design

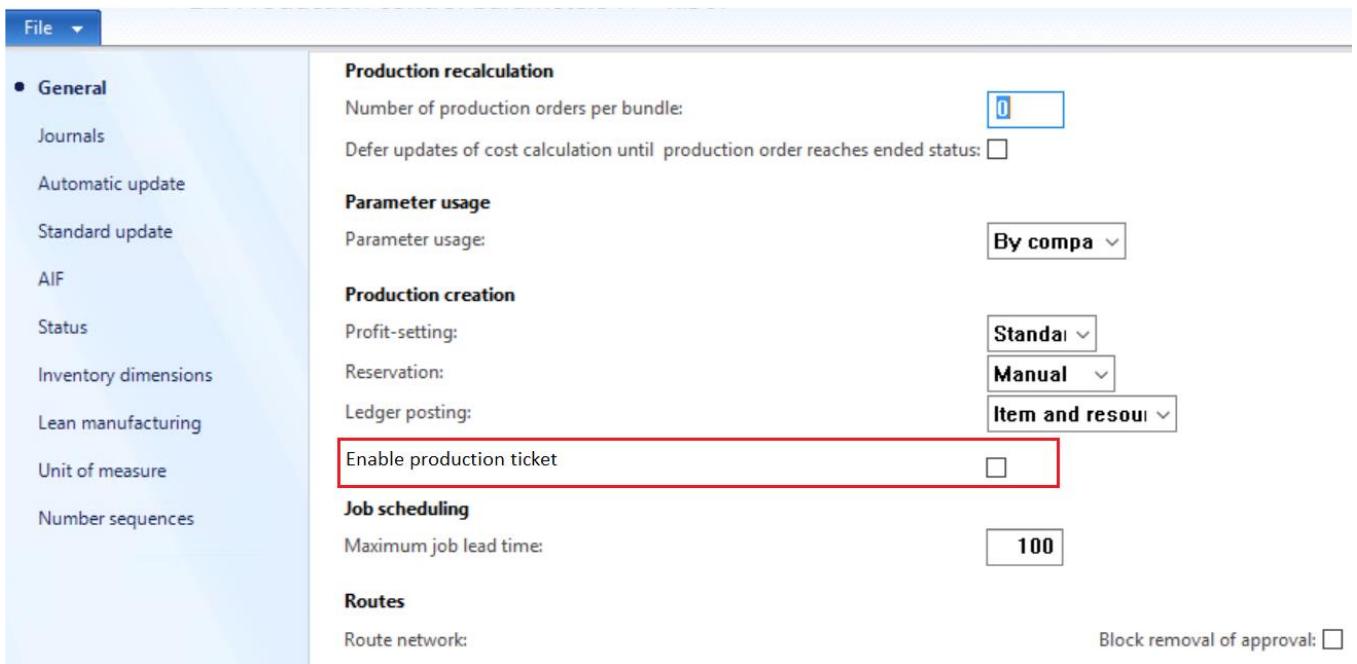
[This section is the basic section of the document and is an illustrated design of the to-be solution. The description focuses on the user interface, the content and the process on the solution, including both the part of the solution that is standard in Dynamics AX and the modifications.].

6.1 Production ticket

6.1.1 Setup

To enable production ticket feature a new parameter is needed. Add new field 'Enable production ticket' to Production control parameters and Production control parameters by site (Path: Production control/Setup) . All customizations related to production feature should be enabled only if Enable production ticket is selected in setup.

 Production control parameters (1 - nl30)



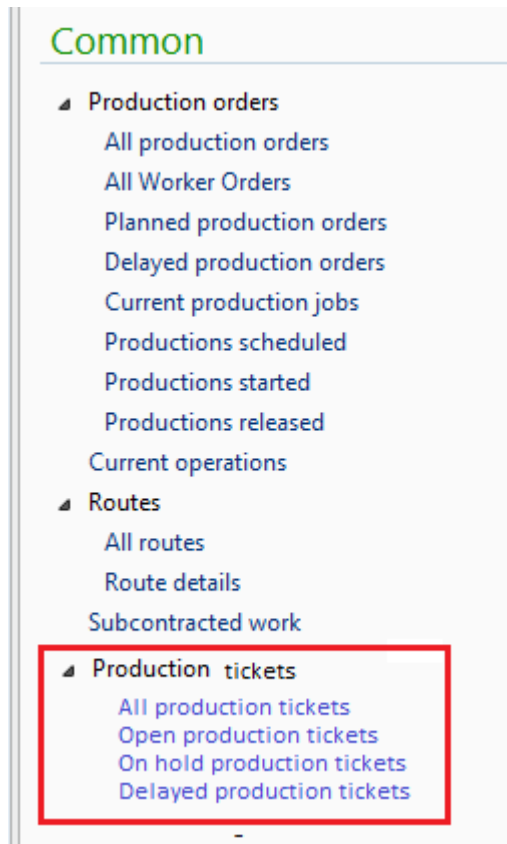
The screenshot shows the 'Production control parameters' setup form. The left sidebar lists various categories: General, Journals, Automatic update, Standard update, AIF, Status, Inventory dimensions, Lean manufacturing, Unit of measure, and Number sequences. The 'General' category is selected. The main area contains several sections: 'Production recalculation' with a text box for 'Number of production orders per bundle' (value 1) and a checkbox for 'Defer updates of cost calculation until production order reaches ended status'; 'Parameter usage' with a dropdown for 'Parameter usage' (value 'By compa'); 'Production creation' with dropdowns for 'Profit-setting' (value 'Standar'), 'Reservation' (value 'Manual'), and 'Ledger posting' (value 'Item and resou'); 'Job scheduling' with a text box for 'Maximum job lead time' (value 100); and 'Routes' with a text box for 'Route network' and a checkbox for 'Block removal of approval'. The 'Enable production ticket' checkbox is highlighted with a red rectangle.

Introduce a new number sequence 'Production ticket' to identify production tickets in system. Place this number sequence in production control parameters/Number sequence.

6.1.2 Production ticket overview

The menu items should be shown only if Enable production ticket is selected in setup.

Create new menu item 'All production tickets', 'Open production tickets', 'On hold production tickets' and 'Delayed production tickets' under Production control/Common/Production tickets.



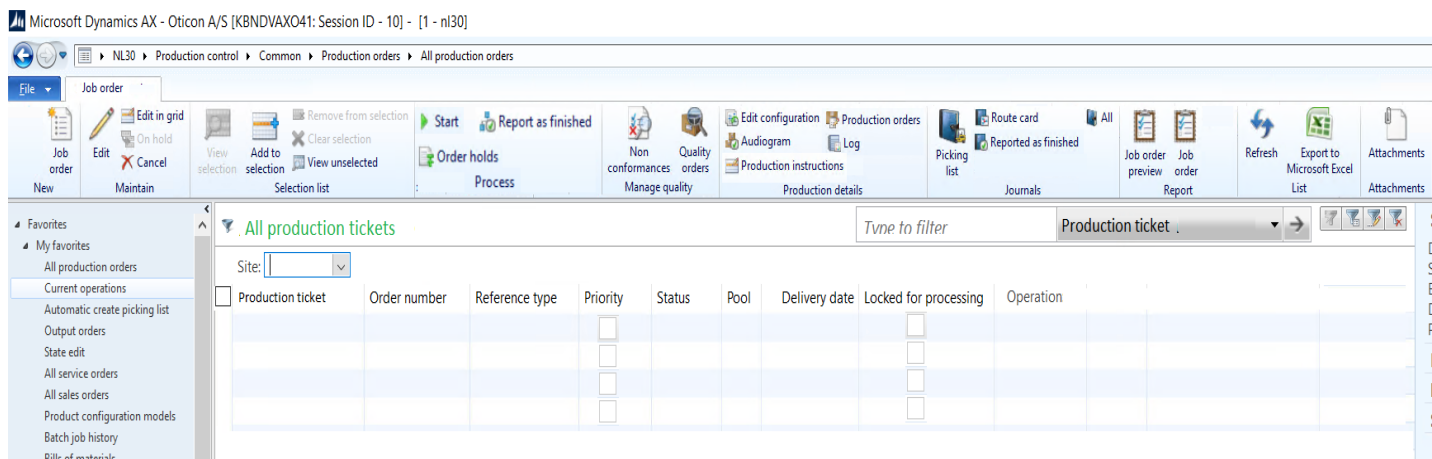
All production tickets: This will show an overview of all production tickets in the system

Open production tickets: This will show an overview of all production tickets, which are not cancelled or complete in the system, i.e. production ticket in status Active or Onhold.

On hold production tickets: This will show an overview of all production tickets, which are in Onhold status in the system.

Delayed production tickets: This will show an overview of all production tickets for which delivery date is less than system date.

Below is representation of the production ticket overview.



Add preview pane to Production ticket list page, the preview pane should include following information (as in red box below)

NL30-000253	Started	prn			47950	A.G. Bellschool
NL30-000252	Created	MOD			862625	IntoEars Hoorkliniek Tilt
NL30-000251	Started	asm			47950	A.G. Bellschool
NL30-000250	Started	PRN			47950	A.G. Bellschool
NL30-000249	Started	ASM			47950	A.G. Bellschool
NL30-000248	Created					
NL30-000247	Started	LAS				

NL30-000252

Production Ticket: NL30-00025

Production ticket status: Created

Order number: SER000043862

Production	Item number	Description	Quantity
PRO000000641	154709		1,00

6.1.3 Production ticket form

Add menu 'Production ticket', which will open the production ticket form. Below is a representation of the production ticket form. For actual list of fields and labels please refer to section New forms

Production Ticket

File Production Ticket

Edit Cancel Start Finish Order holds Non conformances Quality orders Picking list Route card Reported as finished All Edit configuration Log Refresh Export to Microsoft Excel

Maintain Operation Manage quality Journals Production details List

NL30-00000000 Delivery date: 05-02-2019 Operation: QC Assign to:

Scan Error:

General

Identification Production Ticket: NL30-00000000 Production ticket status: Created

Order details Order number: SOR000129383 Reference type: Sales order Order State: SLI

Groupings Pool: Use Impression: Priority: Lock For Processing:

Others Use Impression: Lock For Processing:

Production Ticket Lines

Add line Remove Create BOM Additional consumption Change serial number Colour change Service detail

Include	Production	Item number	Name	Serial number	Quantity	Ear	Operation	Service type
☐						None		

Attachments

Add line Remove

Order number	Reference type	Type	Created date and time	Created by	Name
This grid is empty.					

Service specification
Service order
Service card history
Service consumption

Form status area-

1. Delivery : Sales order header Requested ship date.
2. Operation: This field will show all operations from the route group in the dropdown. User has to scan the operation barcode to indicate the operation..
3. Scan: User can use scan field to scan barcode to retrieve production ticket, start or finish an operation. Alternatively user can also select an operation manually by going to drop down and initiate start and finish action by keyboard
4. Worker: This field can be used to assign a worker to the production ticket.

Tabs-

1. General:
This section will contain general details, order (sales/service) reference and delivery date details, Groupings and Other details. A production ticket can have any of the following status at a given point in time: Created, Started, Onhold, Finished or Cancelled.

Created- This is the default state of production ticket which gets assigned when a production ticket gets created.

Started- This is the status of production ticket when any of the operation has been started.

Finished- This is the status of the production ticket when the production order/s linked to the production ticket are reported as finished. In case production ticket has more than one production order then the status will become Finished only when both production order moves to report as finished state.

Onhold- This is the status of production ticket when onhold code (created from sales order or production ticket) has lock for processing checkbox marked.

Cancelled- This is the status of production ticket when it has been cancelled- either cancelled from production ticket or when linked sales order has been cancelled.

2. Production ticket lines:

This section will contain production orders to be included in the work order. At max it can include two Production orders (binaural case).

Include checkbox will be always be marked by default. In case user wants to update operation only for one of the production order then user has to unmark the other one and then scan the start or finish action (Standard AX creates one production order for one sales line). This can be a case where during quality check only one ear failed quality check, so rework is needed only on one ear.

3. Configuration:

This section will show (view only) attributes for left and right ear.

The view should show only those attributes which are marked for 'Show on production ticket'

Configit integration Id to AX (1 - n30) - Product type: Instrument, Custom

File New Delete

Product type Order

Product type	Order
Instrument	6
Power mould	1
Micro mould	2
Mould	3
Flex mould	4
Lite tip	5

General

Custom/OTS: Custom

Item number: 1823

Ear side: 3447

Left ear value: 3448

Right ear value: 3449

WBI family code: 1891

WBI style code: 1890

WBI product type: HA

Earpiece type:

Attribute integration Ids

Add Remove

Integration ID	Order	Mandatory	Label	Attribute group	Save	Show	Skip when show	No editing	Show on production ticket
1	2	☒	Product Brand	Product Configuration Attributes	☒	☒	☐	☐	☒
10	22	☒	Auto Telecoil	Product Configuration Attributes	☒	☒	☐	☐	☒
11	28	☐	Program Control	Product Configuration Attributes	☒	☒	☐	☐	☒
12	11	☐	Volume Control	Product Configuration Attributes	☒	☒	☐	☐	☒
13	16	☒	Directional	Product Configuration Attributes	☒	☒	☐	☐	☒
17	10	☒	Battery Size	Product Configuration Attributes	☒	☒	☐	☐	☒
1737	33	☐	Earpiece Style retention	Product Physical Attributes	☒	☒	☐	☐	☒
1739	26	☐	Vent	Product Configuration Attributes	☒	☒	☐	☐	☒
18	110	☐	Tamper Resistant	Product Configuration Attributes	☒	☐	☐	☐	☐
1841	106	☐	Hardware PlatformConfiguration	Product Configuration Attributes	☒	☐	☐	☐	☐
1854	109	☐	ProductModelVersion	Product Configuration Attributes	☒	☐	☒	☐	☐
1890	104	☐	WBI Style Code	Product Physical Attributes	☐	☐	☐	☐	☐

4. Attachments:

This section will show comments of type production from sales order header.

5. Work specification:

This section should reflect the work specification from the related service specification form. It should be possible to enter the request codes, failure codes and the resolution codes in the same way as it is done in the service order form.

The buttons (Add, Remove, Work spec wizard, Copy work specification, Reset service codes) should be also visible on the production ticket and the functionality behind then should be exactly the same as for the service specification form.

Work specification									
		Work Spec Wizard	Update service fees and ...	Copy work specification					
Request area	Request area description	Request code	Symptom description	Failure area	Failure area description	Failure code	Failure description	Resolution	Resolution Description
100	Sound experience	111	Feedback	001	Amplifier	001003	Damaged	0003	Replaced
230	Microphone	231	Microphone	004	Battery Drawer	004003	Damaged	0023	Repaired

Buttons:

1. New/Production ticket: This will bring up create production ticket form
2. Maintain/Cancel- This will bring up a dialogue box with message 'Are you sure about cancelling the production ticket' and with options Yes and No. If user click 'Yes' then put the production ticket to Cancelled status. Production order linked to production ticket will be deleted. Any open pick list related to production orders should be deleted. The Kit has to be moved back to warehouse physically so that system stock and physical stock should be same.
If user click 'No' then close the dialogue box without any action on production ticket.
3. Process/Start- This will put current operation for included production order to start state. (add shortcut key Alt 1)
4. Process/Report as finished- This will put current operation for included production order to finished state. (add shortcut key Alt 2)
5. Process/Order holds- This will put the production ticket to On hold status. This will be similar to onhold functionality on sales order. Work order status will be changed to On hold.
6. Manage Quality/Quality orders- This will bring up quality order form showing quality order for production order with included mark and corresponding operation on production ticket. (please refer to Sec 6.4).
7. Manage Quality/Non conformances- This will bring up non conformances form showing non conformances raised on a quality order for included production order and corresponding operation on production ticket (please refer to Sec 6.4).
8. Production details/ Edit configuration- This will bring up Edit configuration form (same as on sales order). User can do changes to configuration from this form.
9. Production details/ Audiogram- This will bring up Audiogram form (non editable) showing audiogram data from sales order.
10. Production details/ Production instructions- This will bring up Production instructions form showing production instructions from sales order (view only).
11. Production details/ Log- This will bring up the Log form. Log form will show start and finish details for all operations.
12. Journals/Picking list- This will bring up the Picking list journal for this production ticket.
13. Journals/Route card- This will bring up the Route card journal for this production ticket.
14. Journals/Report as finished- This will bring up the Report as finished journal for this production ticket.
15. Journals/All- This will bring up all journals for this production ticket
16. Report/Job order preview- This will bring up the job order preview report (same as on sales order)
17. Report/Job order - This will print the job order report (same as on sales order)
18. Buttons- BOM, Route and Log will give access to BOM and Route used for this production ticket. Log will show actual start and finish (date and time) of each operation. (both production orders in case of Binaural)
19. Attachments: This will provide access to view and create attachments for this production ticket. Attachments from sales order line should be shown here. (Confirmation on document indicating if

document is read or not will be in wave2). New button will bring up Document form to create new document.

20. Attachments (header): This will provide access to view and create attachments for this production ticket. Attachments from sales order header should be shown here. (Confirmation on document indicating if document is read or not will be in wave2). New button will bring up Document form to create new document.
21. Audiogram- This will open up Audiogram data.
22. Edit configuration- This button will open up configuration in edit mode. (Later on 'Proposal' button for Order change management will also be added in this section)
23. Print- This will print Production ticket

6.1.4 Changes to Sales order header and line

Below is a representation of changes to be made on All sales orders, All sales orders for order entry, Sales order for order entry overview form and Sales order overview form. For actual list of fields and labels please refer to section 9.1

Sales order form:

Sales order (1 - nl30) - Sales order: SOR000088697, A.G. Bellschool, Sales order: SOR000088697

File | Quick access | Sales order | Sell | Manage | Pick and pack | Invoice | Retail | General | Insurance

Production instruction | Log | Production | Attachments | Quantity discount | Spare- and commercial pa... | Change State | Job order preview | Job order | Internal order print | Order labels | Address label | Document scanning label | Sales order confirmation | Journals

SOR000088697 : 47950 - A.G. Bellschool | State: OE, Order entry | IC State: | Production State: ATS | Oper

Brand information

Commercial brand: Oticon

Sales order header 47950

Sales order lines

Type	Variant number	Item number	Product name	Sales category	Quantity	Unit	Production type	Sales dimension template	Discount percent	Coverage group
		154709	Oticon Alta CIC		1.00	pcs	Central			Man
		800-000-302	Standaard 2 jaar garantie		1.00	pcs				

A new button "Cancel Production ticket" should be added in the header section for both sales order and sales order entry form:

Sales order (1 - de30) - Sales order: SOR001551446, HÖRWERK GmbH, Sales order: SOR001551446

File | Quick access | Sales order | Sell | Manage | Pick and pack | Invoice | Retail | General

Cancel | Production Ticket | Attachments | Quantity discount | Spare- and commercial pa... | Estimated receipt date | Change State | Job order preview | Job order | Internal order

SOR001551446 : 749603 - HÖRWERK GmbH | Production | IC State: | Production State: SLI

Commercial brand: Oticon

Sales order header

Delivery address

Name: HÖRWERK GmbH
Delivery address: HÖRWERK GmbH
Address: Hambürcker Str. 18
76694 Forst
Deutschland

Delivery contact:

Delivery date

Follow-Up date:
Requested ship date: 23-09-2022
Requested receipt date: 26-09-2022
Requested fitting date:
Expected ship date: 23-09-2022
Expected Receipt Date: 26-09-2022
Buffer days: 0

References

Customer reference: Brigitte Hartlieb
Customer requisition:
Sales taker: Katrin Kuhlmann
Retail reference:

Discounts

Total discount % 0,00

Order category

New button "Cancel Production ticket"

Sales order entry 1 - de30) - Sales order: SOR001551446, HÖRWERK GmbH, Sales order: SOR001551446

File Quick access Sales order Sell Manage Pick and pack Invoice Retail General

Edit Cancel Production Ticket Attachments Quantity discount Spare- and commercial pa... Estimated receipt date

Maintain Audioagram Production instructions Order holds Warranties and fees Related instruments

Calculate Change State Workflow Job order preview Job order Order labels Address label Report

SOR001551446 : 749603 - HÖRWERK GmbH

Brand information Commercial brand: Oticon

Sales order header

Customer

Customer account: 749603
Invoice account: 776400
Terms account: 749602
Branch number:
Price group:
Line discount group:
List price group:
Supplementary item group:

Consignment

Consignment type: 03MON
Normal/consignment: Normal
Invoicing consignment sales order: ☐
Consignment return fee order: ☐

Document scanning

Document scanning number: SOR001551446

Misc. delivery info

Partial/complete delivery: Complete
Mode of delivery: UPS A_Standard
Actual fulfillment method: Local delivery
Suggested fulfillment method: Local delivery
Integration prefix:
DisableWBIntegration: ☐
Order handling:
Disable miscellaneous fees: ☐

When “Cancel Production ticket” button is clicked then system should cancel related production ticket according to rules described in section [6.3.10](#)

Sales order overview:

Microsoft Dynamics AX - Oticon A/S [KBNDVAXO41: Session ID - 8] - [1 - nl30]

File Sales order Sell Manage Pick and pack Invoice Retail General Insurance Transportation management

Service order Purchase order Edit in grid Delete Cancel New Maintain Audioagram Payments Copy View Order events Order credit Recap Order holds Functions Production instructions Log Refresh Export to Microsoft Excel List Generate from template Attachments Notes

Favorites My favorites All production orders Current operations Automatic create picking list Output orders State edit All service orders All sales orders

All sales orders Time to filter Sales order

Sales order	Customer account	Name	Invoice account	Channel	Order type	Status	City	Production state	ZIP/post
SOR000088695	47950	A.G. Bellsch...	47950		Sales order	Open order	AMSTERDA...	ATS	1063 E

Production ticket: This should show linked production ticket form.

Log: This should open the linked production ticket log form.

Production state on sales order header will show current operation on production ticket. Production state on sales order line will show current operation of linked production order.

6.2 Processing Production ticket

6.2.1 Create Production ticket through state scan

New action 'createProductionTicket': Action 'createProductionTicket' will create Production ticket and production when called from workflow state. Refer FDS_WP20002, sec 6.4.44 for details .

6.2.2 Changes to Create route detail form

1. Add new field 'Route category' to Route details. This field will indicate if a route is for new production or for remake.
2. Add 'Entry/Exit actions' button on Route (path: Production control/Common/Routes/All routes).
3. Add column 'Mandatory' to indicate if a particular operation is mandatory for report as finished before production order can be marked as report as finished.

Route (1) - Route number: R1, Route 1

File Functions Product change Entry/Exit actions

☐	Oper. No.	Priority	Operation	Mandatory	Scrap percentage	Accumulated	Next	Link type	Hourly rate / piecework rate
☐	10	Primary	MOD		0,00	1,00	20		All
☐	20	Primary	PRN	☐	0,00	1,00	30		All
☐	30	Primary	LAS	☐	0,00	1,00	40		All
☐	40	Primary	ASM	☐	0,00	1,00	50		All
☐	50	Primary	ATS	☐	0,00	1,00	60		All
☐	60	Primary	QC	☐	0,00	1,00	0		All

Applicable resources Maintain resource requir...

Overview General Setup Times Resource requirements Description

☐	Item code	Item relat...	Route c...	Route rela...	Site	Route gr...
☐	Table	154709	Route	R1	SNL01	RG1
☐	All		Route	R1		RG1

The Entry/Exit actions button should bring up Entry/Exit actions form, same as in Workflow setup in Sales and marketing/ Setup/Workflow. This will tie the actions for that particular operation in that route.

Entry/Exit actions (1 - nl30) - New Record

File

☐	Order ...	Entry/Exit acti...	Class name	Method name	Edit
☐	0	Entry action	WDHStateEngine		

Navigation icons: Back, Forward, Search, etc.

Bottom bar: (108356) | EUR | nl30 | Atlas_DEV13 | Close

The actions mentioned against each operation will be called on entry or exit of that operation based upon setup in Entry/Exit actions form.

4. Add button 'Add products' to Route details form. This button will help in linking multiple items to route version at onego.

Route details (1 - nl30) - Route number: R1, Route 1, Route number: R1

File Route

Edit Route Delete Route Approve Create case Associate with case Add to case log Refresh Export to Microsoft Excel List Attachments

Maintain New Process Product change Attachments

R1 : Route 1

General

Route number: R1

Name: Route 1

Item group: CHA

Approved by: 000156

Approved: ☒

Versions

+ Add - Remove Approve Activation Route feasibility Product change **Add products**

Item nu...	Configurat...	Site	From d...	To d...	From q...	Active	Approved...	Approv...
154709		SNL01			0,00	☒	000156	☒
156792		SNL01			0,00	☒	000156	☒

Product change cases

Case ID	Description
This grid is empty.	

More

Date from which the route version applies. | (108356) | EUR | nl30 | Atlas_DEV13 | Close

Add product button should bring up 'Add products to route' form (note- developers can take inspiration from Add products button on sales order lines). On click of Ok, All selected items from available products should get added to route version.

Microsoft D Route details (1 - nl30) - Route number: R1, Route 1, Route number: R1

File Route

File Add products (1 - gb10) - Sales order: SOR000144382, 20:20 Opticians Ltd, Sales order: SOR000144382

File

All categories

Filters

Filter the list of products by category

- HA Main Accessories
 - Hearing Aids (HA)
 - BTE instruments
 - Custom Kits
 - HA Demo Instruments
 - ITE Instruments
 - Custom instrument
 - Fit Instrument
 - Non-WDH custom instrument
 - RITE instruments
 - RITE Kits
- ME Accessories
- New category
- Production Parts
- Reimbursements
- Services

Available products

Item number	Product name	Search name	Translated name	Category
154341	Oticon Acto CIC	Oticon Acto CIC		Custom instru...
154694	Oticon Acto ITC	Oticon Acto ITC		Custom instru...
154695	Oticon Acto ITE	Oticon Acto ITE		Custom instru...
154696	Oticon Acto MIC	Oticon Acto MIC		Custom instru...
154697	Oticon Acto Pro CIC	Oticon Acto Pro CIC		Custom instru...
154698	Oticon Acto Pro ITC	Oticon Acto Pro ITC		Custom instru...
154699	Oticon Acto Pro ITE	Oticon Acto Pro ITE		Custom instru...
154700	Oticon Acto Pro MIC	Oticon Acto Pro MIC		Custom instru...
154701	Oticon Agil CIC	Oticon Agil CIC		Custom instru...
154702	Oticon Agil ITC	Oticon Agil ITC		Custom instru...
154703	Oticon Agil ITE	Oticon Agil ITE		Custom instru...
154704	Oticon Agil MIC	Oticon Agil MIC		Custom instru...
154705	Oticon Agil Pro CIC	Oticon Agil Pro CIC		Custom instru...
154706	Oticon Agil Pro ITC	Oticon Agil Pro ITC		Custom instru...
154707	Oticon Agil Pro ITE	Oticon Agil Pro ITE		Custom instru...

OK Cancel

6.2.2.1 Automatic route version setup from MDM

When custom or mould item releasing from MDM to other legal entity in which production ticket is enabled, that item should be automatically added into route version. For this purpose we need to setup a category field to define the item category and based on that category it should select the route number.

If the route is not matching with item category, the infolog shows the message “The route setup is not matching in this item”.

Route details (1 - it50) - Route number: PTR000000004, Custom hearing instrument, Route number: PTR000000004

File Route

Edit Route Delete Route Approve Create case Associate with case Add to case log Refresh Export to Microsoft Excel List Attachments

Maintain New Process Product change

PTR000000004 : Custom hearing instrument

General

Route number: R000000004

Name: Custom hearing instrument

Item group:

Approved by: 000001

Approved:

Category: Hearing Aids (HA)

Versions

+ Add - Remove Approve Activation Route feasibility Product change Add products

Item number	Configuration	Size	Site	From date	To date	From qty.	Active	Approved by	Approved
101551			SIT01			0,00	✓	000001	✓
101569			SIT01			0,00	✓	000001	✓
154341			SIT01			0,00	✓	000001	✓
154694			SIT01			0,00	✓	000001	✓

The newly added category field should match with the releasing item category or category parent from MDM. For example, when custom item is releasing from MDM, it should select the custom hearing instrument route, If it is mould item, it should select custom mould route.

Based on route selection, the item should be added in the route version. It should automatically approved and activated. But we need to check whether the item is already activated to any other route or not. In such case, we should only automatically approve not activated.

6.2.3 Default setup for start and report as finished

When doing Start or Report as finish of an operation, in standard AX system pops up a form where user has the possibility to choose when and how the route card and picking list should be posted. Since this setup will be same for all production; it can be defaulted. For this purpose we need a setup to define default selections for route card and picking list journal creation and posting on Start and Report as finished. A new parameter ‘Validate mandatory’ has to be added to specify system action if mandatory operation is not reported. This field will have value : None, Warning, Error

6.2.4 Barcode operations

The Scan field on production ticket form can be used to scan and select order number, an operation, to initiate start or report as finished status for an operation. For this purpose we need to create and setup new operations in Barcode operation.

New Barcode operations:

If a barcode starts with character mentioned in operation identifier field in Inventory management/ Inventory and warehouse management parameters then system looks into the Barcode operation setup to identify which operation has to be performed.

1. WDHAssignOperation: Assign the scanned operation to the production ticket.
2. WDHStartOperation: This barcode operation will start the route operation for production order which has included checkbox ticked.
3. WDHReportAsFinishOperation: This barcode operation will report finish the route operation for production order which has included checkbox ticked.

If a barcode doesn't start with character mentioned in operation identifier, then look for sales/service order number. If only one valid (not cancelled) production ticket exists for scanned order number then fetch information of that production ticket in the form. In case of multiple valid production tickets are linked to the order, fetch information of the first production ticket and show information 'More than one production ticket exists for this order, review and select appropriate production ticket from production ticket dropdown'.

6.2.5 Picking list modification

In standard AX it is possible to create pick list when a particular operation has been reached. This setup is done on BOM lines. Since BOM will be maintained out of AX, we need a modification to create picking list based upon call of an action -

'createPickingListProduction'. Also Create a new action 'postPickingList' to post the picking list journal. Refer FDS_WP20002, sec 6.4.44 for details.

6.2.6 Production order Report as finished action

Once all the operations are reported as start and finished, the production order has to be finally Reported as finished and End. This is needed to get physical and financial postings generated. To automatically perform 'Report as finished' and 'End' status on a production order we need an action 'doReportAsFinishedAndEnd'. Refer FDS_WP20002, sec 6.4.44 for details.

6.2.7 Picking list posting

In order to avoid situations where a picking list cannot be posted due to the fact that inventory is closed for physical and financial transactions within a given month, the system must automatically update date for the respective picking list line to today's date.

In below example (Production control > Picking list), it would not be possible to post the picking list, as the picking list was created on June 5. Going forward, the system must therefore update this date automatically once a user tries to post a picking list.

Production journal (1 - it50) - Journal: PCJ0001885, Pick, Posted: NoJournal type: Picking list

File New Delete Lines Validate Post Create new Log Picking list Print

Show: Open

Overview General Setup Blocking History

Production journal lines, picking list (1 - it50) - Date: 6/5/2019, Picking list journal: PCJ0001885

File New Delete Validate Post Log Functions Inventory Print

Production: 0000001128 Quantity: 1.00
Name: MAICO, CAPTO 5 CUSTOM ITC Started: 5.00

Overview General Reference Financial dimensions Inventory dimensions

Date	Lot ID	Item number	Proposal	Consumption	Unit
6/5/2019	LOT000058564	186074	1.00	1.00	pcs

June 2019

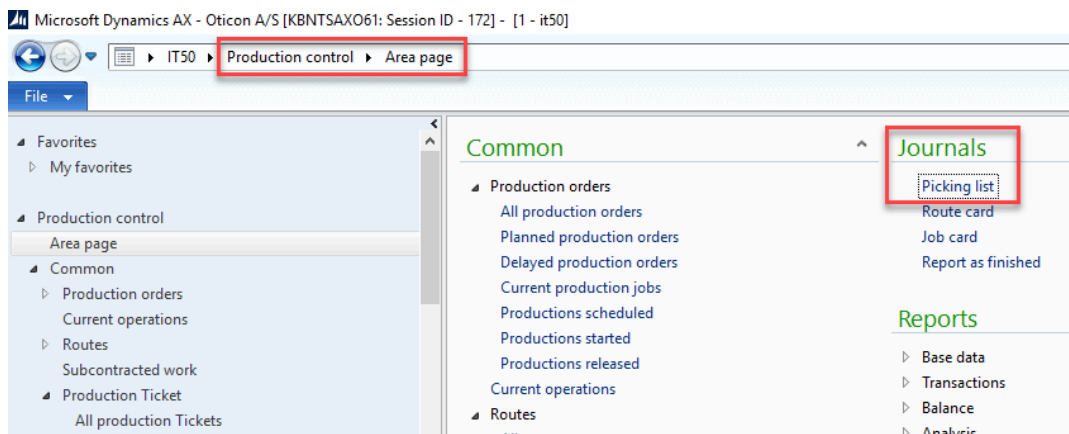
S	M	T	W	T	F	S
26	27	28	29	30	31	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	1	2	3	4	5	6

Pro Today

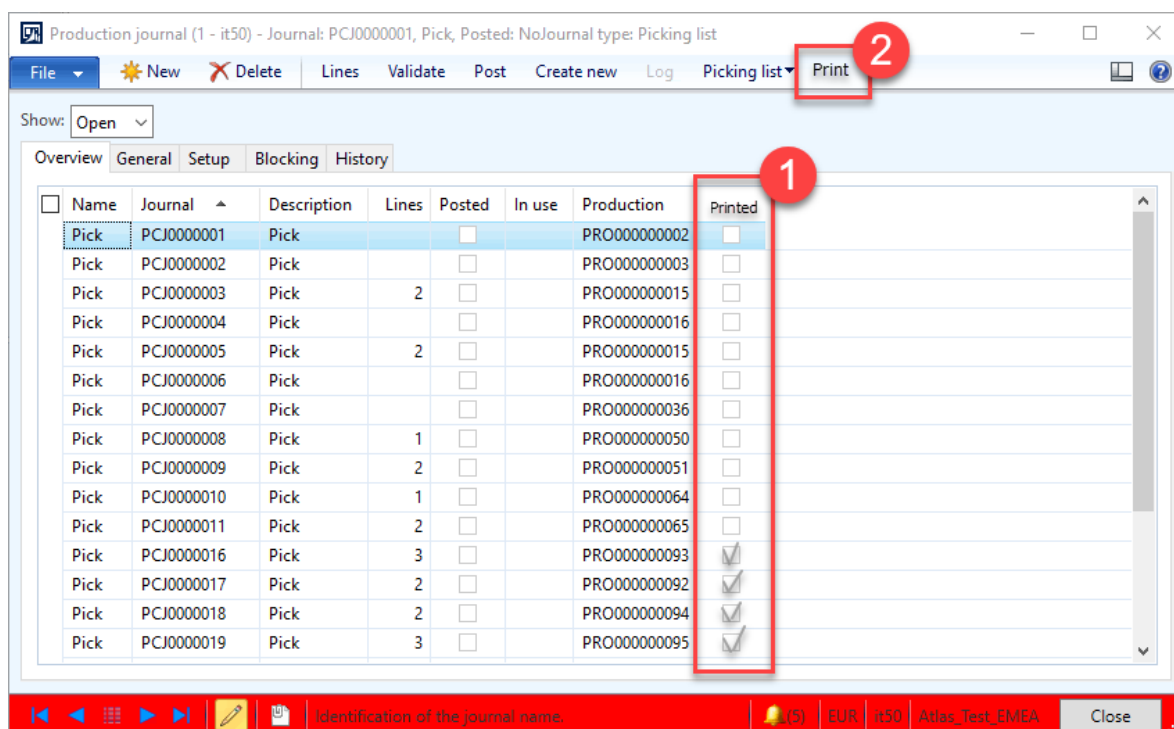
Index: 2.00 Started: 2.00 Released quantit

6.2.8 KIT picking list print

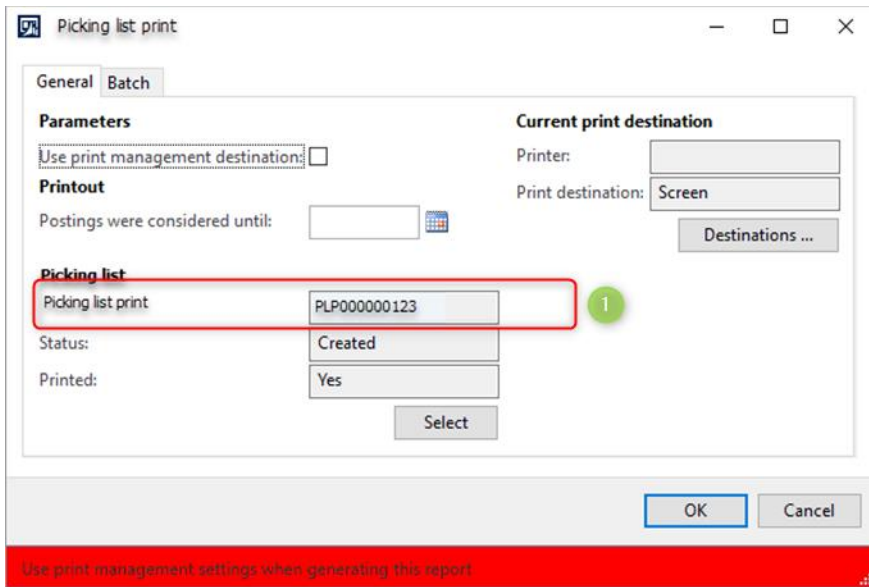
During the flow for local production, an operator will have to go to the warehouse and pick up the KITs that are required for each instrument currently being produced. To do so, each production ticket will have a picking list. The gathered overview of all picking lists can be retrieved from: Production control > Journals > Picking list.



By selecting 'Picking list' below form will appear. The picking list form is standard AX except for the red highlighted areas, which require customization. That is described below.



First, the form needs an additional column named 'Printed'. Once the picking list has been printed, the tick mark for the respective picking list should get ticked off. In above example, four picking lists have been printed. Secondly, the operator should be able to get such print by pressing 'Print'. The print should per default contain all those picking lists where 'printed' has not been ticked off yet unless the operator specifically selects the picking lists he would like to go onto the print. By selecting 'Print', below form should appear. The form is similar to the one generated for the collection letter, which is also standard AX (Accounts receivable > Periodic > Collections > Print/post collection letters). In below form a few fields have been removed and once has been changed, i.e. 'picking list print'.



Picking list print

General Batch

Parameters

Use print management destinations: ☐

Printout

Postings were considered until:

Current print destination

Printer:

Print destination: Screen

Destinations ...

Picking list

Picking list print: PLP000000123 (1)

Status: Created

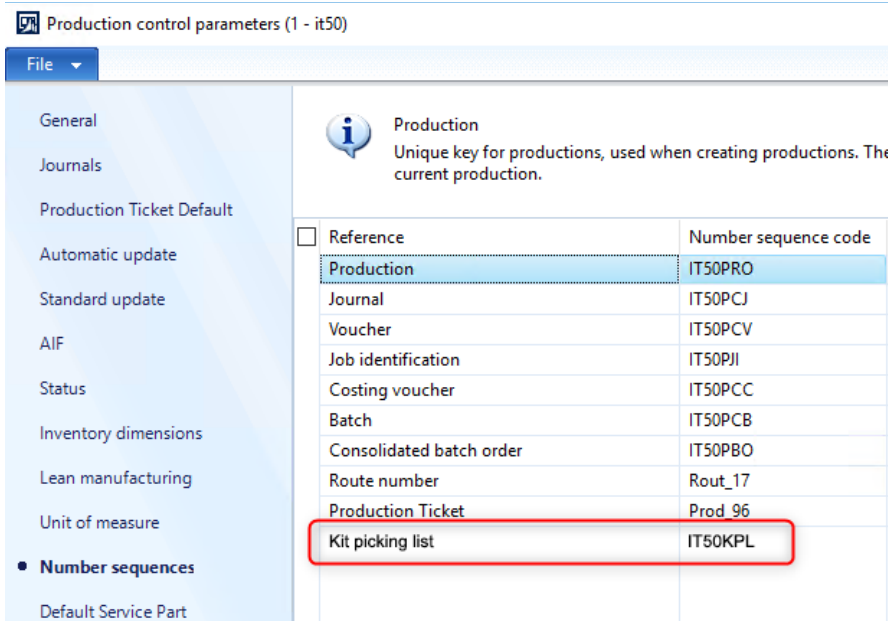
Printed: Yes

Select

OK Cancel

Use print management settings when generating this report

The title of the picking list print should contain a running number (screenshot above (1)). This number needs to be retrieved from a new number sequence “Kit picking list” maintained in Production control / Setup / Production control parameters / Number sequences:



Production control parameters (1 - it50)

File

General

Journals

Production Ticket Default

Automatic update

Standard update

AIF

Status

Inventory dimensions

Lean manufacturing

Unit of measure

• Number sequences

Default Service Part

Production

Unique key for productions, used when creating productions. The current production.

Reference	Number sequence code
Production	IT50PRO
Journal	IT50PCJ
Voucher	IT50PCV
Job identification	IT50PJI
Costing voucher	IT50PCC
Batch	IT50PCB
Consolidated batch order	IT50PBO
Route number	Rout_17
Production Ticket	Prod_96
Kit picking list	IT50KPL

In order to identify, which picking list was on which print , the field “Kit picking list number” on the picking list journal / General tab needs to be updated with the kit picking list number taken from the number sequence at printing. If a journal is printed another time, this value needs to be updated again.

Production journal (1 - it50) - Journal: PCJ0000001, Pick, Posted: NoJournal type: Picking list

File New Delete Lines Validate Post Create new Log Pickir

Show: Open

Overview General Setup Blocking History

Identification

Journal: PCJ0000001

Journal type: Picking list

Name: Pick

Description: Pick

Production: PRO000000002

Kit picking list number: KPL000123

Voucher

Voucher series: IT50PCV

Selection by: Posting

New voucher by: Change date or production

Reference

Purchase order:

Posting

Detail level: Details

Delete lines after posting: ☐

Automatic backflushing

Waiting for creation: ☐

Good quantity:

Error quantity:

Oper. No. 10

End job: ☐

The print should look like this:

The picking list print should contain as many lines as have been selected in the picking list overview. If none has been selected, the print should contain all those picking list lines that have not been printed yet.

All the information on the picking list print can be retrieved from the picking list lines, except for the box coordinate and BC room. These have to be retrieved from the box coordinate table in which each box coordinate location will have a reference to the sales order number. Below is an example of where the box coordinate and sales order reference can be identified.

Production Picking List Report (1)

File

Options

1 of 1 100% Find Next

Picking List Report

Change to BC room

Item id	Kit	Invent ref	Journal Id	Production ticket	Production	Ear	Box coordinate	Row column Display Value
182573	OPN 2, KIT 10 75 C001	SOR000002079	PCJ0002075	PRT000000710	PRO000001123	Left		
182574	OPN 2, KIT IIC 10 75 C005	SOR000001759	PCJ0002076	PRT000000787	PRO000001248	Right		
182574	OPN 2, KIT IIC 10 75 C005	SOR000001759	PCJ0002077	PRT000000787	PRO000001249	Left		

Summary:

182573	OPN 2, KIT 10 75 C001	1
182574	OPN 2, KIT IIC 10 75 C005	2

Sales order link on picking list overview

Production Journal (1 - it50) - Journal: PCJ0001885, Pick, Posted: NoJournal type: Picking list

File New Delete Lines Validate Post Create new Log Picking list Print

Show: Open

Overview General Setup Blocking History

☐	Name	Created date and time	Journal	Description	Lines	Posted	In use	Production	Order number	Printed
☐						No				
☐	Pick	3/19/2018 07:53:58 am	PCJ0001885	Pick	1	☐		PRO000001128		☒
	Pick	3/19/2018 07:53:58 am	PCJ0001886	Pick	1	☐		PRO000001129		☒
	Pick	3/19/2018 07:53:58 am	PCJ0002296	Pick	1	☐		PRO000001424		☒
	Pick	3/19/2018 07:53:58 am	PCJ0002509	Pick	1	☐		PRO000001286		☒
	Pick	3/19/2018 07:53:58 am	PCJ0002510	Pick	1	☐		PRO000001287		☒
	Pick	3/19/2018 07:53:58 am	PCJ0002517	Pick	1	☐		PRO000001357		☒
	Pick	3/19/2018 07:53:58 am	PCJ0002537	Pick	1	☐		PRO000001336		☒
	Pick	3/19/2018 07:53:58 am	PCJ0002538	Pick	1	☐		PRO000001337		☒
	Pick	3/19/2018 07:53:58 am	PCJ0002556	Pick	1	☐		PRO000001427		☒
	Pick	3/19/2018 07:53:58 am	PCJ0002557	Pick	1	☐		PRO000001428		☒
	Pick	3/19/2018 07:53:58 am	PCJ0002566	Pick	1	☐		PRO000001401		☒
	Pick	3/19/2018 07:53:58 am	PCJ0002567	Pick	1	☐		PRO000001402		☒
	Pick	3/19/2018 07:53:58 am	PCJ0002568	Pick	1	☐		PRO000001418		☒

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Production orders (1 - it50) - Production: PRO000001424, AMPLIFON, AMPLI-MINI 2 CUSTOM

File Production order Schedule View

Edit Split Delete Production order Batch order Estimate Report as finished Reset status Release End Start Log

Maintain New Process

PRO000001424 Delivery: 6/11/2019
Item number: 194933 Quantity: 1.00

Route number: PTK000000004 Total cost allocation: ☐

Jobs: ☐

Check route: ☐

Scheduling

Property:

Locked: ☐

Priority: 0

Measurement

Height:

Width:

Depth:

Density:

References

Item reference

Reference type: Sales order

Reference number: SOR000001647

Reference lot: LOT000057156

Origin

Original production:

6.2.8.1 KIT Storage Location on the Production Picking List Report

There is need to add storage location (**WDHInventTable.ReplStorageLocation**) of the KITS on the picking list report, this will enable the users to find the kits easily in the warehouse. Also the column titles for the fields in the "Summary" section should be added.

Production Picking List Report (1)

File

Options

1 of 1 100% Find | Next

Picking List Report

Item Id	Kit	Sales	Journal Id	Production	Ear	BC room
178937	OPN 1, KIT 10 NFM 75 C001	SOR000040765	PCJ0051731	PRO000024091	Right	1.2

Summary:

Item Id	Kit	Storage location	Quantity
178937	OPN 1, KIT 10 NFM 75 C001	XXXXX	1

6.2.9 On hold

Order Onhold should bring up the 'Order hold' form (same as when called from sales order). Once the on hold record is saved the state of production ticket will move to OnHold. Sales order linked to the production ticket should also get on hold flag (as it would have happened when called from sales order form). This should also trigger workflow state on sales order or service specification to go to OnHold status. (Not sure if we are considering this in customer service track)

Even though the production ticket moves to onhold state it will not stop the operator from moving ahead with production, i.e. they can still scan start and report as finished for operations. To stop production from moving ahead 'Lock for processing' checkbox on production ticket should be marked. This can be done only by creating a on hold record with a particular Hold code. For this purpose add new parameter 'Lock for reprocessing' on Hold code.

Order hold codes (1 - nl30)

File New Delete

Hold code	Description

Hold code:

Description:

General

Role:

Default for sales order: ☐

Hold code color:

Remove inventory reservations: ☐

Lock for processing ☐

When 'Ok' action on 'On hold' form is called from Sales order or Production ticket, system should check

1. if there is any open on hold record with on hold code where 'Lock for reprocessing' is true then set the production ticket state as on hold and mark 'Lock for reprocessing' as true.
2. if there are open on hold records but none of them have on hold code where 'Lock for reprocessing' is true then set the production ticket state as on hold and mark 'Lock for reprocessing' as false.
3. if all the on hold records are cleared then set the production ticket state as Active and mark 'Lock for reprocessing' as false.

When a production ticket is put to onhold, assign the mapped state corresponding to onhold code to related sales or service

The “Order holds” button needs to be shown on the “Sales order” form as well as on the “Sales order for order entry form”.

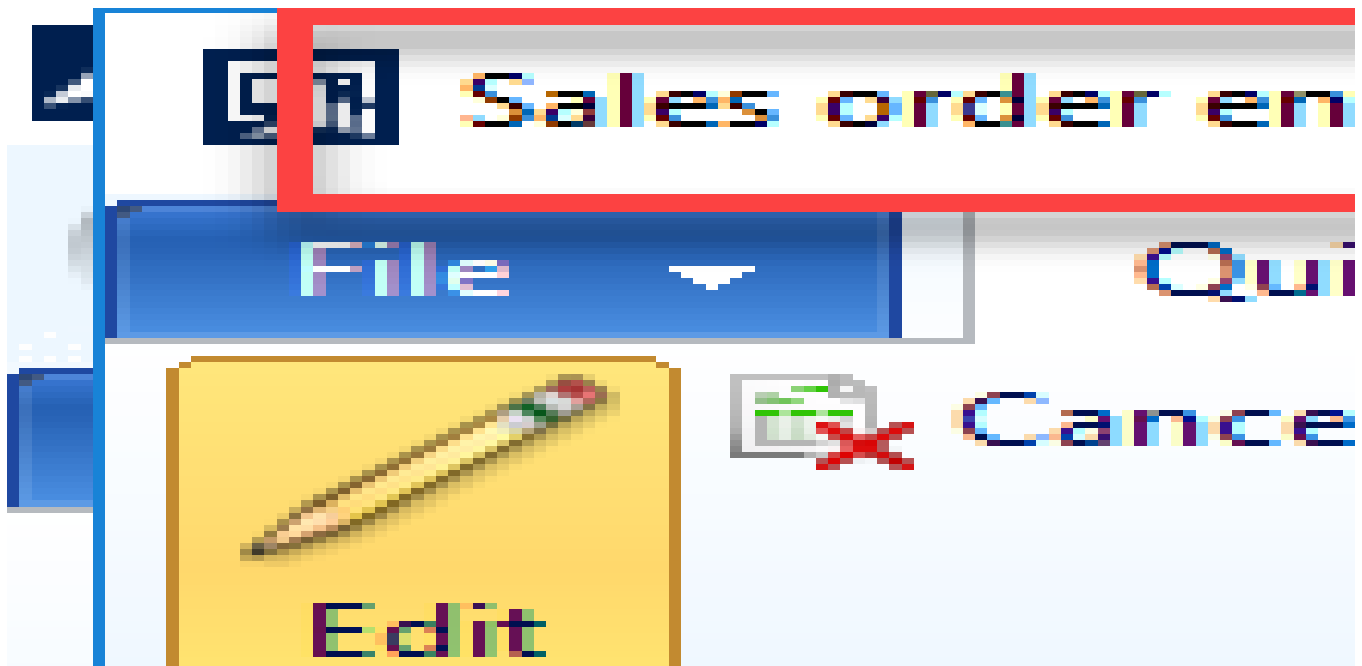
6.2.10 Order holds for sales order entry forms

Standard AX functionality of applying order holds existing on sales order form and all sales order list page (/Sales and marketing/Common/Sales orders/All sales orders and /Accounts receivable/Common/Sales orders/All sales orders). This button should be introduced to salesorder entry list page and details form

This button can be introduced on ‘all sales orders for order entry’ list page as shown below (/Sales and marketing/Common/Sales orders/All sales orders for order entry)



The button should be added to sales order entry form as shown below:



Clicking the button from above mentioned list page and 'sales order entry' form should trigger the order hold form as it does from sales order form:

Add 'Attachments' button to Onhold form. The attachment should open order header attachments for selected record in in order hold form. User should be able to edit records in attachments.

Order hold (1 - it50) - Cleared: NoReference number: SER000041695, New Record

File ▾  New  

Hold details **Override**

Show: **Active** ▾

Check out	Override checkout	Holds ▾	Order ▾	Attachments				
☐	Checked out	Checked out to	Reference number ▾	Hold code	Hold user ID	Days on hold	Checked out date	Hold date
☐			SER000041695	CUSAN ▾		0		

Hold notes **Order notes** Fraud notes Customer notes Customer

6.3 Production ticket maintenance

6.3.1 Transition log

In standard AX, start and finish of an operation creates an entry in log on production order. To get complete visibility of a n order the operation updates should create a record in workflow transition log as well. Add a new coloum 'Production action'- this will be filled in for production logs only- values will be either Start or Finished.

Transition log (1 - gb10) - Created date and time: 27-11-2018 08:51:15, krbo

Overview General															
☐	Document type	Document number	Header/Line	Line number	From workflow ID	FromState	From state name	To workflow ID	ToState	To state name	Action	Operator name	Created by	Created date and time	Batch Log Text
	Production ticket	PRT00034345	Header		R1	LAS	Laser	R1	LAS	Laser	End	Niklas	nijn	29-11-2018 08:10:12	
	Production ticket	PRT00034345	Line	1,0000	R1	LAS	Laser	R1	LAS	Laser	End	Niklas	nijn	29-11-2018 08:10:12	
	Production ticket	PRT00034345	Line	2,0000	R1	LAS	Laser	R1	LAS	Laser	End	Niklas	nijn	29-11-2018 08:10:12	
	Production ticket	PRT00034345	Header		R1	LAS	Laser	R1	LAS	Laser	Start	Niklas	nijn	29-11-2018 08:10:12	
	Production ticket	PRT00034345	Line	1,0000	R1	LAS	Laser	R1	LAS	Laser	Start	Niklas	nijn	29-11-2018 08:10:12	
	Production ticket	PRT00034345	Line	1,0000	R1	LAS	Laser	R1	LAS	Laser	Start	Niklas	nijn	29-11-2018 08:10:12	
	Production ticket	PRT00034345	Header		R1	MOD	Modelling	R1	MOD	Modelling	End	Nirmal	nide	29-11-2018 08:10:12	
	Production ticket	PRT00034345	Line	1,0000	R1	MOD	Modelling	R1	MOD	Modelling	End	Nirmal	nide	29-11-2018 08:10:12	
	Production ticket	PRT00034345	Line	2,0000	R1	MOD	Modelling	R1	MOD	Modelling	End	Nirmal	nide	29-11-2018 08:10:12	
	Production ticket	PRT00034345	Header		R1	MOD	Modelling	R1	MOD	Modelling	Start	Nirmal	nide	29-11-2018 08:10:12	
	Production ticket	PRT00034345	Line	1,0000	R1	MOD	Modelling	R1	MOD	Modelling	Start	Nirmal	nide	29-11-2018 08:10:12	
	Production ticket	PRT00034345	Line	1,0000	R1	MOD	Modelling	R1	MOD	Modelling	Start	Nirmal	nide	29-11-2018 08:10:12	
	Sales order	SOR000139284	Line	1,0000000000	GB10-004	OC	Order confirmed	GB10-004	RFP	Ready for production		Krzysztofa Bor...	krbo	27-11-2018 08:51:15	
	Sales order	SOR000139284	Line	1,1000000000								Krzysztofa Bor...	krbo	27-11-2018 08:51:15	
	Sales order	SOR000139284	Line	1,1000000000								Krzysztofa Bor...	krbo	27-11-2018 08:51:14	
	Sales order	SOR000139284	Header			OC	Order confirmed		RFP	Ready for production		Krzysztofa Bor...	krbo	27-11-2018 08:51:14	
	Sales order	SOR000139284	Line	1,0000000000	GB10-004	OC	Order confirmed	GB10-004	RFP	Ready for production		Krzysztofa Bor...	krbo	27-11-2018 08:51:14	
	Sales order	SOR000139284	Line	1,1000000000								Krzysztofa Bor...	krbo	27-11-2018 08:51:14	
	Sales order	SOR000139284	Header			OE	Order entry		OC	Order confirmed		Krzysztofa Bor...	krbo	27-11-2018 08:51:03	
	Sales order	SOR000139284	Line	1,0000000000	GB10-004	OE	Order entry	GB10-004	OC	Order confirmed		Krzysztofa Bor...	krbo	27-11-2018 08:51:03	
	Sales order	SOR000139284	Line	1,1000000000								Krzysztofa Bor...	krbo	27-11-2018 08:50:21	
	Sales order	SOR000139284	Line	1,0000000000					RFP	Ready for production		Krzysztofa Bor...	krbo	27-11-2018 08:50:20	

6.3.2 State scanning Production ticket

To facilitate bulk reporting of production ticket operation state, create a new form 'State scanning production ticket'. This form can be accessed from menu item 'State scanning production ticket' in Production control/Common.

State scanning production ticket (1 - it50)

User ID:

Scan:

Operation:

Assign to:

☐	Include	Order number	Production Ticket	Production	Operation	Box coordinate	BC room
☒		SOR000001478	IT50-000003	PRO00000...	SLI		
☒		SOR000001478	IT50-000003	PRO00000...	SLI		
☒		SOR000001480	IT50-000004	PRO00000...	MOD		
☒		SOR000001480	IT50-000004	PRO00000...	MOD		

Log:

From this form it should be possible to scan both sales order barcodes and production ticket barcodes. System should add production ticket lines in the grid and it should be possible to add multiple production ticket lines to the grid. The lines scanned into the form should appear in the sequence that they are scanned in. The sequence is currently dependent on the value of the order number where it can be sorted either high to low or low to high (see below picture where it is low to high). Once an operator scans e.g. ten orders in a row, the sequence that they appear in in the form is confusing compared to the sequence he scanned the orders in. To make it more user-friendly, the sequence that the orders appear in should be based on the sequence that the operator scans in, i.e. the first order he scans should be presented in the bottom of all the scanned lines and the last order scanned should be the lines that appear first in the sequence.

State scanning production ticket (1 - it50)

User ID:

Scan:

Operation:

Assign to:

☐	Order number	Production Ticket	Production	Operation	BC room	Box coordinate
☒	SOR000012384	PRT000005571	PRO00000...	QC	G.4	AE
☒	SOR000012384	PRT000005571	PRO00000...	QC	G.4	AE
☒	SOR000012386	PRT000005576	PRO00000...	QC		
☒	SOR000012386	PRT000005576	PRO00000...	QC		
☒	SOR000012431	PRT000005598	PRO00000...	LAC		
☒	SOR000012443	PRT000005604	PRO00000...	LAS	1.2	PRODUCTION
☒	SOR000012443	PRT000005604	PRO00000...	LAS	1.2	PRODUCTION

User can select an operation in the operation field and worker in assign to field. The “Log” field should work exactly as on the production ticket form.

Start button will start the operation, as mentioned on the form, for all the production orders which are included in the line grid.

Finish button will finish the operation for all the production orders in the line grid which are included in the line grid.

Enter button will assign the operation to all the production orders in the line grid which are included in the line grid.

Before system executes Start, Finish and Enter actions, system should verify that all included production orders in the grid are in same operation. In case all production orders are not in the same operation then throw error ‘For bulk update of operations all included production orders should be on same operation’

Clear button should clear all entries from the form, so the user can start all over again..

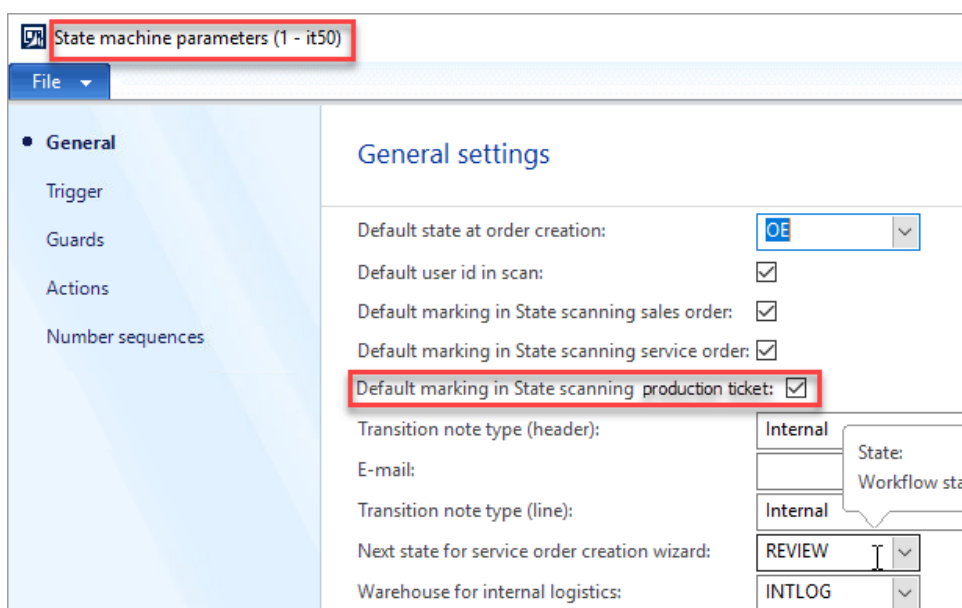
This form is used for changing operation and other relevant information (via barcodes and manual entry) of sales orders/production tickets (one or more sales order/ lines). The form can stay on the screen all the time; the operator always has to enter/scan the following information after each other:

1. User ID of the user
2. Sales order number/Production ticket number (to be scanned from the job order)
3. Operation
4. Assign to (this is optional)

It is explicitly designed for production ticket. It should be possible to change operation for only included production ticket lines. All included lines should be changed instantly to the selected Operation and all actions on that operation should also be executed. (Note: Quality order form should not popup automatically when operations are executed from scan form.)

The fields for ‘box coordinate’ and ‘BC room’ should only be visible when the design for box coordinate is enabled. As other sites might not enable the box coordinate, the fields should not create disturbance for them.

Finally, default marking should be available for the ‘State scanning production ticket’ form – just as it is for the sales and service order state scanning forms. It should be controlled via below setup. When it is ticked, the lines that are scanned in the form should be also ticked by default. If it is unticked, then the opposite. The setup is found here: Sales and marketing > Setup > Workflow > State machine parameters > General.



The screenshot shows the 'State machine parameters (1 - it50)' configuration window. The 'General settings' tab is active. The 'Default marking in State scanning production ticket' checkbox is checked and highlighted with a red box. Other settings include:

- Default state at order creation: OE
- Default user id in scan: ☒
- Default marking in State scanning sales order: ☒
- Default marking in State scanning service order: ☒
- Transition note type (header): Internal
- E-mail:
- Transition note type (line): Internal
- Next state for service order creation wizard: REVIEW
- Warehouse for internal logistics: INTLOG

6.3.3 Modifications on Sales order form

Add new field 'Production state' on sales order header and sales line to show the current operation state. Header production state should show current production ticket operation. Sales line production ticket state should show operation on corresponding production orders linked to that production ticket.

Add 'Production ticket' button on sales order header. This should open up production ticket linked to this sales order.

Sales order (1 - nl30) - Sales order: SOR000088697, A.G. Bellschool, Sales order: SOR000088697

File | Quick access | Sales order | Sell | Manage | Pick and pack | Invoice | Retail | General | Insurance

Maintain | Audioqram | **Production ticket** | Attachments | Quantity discount | Spare- and commercial pa... | Change State | Job order preview | Job order | Internal order print | Order labels | Address label | Document scanning label | Sales order confirmation | Journals

SOR000088697 : 47950 - A.G. Bellschool | State: RFP, ready for production | IC State: | **Production State: ATS** | Oper

• Brand information

Commercial segment: Oticon

• Sales order header | 47950

• Sales order lines

Type	Variant number	Item number	Product name	Sales category	Quantity	Unit	Production type	Sales dimension template	Discount percent	Coverage group
✓		154709	Oticon Alta CIC		1,00	pcs	Central			Man
		800-000-302	Standaard 2 jaar garantie		1,00	pcs				

Add new field 'Production state' on sales order overview to show the current operation state. Production state should show current production ticket operation.

Add 'Production ticket' button on sales order overview header. This should open up production ticket linked to this sales order.

Microsoft Dynamics AX - Oticon A/S [KBNDVAX041: Session ID - 8] - [1 - nl30]

File | Sales order | Sell | Manage | Pick and pack | Invoice | Retail | General | Insurance | Transportation management

Sales order | Edit in grid | Audioqram | From all | From journal | Totals | Order events | Order credit | **Production instructions** | **Production ticket** | Refresh | Export to Microsoft Excel | Generate from template | Attachments | Notes

Production

• Favorites

- My favorites
- All production orders
- Current operations
- Automatic create picking list
- Output orders
- State edit
- All service orders
- All sales orders

All sales orders

Time to filter

Sales order

Sales order	Customer account	Name	Invoice account	Channel	Order type	Status	City	Production state	ZIP/post
SOR000088695	47950	A.G. Bellsch...	47950		Sales order	Open order	AMSTERDA...	ATS	1063 B

The 'Production ticket' button should bring up production ticket form. If only one valid (not cancelled) production ticket exists then bring up production ticket form with information of that production ticket. In case of multiple valid production tickets are linked to the sales order, bring up production ticket form with information of first production ticket and show information 'More than one production ticket exists for this order, review and select appropriate production ticket from production ticket dropdown'.

6.3.4 Update configuration Changes and BOM on Production ticket

Configuration for a custom instrument can be changed from sales order or from production ticket. Changes in configuration can potentially result in changes in BOM as well. If 'Production ticket' is enabled then system should not create new config id when configuration is edited- either from sales order or from production ticket or from service order/card, it should only save attribute values in relevant tables. Note: On production ticket, for edit configuration the history should be maintained in the same way as on the Sales Order.

Based upon modified configuration system can propose a new BOM. At present we have two sources of BOM- EPDS BOM service, and AX BOM on product model. System should check if the selected BOM has different items than on the pick list that already got created for production orders. In such case system should provide infolog to review the BOM, " Production BOM components are different than proposed BOM. Please review the production components on pick list".

Add 'Create BOM' button on Production ticket/ Production ticket lines. This button should bring up production BOM when reference type on production ticket is 'sales order' and Service BOM if the reference type on production ticket is 'service order'.

Production BOM: If the source of BOM is EPDS BOM (i.e Production BOM is marked, path-Releaed product/ Engineer/BOM service/), then bring up the BOM proposal form (same as we get on clicking Create BOM on sales order lines/ product and supply). If the source of BOM is not EPDS then this button should be disabled.

Add 'Additional Consumption' button on Production ticket/Production ticket lines. This button should bring up the Picking list journal form if reference type on Production ticket is 'Sales order'.

File Production Ticket

Edit Cancel Start Finish Order holds Maintain Operation

Non conformances Quality orders Picking list Route card Reported as finished All Journals Manage quality

Edit configuration Log Refresh Export to Microsoft Excel Production details List

NL30-00000000 Delivery date: 05-02-2019 Operation: QC Assign to:

Scan Error:

General
Identification Production Ticket: NL30-00000000 Production ticket status: Created
Order details Order number: SOR000129383 Reference type: Sales order Order State: SLI
Groupings Pool: Use Impression: Priority: Lock For Processing:
Others

Service specification
 Service order
 Service card history
 Service consumption

Production Ticket Lines

Add line Remove Create BOM Additional consumption Change serial number Colour change Service detail

Include	Production	Item number	Name	Serial number	Quantity	Ear	Operation	Service type
☐	~~~~~					None		

Attachments
 Add line Remove

Order number	Reference type	Type	Created date and time	Created by	Name
This grid is empty.					

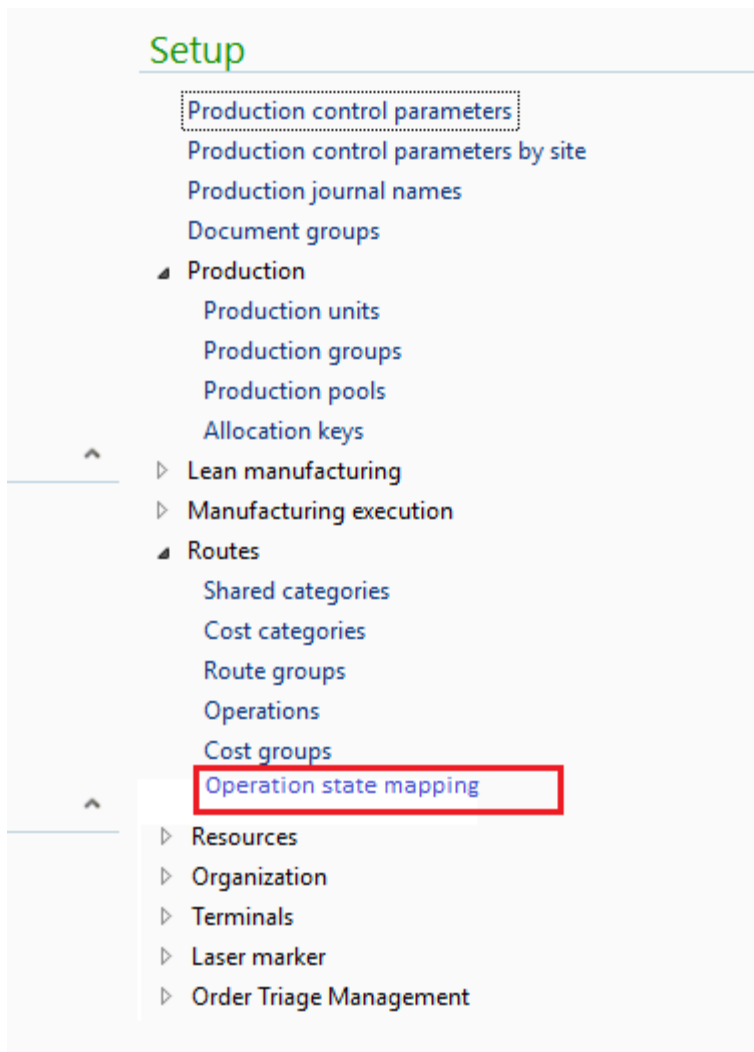
6.3.5 Actions on Production ticket

Following validations and actions are needed to ensure that a user doesn't miss any step from reporting :

Actions for validating different kind of checks :

1. validateQualityorder: This action should validate if there are any open quality orders for that production order. Refer FDS_WP20002, sec 6.4.44 for details
2. validateNonconformance : This action should validate if the non-conformances are raised against failed quality orders for a production order and if non-conformances raised are completed or not. Refer FDS_WP20002, sec 6.4.44 for details
3. checkProdMandatoryOperation : This action should validate if all mandatory operations are reported or not. Refer FDS_WP20002, sec 6.4.44 for details
4. setStateOnOrder: On QC control we should trigger workflow state RFPICK on sales order or service order. To make it possible

Add new setup 'Operation state mapping' in production control/Setup.



In Operation state mapping, we can map a workflow state for an operation in a route.

☐	Route	Operation	Mapped state ▲
			▼

The action `setStateOnOrder` will be called from an operation on a production order and system will put corresponding sales order line/service order line to workflow state as set in Operation State mapping setup. Refer FDS_WP20002, sec 6.4.44 for details.

6.3.6 Option based pricing

6.3.7 Service and remake

Production features like scheduling, capacity planning, etc. are much needed for service as well, like for new production. This created the need of linking service orders to production order as well. The need to combine service and sales shipment together in picking put forward the suggestion of using a notional stock item to represent service. The stock item representing service will be maintained at 0 cost price and will have a separate warehouse to separate it from other stocked items to omit any confusion during counting or material planning. These items will also be blocked for sale or purchase. Add new checkbox field 'Service stock part' on Released product/ General/Administration, this field is to identify such products which will be used to create production order for items came in for service.

Since different kind of service like repair, remake, modification, swap etc. follow different routes, we need to create stock items to represent each route in service. Add 'Default service type' to Production control parameters. This setup will hold items to be used for production order creation for each kind of service type (Repair, remake, SWAP etc). and product type (Instrument, Flex mold, Lite tip etc) and OTS/Custom combination. Default service item is a fall back setup which will be used if system will not find any valid record in 'Default service item per service type'.

Production control parameters (1 - gb10)

File ▾

- General
- Journals
 - Production Ticket Default
 - Automatic update
 - Standard update
 - AIF
 - Status
 - Inventory dimensions
 - Lean manufacturing
 - Unit of measure
 - Number sequences
 - Default Service part**

Set up default production items for service and repair order

Default service item per service type

✚ Add ✖ Remove

Service type	Category hierarchy	(Category)	Item

Default service item

Create an action 'createServiceProductionTicket', this action will create production order and production ticket for service specification. The action createServiceProductionTicket will be called from service order state. This will create production ticket and production order. System should identify service stock item from Default service part setup in production control parameters. Action should create production order for each service specification using item from Default service part setup and all items having same route should be added in same production ticket.

User will be able to perform following actions from production ticket form- edit configuration, change audiogram, change serial number, change colour, BOM and additional consumption. Users will be able to access service order, service specification, service card history from production ticket.

Production ticket should reflect changes done in the service order or service specification. If product configuration has been changed on the service order form or service specification form then those changes should also be visible on the production ticket.

Additionally user will also be able to open the production ticket from the service order.

File Production Ticket

Edit Cancel Start Finish Order holds Maintain Operation Manage quality Non conformances Quality orders Picking list Journals Route card Reported as finished All Edit configuration Log Production details Refresh Export to Microsoft Excel List

NL30-00000000 Delivery date: 05-02-2019 Operation: QC Assign to:

Scan Error:

General

Identification Production Ticket: NL30-00000000
 Production ticket status: Created

Order details Order number: SOR000129383
 Reference type: Sales order
 Order State: SLI

Groupings Pool: Use Impression: Lock For Processing:

Others Service specification
 Service order
 Service card history
 Service consumption

Production Ticket Lines

Add line Remove Create BOM Additional consumption Change serial number Colour change Service detail

Include	Production	Item number	Name	Serial number	Quantity	Ear	Operation	Service type
☐	~~~~~						None	

Attachments

Add line Remove

Order number	Reference type	Type	Created date and time	Created by	Name
This grid is empty.					

When the reference type on production ticket is service order, then :

1. 'Create BOM' button on Production ticket/ Production ticket lines should bring up Service BOM. 'Create BOM' for reference type service should call the same logic as if user has clicked Service BOM button on service consumption /service specification form. Any action taken on service BOM form should get applied to the respective service specification.
2. 'Additional consumption' button on Production ticket/ Production ticket lines should bring up Additional consumption form as if user has clicked Additional consumption button on service consumption /service specification form. Any action taken on Additional consumption form should get applied to the respective service specification.
3. 'Edit configuration' button on Production ticket should bring configuration from related service specification and when modifications are done it should save data back to service specification.
EPDS should be called with the service assortment instead of the sales assortment (as defined in the fields "Service assortment slot id" and "Service assortment unique Id" (path: Product information management / Setup / Product information management parameters per legale entity)
4. 'Audiogram' button on Production ticket should bring the audiogram form for the selected production ticket line.

The production order created should display the 'Service item' that is being serviced.

Production orders (1 - n30) - Production: PRO000000055, Repair item

File Production order Schedule View

Edit Split Delete Production order Batch order Estimate Report as finished Reset status Transactions Trace On-hand Update registrations Lot Inventory BOM Formula All jobs Service BOM Order details Consolidated on-hand Add to Remove from Refresh Export to Microsoft Excel List Attachments Attachments

PRO000000055 Delivery: 14-12-2018
Item number: 800-000-301 Quantity: 1,00

General

Identification
Production: PRO00000001
Item number: 800-000-301
Name: Repair item
Service Item 154726
Production
Type: Standard
Quantity: 1,00
Delivery: 14-12-2018
Time: 10:00
Color:

Status
Status: Scheduled
Scheduling status: Operations scheduled
Remain status: Route consumption
Quality order status:

Groupings
Pool:
Production group:

Date scheduled
Start date: 14-12-2018 Start time: 07:00
End date: 14-12-2018 End time: 13:00

Setup

References

Item reference
Reference type:
Reference number:
Reference lot:
Origin
Original production:

Summary reference
Reference production: PRO000000055
Reference level: 0

Planned-order reference
Master plan:

There should be a new columns "Serial number used" and "Item number used" added to the Production Ticket Lines fast tab. Values in these columns should be updated accordingly:

- "Serial number used" - In case of having service type "Swap" the value from this column should be transferred to the service specification lines and the update the "Serial number used" column located on the service specification lines. Based on that swap consumption journal will be updated with the new serial number for swapped instrument. In case when the change of the serial number has been done using the "Change serial number" button (for all service types except for swap) the column "Serial number used" on the production ticket should be updated with new serial number for the item.
- "Item number used" – in case of the swap scenario the value in this column should be updated with the value from the column "Item number used" located on the service specification form (for the scenario when we swap the instrument with the different item number). When color of the instrument has been changed using the "change color" button, the value in the "Item number used" column located on the production ticket should be updated with the new value for item number and this new value should be also updated in the "Item number used" column on the service specification form.

In case when service type of the service specification related to the production ticket line is swap then column "Serial number used" should be possible to edit, so user can update it with the new serial number for swapped instrument.

Production Ticket (1 - it50) - Production Ticket: PRO000000177, Completed

File Production Ticket

Edit
 Cancel
 Start
 Finish
 Order holds
 Non conformances
 Quality orders
 Current quality orders
 Current non conformances
 Picking list
 Route card
 Reported as finished
 All
 Edit configuration
 Audiogram
 Refresh
 Export to Microsoft Excel

Maintain Operation Manage quality Journals Production details

PRO000000177

Delivery date: 04.04.2019 Operation: SWAP Assign to:

Scan: Infolog:

General

Production Ticket: PRO000000177
 Production ticket status: Completed
 Box coordinate:
 BC room:

Order details

Order number: SER00000000
 Reference type: Service order

Groupings

Pool:
 Priority: ☐

Others

Use Impression: ☐
 Hold Production: ☐

Production Ticket Lines

Add line
 Remove
 Service BOM
 Additional consumption
 Configuration history
 Change serial number
 Colour change
 Service Detail

☐	Include	Production	Item number	Name	Serial number	Quantity	Ear	Operation	Service type
☒			PRO000000328	190469	SIYA 2, MRITE 312 2.4G NFM PB C091	54737092	1,00	None	SWAP

Line details

Attachments

Add line
 Remove

☐	Order number	Reference type	Type	Created date and time	Created by	Name
This grid is empty.						

AX - Oticon A/S [KBNTSAX061: Session ID - 133] - [1 - it50]

IT50 Service and repair Common Service orders & quotations All service

File Service order Retail General

Service specification (1 - it50) - Service order: SER000000008, AUDIOCLINIC S.R.L.

File Service specification Recalculate

New
 Delete
 Change
 Totals
 Sales line shared comment line
 Add Communication codes
 Communication code
 Production instructions
 Proforma service quotation Send
 Change State
 State history
 Service card
 Show scanned document
 Attachments
 Change serial number
 Colour change
 Service card history
 All history
 Audiogram
 Retrieve battery serial number
 Configure line
 Edit configuration
 Configure line

Maintain View Communication Service card Audiogram Battery Configure line

Service specification

☐	Available physical	Serial number	Ear	Batch number	Battery serial number	Item number	Product name	Search name	Serial number used	Batch number used	Item number used	Communication codes
		80366277	Right			192304	Oticon Siya 2 Custom IIC	Oticon Siya 2 Custom IIC				

Buttons “Add” and “Add additional” related to service consumption should be added to the production ticket form. Button “Additional consumption” should be renamed into “Service BOM-Additional”. All buttons related to the service consumption should be gruppued into one section called “Service consumption”.

When “Add” button is pressed then a form with service consumption should appear, so user can select one item from the list and for this selected item a consumption journal will automatically be created in the same way as consumption journal is created on the service specification form. When “Add additional” button is pressed then a form should appear with service consumption and user can select one item from the list and for this selected line consumption journal will automatically be created in the same way as additional consumption journal is created on the service specification form (such journal has blank financial dimension for customer).

In case when service consumption is done for serialized item there should be additional field “Serial number” visible on the Add and Add additional form to register serial number for the service part. Serial number of the service part should be updated accordingly in the column “Serial number used” on the service consumption journal.

~~A new Boolean field “Stock consumption” should be added to the Add and Add additional forms. Default value for this field should be “Yes” and this value should update accordingly value in “Stock consumption” column on service consumption grid of related service order. Same validations need to take place, as if the checkmark is updated on the service consumption itself on the service specification form.~~

The Production ticket button on service order should bring up the production form. If only one valid (not cancelled) production ticket exists then bring up production ticket form with information of that production ticket. In case of multiple valid production tickets are linked to the service order, bring up production ticket form with information of first production ticket and show information ‘More than one production ticket exists for this order, review and select appropriate production ticket from production ticket dropdown’

A new button “Order holds” should be added to the service order action pane. Functionality behind this button should be the same as is for the sales order.

The screenshot shows a software interface for a service order. At the top, a header bar displays the order details: "SER000000211 - ACUSTICA TRENTINA S.R.L., Sales order: SER000000211". Below this is a menu bar with tabs for "File", "Service order", "Retail", and "General". The "Service order" tab is active, showing a toolbar with various icons. The "Order holds" button, represented by a document icon with a red 'X', is highlighted with a red rectangle. Below the toolbar, a status bar shows "SER000000211 : CLO2205 - ACUSTICA TRENTINA S.R.L." and "State: RFREP, Ready for repair". The main form area is divided into sections: "Customer information", "Order information", "Financial dimensions", and "Service information". The "Service information" section is expanded, showing fields for "General", "Service quotation", "Administration", "Product", and "Vendor information".

New column called “New item” should be added to the production ticket lines. Type of this column should be boolean and it should be active only if service type of related service order is “Swap”. Value in this field should be inherited from the “New item” column on the service specification and updated accordingly, no matter on which form update has been done (production ticket or service specification). Whenever the value of the “New item” column is changed on the production ticket then the column “New item” on the service specification consumption journal should be updated with the same value (for the line where field “Serial number used” has the same value as “Serial number used” on the production ticket). In the same way if value of the column “New item” has been changed in service consumption journal then the value in column “New item” on the production ticket should be updated with the same value.

6.3.7.1 Add Battery serial number & Battery serial number used field on the production ticket form for service orders

There is a need to add new columns “Battery serial number” and “Battery serial number used” added to the Production Ticket Lines fast tab. Values in these columns should be updated accordingly:

- “Battery serial number” – If “Exist in global device card” is set to TRUE on the product, then this field should be updated with the battery serial number from the service card (WDHServiceCardDevice. OrigBatterySerialNum). Should be non editable
- “Battery serial number used” – In case of Swap/ Substitute scenarios for products with Li- Ion battery, if the users updates the battery serial number used field on the Production ticket, then same should be updated in the Service specification form of the associated service order.

In case of service type Repair/Swap/Substitute, where a battery is consumed separately in the Service consumption form, then the serial number updated for the battery in the Serial number used field on the service consumption form should be updated in the battery serial number used field on the production ticket as well as Service specification form. Battery item for Li- Ion products will have “Update in global device card” set to TRUE.

PRT000014132 Delivery date: 05-02-2021 Operation: QC Assign to: Scan: Infolog:

General

Production Ticket: PRT000014132 Order details: Order number: SER000104440 Pool: Others: Use Impression: Hold Production: BC room: Reference type: Service order Route number: PTR000000001 Name: Service - Repair

Production Ticket Lines

Add line Remove Configuration history Change serial number Colour change Service Detail Maintain Consumption

Include	Available physical	Production	Item number	Name	Serial number	Battery serial number	Serial number used	Battery serial number used	New item	Q
☒		PRO0000206...	156794	Oticon Alta2 Pro IIC	U194357657					

Line details

Attachments

Add line Remove

Order number	Reference type	Type	Created date and time	Created by	Name
SER000104440	Service order	Cen...	02-02-2021 10:17:40	ancl	HML...
		Inte...	03-02-2021 16:57:43	Imcg	new ...

6.3.7.2 Production bom journal for service part consumption

At present when spare parts are added to service consumption, system creates a service consumption journal and add items to the journal which later on gets posted on order release state. There are multiple ways of adding spare parts to service consumption- 'Service BOM', 'Service BOM-additional', or adding consumption lines manually using 'Add' or 'Add additional' button. With this change, when a service order has a production ticket linked to it then for each service specification system should add consumption lines to corresponding production order BOM (instead of creating service consumption journal). The BOM will get posted when production order will be reported as finished.

Add new field 'Operation for picking list' (path: Production control/ production control parameters/ Default service setup) to mention the operation at which system should create picking list.

File

General Journals Production Ticket Default Automatic update Standard update AIF Status Inventory dimensions Lean manufacturing Unit of measure Number sequences

Default Service Part

Setup default production items for service and repair order

+ Add - Remove

Service type	Route number	Name	Operation for picking list
Repair	PTR000000001		REP

Default service item: S-SWAP

Enable route based service setup: ☒

Mapping of fields in production BOM is as follows-

1. Add item number as mentioned in service consumption,
2. Set quantity and estimated field in bom journal as quantity mentioned in service consumption.

- Set serial number on BOM line/picking list line as mentioned in service consumption.

The screenshot shows the 'Service specification' window with the 'Service consumption' table highlighted. The table has columns: Item number, Product name, Serial num..., and Quantity. The data rows are:

Item number	Product name	Serial num...	Quantity
M51734	PONTO PLUS POWER, L, CBE		1.00
M51732	PONTO PLUS POWER, L, MBR	45886518	-1.00
M52636	PONTO 3 POWER, L MBR	ND663344	1.00

- Set Oper. No. field in 'General' tab as operation number for the corresponding service type mentioned in Default service part > 'Operation for picklist'.
- Financial dimension logic should be copied as it is on service consumption journal creation. Customer dimension should be blank when lines are added through Add additional or Service BOM additional buttons.

6.3.7.3 Default warehouse for the production ticket

A new parameter "Default warehouse for service" should be added in the production control parameters (Path: Production control/Setup/Production control parameters/Default Service Part) to specify the warehouse that should be used for the transactions on the production ticket with the specified service type – product category – service stock part relation. Default warehouse should be used only for the production tickets having reference type of "Service order". List of the available warehouses should be filtered with the warehouses which are not marked as "Blocked", "Allow return" or "Consignment/Loan".

The screenshot shows the 'Production control parameters (1 - no10)' window. The 'Default Service Part' section is expanded, showing a table of service types and their corresponding categories and item numbers. The 'Default warehouse for service' field is highlighted with a red box and set to '01'.

Service type	Category hierarchy	Category	Item number
Repair	Product hierarchy	BTE instrument	S-REPAIR
Substitute	Product hierarchy	BTE instrument	S-SUBSTITUTION
Swap	Product hierarchy	BTE instrument	S-SWAP
Repair	Product hierarchy	Custom instrument	S-REPAIR
Repair	Product hierarchy	RITE instrument	S-REPAIR
Substitute	Product hierarchy	RITE instrument	S-SUBSTITUTION
Swap	Product hierarchy	RITE instrument	S-SWAP
Repair	Product hierarchy	Streamer - HA	S-REPAIR
Substitute	Product hierarchy	Streamer - HA	S-SUBSTITUTION

Default service item: S-SWAP

Enable route based service setup: ☐

Default warehouse for service: 01

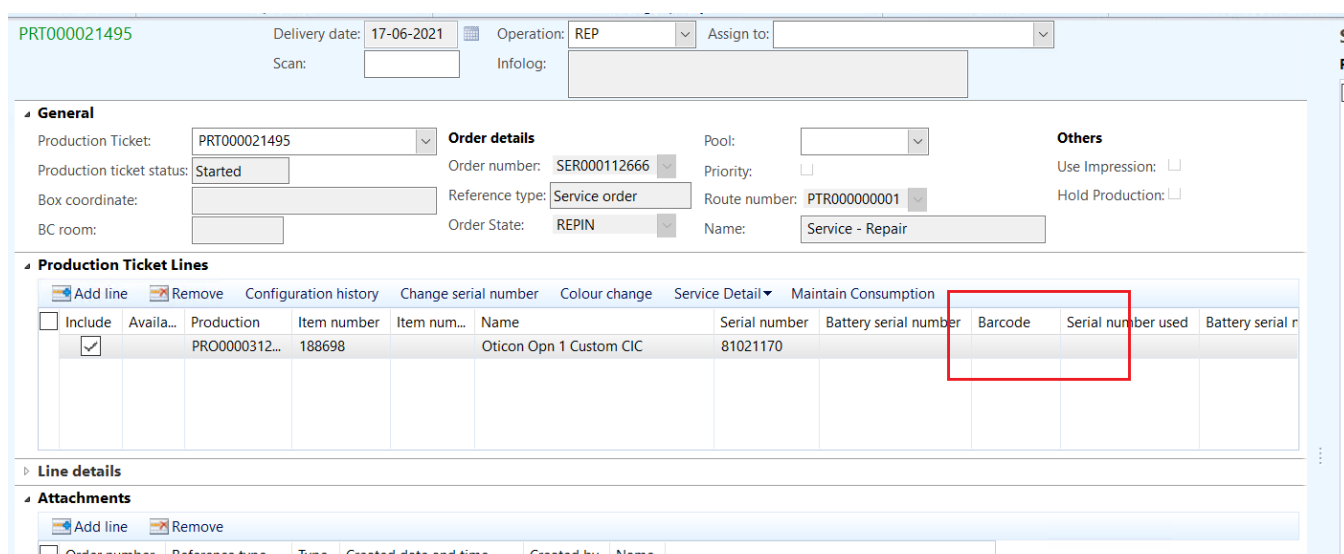
While creating a new production ticket for the service order system should always check first the setup for default warehouse and only if this setup is empty try to retrieve the site and warehouse from the service card. This will help to avoid issues with the empty site and warehouse for service orders created with non-serialized items.

6.3.7.4 Enabling “Barcode” field to Scan the Battery serial number

On the Production ticket form for service orders, the Barcode field should be enabled for Service type – Repair, Swap, Substitute & Express swap.

When barcode for product(hearing aid) with Li – Ion battery (i.e. products where “Exist in global device card” is set to TRUE on the product master), there is a need to extract the battery serial number and update it on “Battery serial number used field. This is valid for scenarios where service type is either Swap, Substitute or Express Swap and defective instrument is replacement by a new/refurbished product with a battery.

For service type repair, swap, substitute, where a new battery is consumed and a consumption line is created in the Service consumption tab on the Service specification form/Maintain consumption form following logic needs to be used to update the battery serial number in the Battery serial number used field. When a barcode is scanned in the Barcode field, system should check if “Update global device card” is set to TRUE, then the serial number of such product should be updated in the Battery serial number used field, if set to FALSE then serial number should be updated as per logic described in section 6.3.14.3 (FDS WP11003).



General

Production Ticket: PRT000021495

Production ticket status: Started

Box coordinate:

BC room:

Order details

Order number: SER000112666

Reference type: Service order

Order State: REPIN

Pool:

Priority:

Route number: PTR000000001

Name: Service - Repair

Others

Use Impression:

Hold Production:

Production Ticket Lines

Include	Availa...	Production	Item number	Item num...	Name	Serial number	Battery serial number	Barcode	Serial number used	Battery serial n
☒		PRO0000312...	188698		Oticon Opn 1 Custom CIC	81021170				

Line details

Attachments

☐ Order number ☐ Reference type ☐ Type ☐ Created date and time ☐ Created by ☐ Name

6.3.8 Serial number registration

KIT and other serialised component for production should be registered before they can be picked. Intent of this design is to make the serial number registration, picking and unpicking a bit simpler. Registration overview form will be used to register serial number of kits. The Registration overview form will show all picking list lines which have serialized item with a reference to production order and are not picked (i.e not posted). The line view will show Box coordinate and ear as well beside other general item details as mentioned in sec 9.2. The Unregistered lines tab will show (filter) all picking list lines which have a reference of production order and for which serial number is not registered. Whereas ‘Registered lines’ will show (filtered) records for which serial numbers are registered.

The screenshot shows the 'Registration overview' window with the following details:

- Registration section:** Includes fields for 'Order No.' and 'Concatenated barcode'.
- Lines section:**
 - Unregistered lines:** A table with columns: Item num..., Product name, Site, Serial nu..., Quant..., OTS/Custom, Storage locations, Order No., Order t..., Production t..., BC Coord..., BC room, Ear. It contains one entry: 171140, OPN 1 BTE POWER PLUS 13 ARGENTO, SIT01, 1.00, Off the shelves, SOR000000085, Sales order, PRD0000172.
 - Registered lines:** A table with the same columns as above, but it is empty with the message 'This grid is empty.'

User can then scan concatenated barcode (barcode having item number + serial number).

Register serial number form should be accessible from production control >Common and from production ticket form as well. Add button 'Register serial number' on production ticket form, system should automatically filter records on order number referenced in production ticket when called from a production ticket.

The screenshot shows the 'Production Ticket' form for ticket PRT000027353. Key elements include:

- Toolbar:** Includes buttons like 'Cancel', 'Start', 'Finish', 'Order holds', 'Maintain', 'Operation', 'Quality Acceptance', 'Non conformances', 'Quality orders', 'Current quality orders', 'Current non conformances', 'Picking list', 'Route card', 'Reported as finished', 'All', 'Journals', 'Edit configuration', 'Audiogram', 'ATS Result/Report', 'Production details', 'Refresh', and 'Export to Microsoft E List'.
- Form Fields:**
 - Delivery date: 27/07/2020
 - Operation: [Dropdown]
 - Assign to: [Dropdown]
 - Scan: [Input field]
 - Infolog: [Text area]
- General section:**
 - Production Ticket: PRT000027353
 - Production ticket status: Created
 - Box coordinate: [Input field]
 - BC room: [Input field]
 - Order details: Order number: SER000041695, Reference type: Service order, Order State: REPIN
 - Groupings: Pool: [Dropdown], Priority: [Checked]
 - Others: Use Impression: [Unchecked], Hold Production: [Unchecked]
- Production Ticket Lines section:**
 - Buttons: Add line, Remove, Service consumption, Configuration history, Change serial number, Colour change, Service Detail
 - Table with columns: Include, Production, Item number, Item number used, Name, Serial number, Barcode. It contains one entry: [Checked], PRO000035565, 163280, Oticon Lite Tip, 2017869.
- Left configuration details:** A list of attributes and values:

Attribute type	Value
Fitting Level	60
Power Level	N/A
Material text...	Hard
Mould color	Transparent
Faceplate Co...	N/A
Receiver Prot...	ProWax
Pull out string	Short (5 mm)
Nail Grip	No
VentType	Lite Tip Vent

If there is a particular Order number selected then system will register the serial number (by comparing item number) for the picking lines related to production order for that sales order. If there are more than one match then use the sequence of picking list number for the same item. If scanned item number does not exist in picking list lines for this sales order, then system should throw error 'Item number does not exist'.

If Order No. is not mentioned then system will assign serial number (by comparing item number) based upon sequence of picking list number. If scanned item number does not exist for any of the picking list lines in unregistered tab, then system should throw error 'Item number does not exist'.

Existing action 'PostPickingList' should then do posting of picking list when called from a particular state or operation.

Additional consumption: Operators may need to consume additional component as the one picked earlier is damaged. At this point user need to create a picking list for any additional consumption. When Additional consumption is called for sales order from production ticket line , system will show the 'Additional consumption' form. Additional consumption form is a trim down version of 'Create lines' form (Production control/Journals/Picking list/ Picking list button/ Create lines). Function of additional consumption form is same as of create lines form for the production order line in production ticket. Field 'Reverse to warehouse' indicates the warehouse to which the component/kit should be reversed, like to a return warehouse or scrap warehouse or main warehouse. On click of OK,

- 1) If Reverse to warehouse is kept blank then system should create a picking list with the line item (as standard based upon the 'Proposal' field value).
- 2) If Reverse to warehouse has a value same as warehouse of the original picking plst line, then in addition to pick list created for proposal as quantity, system should create an additional picking list journal with proposal as 'Full reversal' and warehouse on picking lines should be the warehouse from 'Reverse to warehouse'.
- 3) If Reverse to warehouse has a value which is different from the warehouse of the original picking plst line, then in addition to pick list created for proposal as quantity, system should create an additional picking list journal with proposal as 'Full reversal' and warehouse on picking lines should be the warehouse from 'Reverse to warehouse'. As per standard AX, reversal always happens to same warehouse from which picking was done. So for cases where reversal warehouse is different from the warehouse item was picked initially (say the picked kit is faulty and has to be returned to vendor for a credit), system should create a transfer journal to put the stock from warehouse from where the item was picked initially to reversal warehouse.

(For companies where FIFO is applied on finished products, standard AX put additional consumption cost on the finished goods. Should the company wish to track the additional consumption separately and not as loaded on finished good, then it is not possible because of standard FIFO behavior. For such case we need development to post the additional consumption into a production variance account. To make it possible it has been suggested to post a movement journal with offset account as production variance account. So to support this we need following development)

System should further check if the reverse to warehouse matches with the 'Scrap warehouse' on production control parameters. If there is a match then system should create a movement journal with negative quantity and with inventory dimension same as on 'To' section of transfer journal from previous step. To create the the movement journal use the journal mentioned in the Production control parameters/Journal/ Default journal names/Movement. This journal should get posted through action 'postPickingList' after the transfer journal has been posted through the same action.

All the newly created picking list, transfer journal and movement journal will get posted through action 'postPickingList'. In addition to this, before creating additional consumption for proposal quantity system should check if there is sufficient stock of the item, for that check Total available in onhand, if Total available is less than quantity mentioned then throw error 'Not enough stock for additional consumption' and system should not create any picking or transfer journal.

 Additional consumption lines (1) X

Create lines

Production: PRO000000571

Date: 06-05-2019

Quantity:

Proposal:

Reverse warehouse: Quantity
Full reversal

OK Cancel

6.3.9 Cancellation of sales order/lines

When we cancel the Sales order(Header level), the whole Production Ticket also should get to cancel status.

When we remove/cancel the sales line and add new line or change item via product configuration and do RFP, the existing/new production ticket should get updated as per sales lines.

- a. Ex1 : If we remove or cancel the right ear line in Binaural Sales order, then when we create Production ticket, The right ear Prod ticket lines also should get deleted.
- b. Ex2: If we change the item No via product configuration and creating Production ticket, the old item line should get deleted & new Item line should get added in Prod ticket lines.

6.3.10 Cancellation of Production ticket

The Cancel button on production ticket (list page and form) should only be enabled, if the Production ticket status is not 'Completed' or 'Cancelled'. If related picking list has been posted it should not be possible to cancel production ticket and following InfoLog should be shown: "Production ticket cannot be canceled due to posted picking list [Picking list number]".

Cancellation of production ticket should result in deletion of the production order linked to the production ticket. Production order can only be deleted if it is in status 'Created' and the BOM is not being picked. So when a production ticket is being cancelled system should first reverse any picked BOM line, then reset status of production order to Created and delete the production order. In standard a picked BOM line transaction can be reversing the picking (as shown below)

Untitled - Paint

View

File

Select handling unit to pick from: None

Transactions

Add picking list updates Update reservations Split Dimensions display

Picking status	Update	Issue status	Reference	Number	Item number	Site	Warehouse	Serial number	Quantity
✓	Added	Picked	Production line	PRO000003162	192028	SIT01	01	51285886	1

Picking list updates

Add Remove Register all Dimensions display

Item number	Site	Warehouse	Serial number	Reference	Number	Picked quantity	Unit
192028	SIT01	01	51285886	Production line	PRO000...	-1.00	pcs

Production order status can be reset back to Created by using 'Reset status' function.

Production orders (1 - it50) - Production: PRO000003162, Bernafon Zerena 1 Custom CIC

File Production order Schedule View

Edit Split Production order Batch order Estimate Report as finished Reset status Transactions Trace

Maintain New Process Inventory

PRO000003162 Delivery: 6/26/2019

Item number: 192311 Quantity: 1.00

General

Identification

Production: PRO000003162

Item number: 192311

Name: Bernafon Zerena 1 Custom CIC

Service Item:

Production

Status

Status: Started

Scheduling status: Operations scheduled

Remain status: Material consumption

Groupings

Pool:

Reset status (1 - it50)

Overview General Batch

Production	Name	Desired status
PRO000003162	Bernafon Zerena 1 Custom CIC	Released

Created

Estimated

Scheduled

Released

Started

Reported as finished

Ended

Select Default values Late selection: ☐

So to ensure that system reverses all transactions properly, the development on Production ticket cancellation should do reversal of picked BOM transaction (if any), then reset status of the production order back to Created and then delete the production order.

As a result of production ticket cancellation all related quality orders should be automatically deleted in order to remove inventory blockings on the items.

6.3.11 Maintain consumption

Users should be able to modify proposed consumption during production /service process. For this purpose the consumption lines should be easily accessible from production ticket form. At present to modify any consumption, users have to go to production order and from there they have to access production BOM and then need to modify items to be consumed and need to fill in few other fields as well. Similarly for service users first need to access service specification form and from there they can modify service part consumption.

To improve usability add new button 'Maintain consumption' on production ticket lines. Maintain consumption will bring up the production BOM for selected production order. Existing service consumption button on production ticket also need to be modified. The Service consumption button will bring up a new form showing service consumption for selected service line.

Production Ticket (1 - it50) - Production Ticket: PRT000027321, Started

File | Production Ticket

Cancel | Start | Finish | Order holds | Operation | Quality Acceptance | Non conformance | Quality orders | Manage quality | Current quality orders | Current non conformances | Route card | Reported as finished | All | Journals | Edit configuration | Audio diagram | ATS Result/Report | Production details | Refresh | Export to Microsoft Excel List | Transition log Workflow | Attachments

Maintain | Operation

PRT000027321

Delivery date: 10/07/2020 | Operation: SLI | Assign to: | Scan: | Infolog: The Production order PRO000035526, PRO000035527 is started for SLI operation

General

Production Ticket: PRT000027321 | Order details: Order number: SOR000060341 | Groupings: Pool: | Others: Use Impression: | Production ticket status: Started | Reference type: Sales order | Priority: | Hold Production: | Box coordinate: PRODUCTION | Order State: RFP | BC room: 1.1

Production Ticket Lines

Add line | Remove | Create BOM | Additional consumption | Configuration history | **Maintain consumption**

Production	Item number	Name	Serial number	Quantity	Ear	Operation
☒ PRO000035526	154873	Bernafon Inizia 1 CIC	W20109893	1.00	Right	SLI
☒ PRO000035527	154873	Bernafon Inizia 1 CIC	W20109894	1.00	Left	SLI

Line details

Style: | Fitting level: | Omni-directional or directional: | Ventilation size: | Mould/Shell size: | Wireless and Telecoil: | Ventilation code: | Maximize reserved output: |

Change permission

Attachments

Add line | Remove

Order number	Reference type	Type	Created date and time	Created ...	Name
This grid is empty.					

Right configuration details

Show all: ☐

Attribute type	Value
Custom Shell...	Beige
Fitting Level	N/A
WirelessRF	N/A
Auto Telecoil	N/A
Vent	Collection Vent 2.4 n
Receiver Prot...	ProWax
Microphone ...	Protection Grid
Program Co...	No
Pull out String	Standard (8 mm)
Large Ball Re...	No
Pull Out Clot...	N/A
Raised Push ...	N/A

Maintain consumption for sales process:

Maintain consumption should be shown only for production tickets with reference type sales order, this button should bring production bom form.

BOM (1 - it50) - Production: PRO000035526, Bernafon Inizia 1 CIC

File | New | Delete | Copy | Inventory | Inquiries

Overview | General | Setup | Update | Quantity | Reference | Financial dimensions | Inventory dimensions

Item number	Quantity	Per series	Unit	Item type	Product name
102118	1.0000	1	pcs		IN1 CIC, BE INIZIA 1

For ease of use we need to automate following things when a new line is added to the production BOM form (and when 'Enable production ticket' is enabled).

1. Set Oper. No. field in General tab as operation number for the operation mentioned in Production ticket default> Operation for picklist creation.

BOM (1 - it50) - Production: PRO000035526, Bernafon Inizia 1 CIC

File ▾ New ✨ Delete ✕ Copy Inventory ▾ Inquiries ▾

Overview **General** Setup Update Quantity Reference Financial dimensions Inventory dimensions

Identification

Production:

Item number:

Position:

Subcontractor

Vendor account:

BOM

Line type:

Reservation:

Calculation: ☒

Sub-BOM:

Subroute:

Operation

Oper. No.

End: ☐

Consumption

Resource consumption: ☐

2. Set Estimated filed in Quantity tab as value in quantity field on overview tab of BOM form.

BOM (1 - it50) - Production: PRO000035526, Bernafon Inizia 1 CIC

File ▾ New ✨ Delete ✕ Copy Inventory ▾ Inquiries ▾

Overview General Setup Update **Quantity** Reference Financial dimensions Inventory dimensions

☐	Item number	Estimated	Remaining quantity	Started	Released quantity	Unit
	102118	1.0000	1.00	1.00		pcs

Service consumption for service process:

Existing Service consumption button need to be modified. This button is enabled only for production tickets with reference type service order, and it shows further options for selection like service bom, add additional etc. options. These options need to be removed, instead on clicking of service consumption button system should bring up a new form Service consumption. This form layout, buttons and fuctions should be same as service consumption section in service specification form. Any changes done on service consumption form whould be reflected in service consumption section on service specification form.

New form: Service consumption

Item number	Product name	Size	Quantity	Storage location	Journal	Warehouse	N.	P.	P...	Serial
198638	AMPLIFIER, OPN S 3 BTE 13 PP 2....		1.00	RED_O	IWJ0030019	01	☐	☐	☐	
156927	INLET SET, FOR MIC INCL O-RIN...		1.00	RED_O	IWJ0030019	01	☐	☐	☐	
137900	PIN, 0.5 x 6.7 mm		1.00	AX0406S1B6	IWJ0030020	01	☐	☐	☐	

6.3.12 Subcontracting of operation

As we know, wholesale companies beside being a distributor they are also equipped with skills and tools to carry out local production for custom instruments and to perform service. There are few situations, like lack of capacity, tool or skill due to which wholesale would like to subcontract few operations to main production at PL. To cater for this requirement users should have option to change route on already created production orders for production and service. AX provides subcontracting and change of route as standard feature. We would like to utilize the standard functionality with few developments to make the standard solution more user friendly and to suit the process.

6.3.12.1 Change of route/ operation

It should be possible to change suggested route or to subcontract an operation during production. For this purpose add new button 'Change route' on production ticket form and 'All production tickets' overview.

Production Ticket (1 - it50) - Production Ticket: PRT000034869, Created

File | Production Ticket

Buttons: Edit, Cancel, Start, Finish, Change Route, Order holds, Quality Acceptance, Non conformances, Quality orders, Current quality orders, Current non conformances, Route card, Reported as finished, Picking list, All, Journals, Edit configuration, Register serial number, Audio diagram, ATS Result/Report, Change Route, Attachments

Production Ticket: PRT000034869
 Delivery date: 29/10/2020
 Scan:
 Operation:
 Assign to:
 Infolog:

General

Production Ticket:
 Production ticket status:
 Box coordinate:
 BC room:

Order details

Order number:
 Reference type:
 Order State:

Groupings

Pool:
 Priority: ☐

Others

Use Impression: ☐
 Hold Production: ☐

Production Ticket Lines

Include	Production	Item number	Name	Serial number	Quantity	Ear	Operation
☒	PRO000045221	195299	MAICO, CAPTO 3 CUSTOM IIC	81878366	1.00	Left	

Line details

Attachments

Order number	Reference type	Type	Created date and time	Created by	Name
This grid is empty.					

Show all: ☐

Left configuration details

Attribute type	Value
Custom Shell...	Black
Fitting Level	75
WirelessRF	No
Auto Telecoil	No
Receiver Prot...	ProWax miniFit
Microphone ...	T-Cap Filter
Program Co...	No
Pull out String	Measured length
Large Ball Re...	No
Pull Out Clot...	N/A
Raised Push ...	N/A

Local production:

'Change route' button should popup a new form 'Change route'.

Change route (1 - it50)

☒ **Route**
 Route number:

☐ **Operation**
 From Oper. No.
 To Oper. No.
 New Oper. No.

Include	Production	Item number	Name	Serial number	Service type	Ear
☒	PRO000041312	154712	Oticon Alta Pro CIC	I15103182	Repair	Left
☒	PRO000041313	154712	Oticon Alta Pro CIC	I15103183	Repair	Right

This form will allow user to select a different route for selected production orders. The 'Route number' drop down should show all valid routes for the products in the production ticket. It is also possible to select a range of operation and subcontract them.

Note: Consider first production order item to show the valid routes.

Change route (1 - it50)

☒ **Route**
Route number: PTR000000004

☐ **Operation**
From Oper. No. New Oper. No.

Include	Product	Site	Route number	Name	Active	Approved
☒	PRO0000	SIT01	PTR000000004	Custom hearing instrument	☒	☒
☒	PRO0000	SIT01	PTR000000010	SubcontractModelSLI	☐	☒

OK Cancel

Change route (1 - it50)

☒ **Route**
Route number: PTR000000004

☐ **Operation**
From Oper. No. New Oper. No.

Include	Production	Site	Route number	Name	Active	Approved
☒	PRO000041312	SIT01	PTR000000001	Service - Repair	☒	☒
☒	PRO000041313	SIT01	PTR000000002	Service - Remake	☐	☒
		SIT01	PTR000000003	Service - Swap	☐	☒

OK Cancel

System will change route or operation based upon option selected. Selecting 'Route' radio button will disable the Operation section and vice versa.

Change route (1 - it50)

☐ Route
 Route number:

☒ **Operation**
 From Oper. No. New Oper. No.
 To Oper. No.

Include	Production	Item number	Name	Serial number	Service type	Ear
☒	PRO000041312	154712	Oticon Alta Pro CIC	I15103182	Repair	Left
☒	PRO000041313	154712	Oticon Alta Pro CIC	I15103183	Repair	Right

OK Cancel

Note: Consider first production order route to show operations in 'From Oper. No.' and 'To Oper. No.' dropdown. 'New Oper. No.' dropdown should show all operations which are part of approved routes for the item and having route type 'Vendor' in the route setup.

Change route (1 - it50)

☐ Route
 Route number:

☒ **Operation**
 From Oper. No. New Oper. No.
 To Oper. No.

Include	Production	Item number	Name	Serial number	Service	Ear
☒	PRO000041312	154712	Oticon Alta Pro CIC	I15103182	Repair	Left
☒	PRO000041313	154712	Oticon Alta Pro CIC	I15103183	Repair	Right

OK Cancel

Operation dropdown menu:

Operation	Name
S-MS	Subcontract MOD,SLI
S-SLI	Subcontract,SLI

Radio button Route:

Use this option when the production hasn't started yet and we want to change the route. This button should be marked automatically if 'Change route' is accessed for a production ticket with state 'Created'.

If user tries to select radio button 'Route' for a production ticket with state 'Started' then system should show infolog 'Production orders are in started state. A change of route means all operations in new route need to be started and finished.'

On click of OK system should change the route on selected production orders. If not all production orders are selected for change, then system should do the changes and create a new production ticket and assign the selected production orders to that production ticket. This is because production orders linked to a production ticket can follow only one route.

In standard we have a functionality to copy route on 'Copy bom/route' (path: production order/Route/Functions/Copy). Call the same code to replace the route on selected production orders.

1. In standard system allows to change route only when the production order is in Create state, hence system should first do reset status to 'Created' for selected production order and then execute the code to replace the route.
Note: use standard 'Reset status' option on production order to do reset.
2. System will delete associated quality orders with the production orders
3. Use the standard options on Copy bom/route form (as shown below) to replace the route number on production orders.
4. 'Release' the production order/s.

Standard Copy bom/route form:

Oper. No.	Priority	Operation	Rout.	Next	Quality order status
10	Primary	S-MS	Vendor	1.00	20
20	Primary	LAS	Standard	1.00	30
30	Primary	ASM	Standard	1.00	40
40	Primary	LAC	Standard	1.00	50
50	Primary	ACTI	Standard	1.00	60
60	Primary	QC	Standard	1.00	0

Radio button Operation:

Use this option when the production has started already, and we want to subcontract one/more operations. This button should be marked automatically if 'Change route' is accessed for a production ticket with state 'Started'.

Change route (1 - it50)

☐ Route
 Route number:

☒ Operation
 From Oper. No. To Oper. No. New Oper. No.

Include	Production	Item number	Name	Serial number	Service	
☒	PRO000041312	154712	Oticon Alta Pro CIC	I15103182	Repair	Left
☒	PRO000041313	154712	Oticon Alta Pro CIC	I15103183	Repair	Right

Operation	Name
S-MS	Subcontract MOD,SLI
S-SLI	Subcontract,SLI

On click of OK system should do the following for selected production orders. If all production orders are not selected for change, then system should create a new production ticket and assign the selected production orders to that product ticket and then do the changes to route. This is because production orders linked to a production ticket can follow only one route.

1. In standard system allows to change route only when the production order is in 'Create' state, hence system should first do reset status to 'Created' for selected production order and then execute the code to replace the operation.
2. Delete the operations between (inclusive) From and To operation numbers.
3. Add the operation mentioned in the 'New Oper. No.' to the route. Put Oper. No. same as mentioned in From Oper No. field. Copy all other details of the operation from the route it is linked to.
4. 'Start' the production order/s.

Changes to production ticket for service order-

Changes to Default service part setup-Add 'Route number' and 'Name' field to the form. Remove other existing fields "Category hierarchy", "Category" and "Item number" except 'Service type' field. Add new fields Route number and Name.

Production control parameters (1 - it50)

File

General

Journals

Production Ticket Default

Automatic update

Standard update

AIF

Status

Inventory dimensions

Lean manufacturing

Unit of measure

Number sequences

• **Default Service Part**

Setup default production items for service and repair order

+ Add - Remove

☐ Service type	Route number ▲	Name
Repair	PTR000000001	Service - Repair
Remake	PTR000000002	Service - Remake
Swap	PTR000000003	Service - Swap

Default service item: Service ▼

Delete the selected record

Close

Changes to production ticket creation for for service order- At present system creates a production order with a dummy item number for service, and that item number is specified as a relation between service type, item category.

With this change system should always create production order with item number mentioned in field 'Default service item' and with route based upon service type and route number relation mentioned in the setup.

Changes to Production ticket:

Add field "Route number" and "Name" on 'Production ticket' form, 'All Production tickets' overview form

Production Ticket (1 - it50) - Production Ticket: PRT000031834, Started

File | Production Ticket

Tools: Edit, Cancel, Start, Finish, Order holds, Quality Acceptance, Non conformances, Quality orders, Current quality orders, Current non conformances, Route card, Reported as finished, Edit configuration, Register serial number, Picking list, All, Audiogram, ATS Result/Report, Journals, Production details.

Production Ticket: PRT000031834

Delivery date: 22/09/2020 | Operation: SLI | Assign to: | Scan: | Infolog: |

General

Production Ticket: PRT000031834 | Order details: Order number: SOR000070906 | Groupings: Pool: | Others: Use Impression: |
 Production ticket status: Started | Reference type: Sales order | Priority: | Hold Production: |
 Box coordinate: | Order State: RFP | Route number: PTR000000004 | Name: Custom hearing instrument

Production Ticket Lines

	Add line	Remove	Additional consumption	Configuration history	Maintain Consumption			
☐	Include	Production	Item number	Serial number	Ear	Name	Quantity	Operation
☒			PRO000041246	156806	W20111228	Right	Oticon Ria2 ITC	1.00 SLI

At present the 'operation' dropdown on production ticket form is fetched from operations in route master. Operations should come from production order route (consider first production order to get the route operations)

Add Production ticket and order number on Subcontracted work form (Path: Production control/Common/Subcontracted work)

Add 'IC state' of subcontracted PO lines to Subcontracted work form.

IT50 > Production control > Common > Subcontracted work

Production order

Tools: Route, Transaction, BOM, Lot, Refresh, Export to Microsoft Excel List, Attachments, Attachments.

Subcontracted work

Production	Oper. No.	Reference	Number	Vendor account	Operation status	Purchase order status	Production Ticket:	Order number:
This grid is empty.								

6.3.12.2 Placing purchase order for subcontracted operation

Standard AX generates purchase order for subcontracted operation when production order is estimated. The purchase order is generated for an item (item type service with item model group stocked product) in BOM which is linked to the subcontracted operation in route. A BOM setup in AX is not possible as BOM is generated on runtime from EPDS. For this reason we need customization to add item representing subcontracted operation in BOM.

Changes to production BOM:

Route (1) - Route number: PTR000000010, SubcontractModelSLI

Oper. No.	Priority	Operation	Mandatory	Scrap percentage	Accumulated	Next	Link type	Hourly rate / piecework rate
10	Primary	S-MS	☐	0.00	1.00	20		All
20	Primary	LAS	☐	0.00	1.00	30		All
30	Primary	ASM	☐	0.00	1.00	40		All
40	Primary	LAC	☐	0.00	1.00	50		All
50	Primary	ACTI	☐	0.00	1.00	60		All
60	Primary	QC	☒	0.00	1.00	0		All

 BOM (1 - it50) - Production: PRO000041380, HOUSING, VE7 SIL SER

Overview	General	Setup	Update	Quantity	Reference	Financial dimensions	Inventory dimensions
----------	---------	-------	--------	----------	-----------	----------------------	----------------------

Route (1) - Route number: PTR000000010, SubcontractModelSLI

File Functions Product change Entry/Exit actions

Oper...	Priority	Operation	M..	Scrap ...	Accu...	Next	Link ty...	Hourly rate / pie...
10	Primary	S-MS	☐	0.00	1.00	20		All
20	Primary	LAS	☐	0.00	1.00	30		All
30	Primary	ASM	☐	0.00	1.00	40		All
40	Primary	LAC	☐	0.00	1.00	50		
50	Primary	ACTI	☐	0.00	1.00	60		
60	Primary	QC	☒	0.00	1.00	0		

Resources (1 - it50) - Resource: V1, V1, Individual resource: Yes

File Resource

Resource Copy resource Bulk edit Add capability Add to resource group Delete Calendar deviations Dispatching Gantt chart Route transaction

Resource

Resource	Description	Type
A	A	Machine
HR	Human resources	Human resource
V1	V1	Vendor

Resource: V1
Description: V1
Type: Vendor

General Resource

Vendor: PL30
Worker:
Worker name:

Resources

Quantity:
Load: 100.00

Resource requirements

New Delete

Requirement type	Requirement	O...	J...
Resource	V1	☒	☐

As part of standard, on production order estimation system will create purchase order for the subcontracted operation. Add field 'Subcontract item' to show the item for which the operation is subcontracted. 'Subcontract item' is the item for which production order is created. System should update this field when purchase order is created for subcontracted operation.

Purchase order (1 - it50) - Purchase order: POR000039854, Oticon A/S - ITE Production

File Purchase order Purchase Manage Receive Invoice Retail General

Maintain charges Multiline discount Total discount Prepayment Remove prepayment Purchase inquiry Pro forma confirmation Confirm Finalize Distribute amounts View distributions Purchase inquiry Purchase order confirmations

Credit note Create Charges Sales tax Calculate Prepay Generate Accounting Journals

POR000039854 : DK50 - Oticon A/S

Purchase order header

Purchase order lines

Add line Add lines Add products Remove Purchase order line Financials Inventory Product and supply Update line

Type	B...	Line n...	Item number	Product name	Subcontract item	Quantity	Unit	Unit price	IC State	IC state description
		10	Subcontract	Subcontract operation	194934	1.00	pcs	144.00	DGS IN	DGS IN

Purchase order for subcontracted operation should be confirmed when operation having assigned vendor item is started on the production ticket.

Route (1) - Route number: PTR000000016, Subcontract modelling for remake

File Functions Product change Entry/Exit actions

Oper. No.	Priority	Operation	Mandatory	Scrap percentage	Accumulated	Next	Link type	Hourly rate / piecework
10	Primary	SMOD	☒	0.00	1.00	20	All	
20	Primary	SLI	☒	0.00	1.00	30	All	
20	Secondary 1	PASM	☐	0.00	1.00	30	All	
30	Primary	LAS	☒	0.00	1.00	40	All	
40	Primary	ASM	☒	0.00	1.00	50	All	
50	Primary	LAC	☒	0.00	1.00	60	All	
60	Primary	ACTI	☐	0.00	1.00	65	All	
65	Primary	ATS	☐	0.00	1.00	70	All	
70	Primary	QC	☒	0.00	1.00	0	All	

Applicable resources Maintain resource requir...

Overview General Setup Times Resource requirements Description

Identification

Item code: All
 Item relation:
 Configuration:
 Route code: Route
 Route relation: PTR000000016
 Operation: SLI
 Site: SGB01

Route

Property:
 Route group: RG
 Route type: Vendor
 Vendor item: S-SubContModeling

6.3.12.3 Changes to WBI

NAV has a different way of handling order for subcontracted operation. For this reason we need to make certain changes in the wbi message for subcontracted purchase order. Subcontracted purchase order can be identified from reference type as 'Production order'

POR000039867 : DK50 - Oticon A/S

Purchase order header

Purchase order lines

Add line Add lines Add products Remove Purchase order line Financials Inventory						
☐	Warehouse	Quantity	Unit	Unit price	IC State	IC state description
	01	1.00	pcs		DGS IN	DGS IN

Line details

Product dimensions

Configuration: CFI0956532
 Size:
 Color:

Storage dimensions

Site: SIT01
 Warehouse: 01
 Location:

Item reference

Reference type: Production order
 Reference number: SOR0000
 Reference lot: LOT0014

For sales order subcontracted purchase order system should have following modifications.

```

<IsSubcontract>Yes</IsSubcontract>
<OrderLines>
  <OrderLineDTO xsi:type='CustomHAOrderLineD
    <LineId>1001416054</LineId>
    <Description>Subcontract operation</Description>
    <ProductName>Subcontract operation</ProductName>
    <Quantity>1</Quantity>
    <UnitOfMeasure>pcs</UnitOfMeasure>
    <RequestedDeliveryDate>2020-10-22</Requeste
    <SubcontractItem>Subcontract</SubcontractItem>
  <Comments>
    <CommentDTO>
      <CommentNo>1</CommentNo>
      <Text>Change permission (flow: Left Ea
      <Date>2020-10-16</Date>
    </CommentDTO>
  </Comments>
  <DGSPackingSlipLine>
    <ItemNo>'.....'</ItemNo>
    <ExternalItemNo />
    <InsuranceNumber />
    <CustomerRefNo />
    <GTIN />
  </DGSPackingSlipLine>
  <IsBinaural xsi:nil='true' />
  <TrackingDimensions>
    <TrackingDimensionDTO>
      <SerialNo>80070435</SerialNo>
      <BatchNo />
      <ExpirationDate xsi:nil='true' />
      <Quantity xsi:nil='true' />
      <WarrantyStartDate xsi:nil='true' />
      <WarrantyEndDate xsi:nil='true' />
    </TrackingDimensionDTO>
  </TrackingDimensions>
  <ProductType>HA</ProductType>
  <ProductFeatures>
    <Features>
      <ProductFeatureDTO>
        <Name>Audiological Style</Name>
        <NameUniqueId>23</NameUniqueId>
        <Value>Custom IIC</Value>
      </ProductFeatureDTO>
    </Features>
  </ProductFeatures>

```

1. Add <IsSubcontract> to the outbound wbi message. This should be set to 'Yes' for subcontracted purchase order.
2. Add <Subcontract item> to order line dto. This is mapped to the 'item' on purchase line.
3. <DGSPackingSlipLine> should be left blank.
4. <ProductFeature> section should be populated from sales order line (sales order is linked to production order reference on PO line). The item on purchase order is not the actual custom item, hence we need to get the configuration from sales line.

```

- <OrderDTO xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="DGS.Common.DTO">
  <IntegrationPrefix>R</IntegrationPrefix>
  <PurchaseOrderNo>0065372</PurchaseOrderNo>
  <SalesOrderNo>SOR000118958</SalesOrderNo>
  <OrderHandling />
  <HasOrderRelation>true</HasOrderRelation>
  <OrderDate>2021-06-22</OrderDate>
  <RequestedDeliveryDate>2021-06-25</RequestedDeliveryDate>
  + <ShippingInformation>
  + <DGSPackingSlip>
  <BusinessClassifications />
  <Comments />
  <IsSubcontracting>No</IsSubcontracting>
- <OrderLines>
  - <OrderLineDTO xsi:type="CustomHAOrderLineDTO">
    <LineId>2002275962</LineId>
    <Description>Oticon Flex Mould</Description>
    <ProductName>Oticon Flex Mould</ProductName>
    <Quantity>1</Quantity>
    <UnitOfMeasure>pcs</UnitOfMeasure>
    <RequestedDeliveryDate>2021-06-25</RequestedDeliveryDate>
    + <Comments>
    + <DGSPackingSlipLine>
    <IsBinaural xsi:nil="true" />
    - <TrackingDimensions>
      - <TrackingDimensionDTO>
        <SerialNo>2029737</SerialNo>
        <BatchNo />

```

5. <SalesOrderNo> should be picked from 'reference number' field on production order/References tab. Production order number can be found in PO line details/product tab/Reference number
6. <SerialNo> should be picked from sales order line for which Production order is created.
7. <RequestedDeliveryDate> should be picked from production order delivery date.

```

  </BusinessClassifications>
- <Comments>
  - <CommentDTO>
    <CommentNo>1</CommentNo>
    <Text>Production instruction (flow: Header)</Text>
    <Date>2022-01-18</Date>
  </CommentDTO>
  - <CommentDTO>
    <CommentNo>2</CommentNo>
    <Text>Touch the canal outlines</Text>
    <Date>1900-01-01</Date>
  </CommentDTO>
</Comments>
<IsSubcontracting>Yes</IsSubcontracting>
- <OrderLines>
  - <OrderLineDTO xsi:type="CustomHAOrderLineDTO">
    <LineId>2009184172</LineId>
    <Description>Subcontract modelling</Description>
    <ProductName>Oticon Mould</ProductName>
    <Quantity>1</Quantity>
    <UnitOfMeasure>pcs</UnitOfMeasure>
    <RequestedDeliveryDate>2022-01-21</RequestedDeliveryDate>
    - <Comments>
      - <CommentDTO>
        <CommentNo>1</CommentNo>
        <Text>Change permission (flow: Right Ear from sales order line)</Text>
        <Date>2022-01-19</Date>

```

8. <Comments> should be mapped from sales order (same logic when we do SOTAS)
9. <Description> mapping should be changed to item description on sales line for which the subcontracted order got created.

For subcontracted purchase order created for service and repair, the wbi file should have following modifications. (Refer section [Placing purchase order for subcontracted operation](#) to know how a subcontracted purchase order will get created from production ticket.)

```

- <MessageParts xmlns="http://schemas.microsoft.com/dynamics/2011/01
- <ServiceOrderDTO xmlns:xsi="http://www.w3.org/2001/XMLSchema-inst
  <PurchaseOrderNo>0120366</PurchaseOrderNo>
  <ServiceOrderNo>SER000120366</ServiceOrderNo>
  <ServiceOrderDate>2022-07-26</ServiceOrderDate>
  <RequestedDeliveryDate>2022-08-02</RequestedDeliveryDate>
- <ShippingInformation>
  <DirectDelivery>false</DirectDelivery>
- <ShippingAddress>
  <NameLine1>Demant Italia S.r.L.</NameLine1>
  <NameLine2 />
  <AddressLine1>Via Panciatichi 94</AddressLine1>
  <AddressLine2 />
  <PostalCode>50127</PostalCode>
  <City>Firenze</City>
  <County>FI</County>
  <Country>IT</Country>
</ShippingAddress>
</ShippingInformation>
<Comments />
- <Quotation>
  <QuoteNeeded>false</QuoteNeeded>
  <QuoteLimit>0</QuoteLimit>
  <Quote>1</Quote>
</Quotation>
<IsSubcontracting>No</IsSubcontracting>
- <ServiceOrderLines>
- <ServiceOrderLineDTO xsi:type="CustomHAServiceOrderLineDTO">
  <LineId>2000140063</LineId>
  <ServiceType>Remake</ServiceType>
  <Description>Oticon Opn 1 Custom CIC</Description>
  <RequestedDeliveryDate>2022-08-02</RequestedDeliveryDate>
  <Subcontract item> Subcontract <Subcontract item>
- <ServiceOrderLines>
- <ServiceOrderLineDTO xsi:type="CustomHAServiceOrderLineDTO">
  <LineId>2000140063</LineId>
  <ServiceType>Remake</ServiceType>
  <Description>Oticon Opn 1 Custom CIC</Description>
  <RequestedDeliveryDate>2022-08-02</RequestedDeliveryDate>
  <Comments />
- <TrackingDimensions>
- <ServiceTrackingDimensionDTO>
  <SerialNo>22700404</SerialNo>

```

1. Add <IsSubcontract> to the outbound wbi message. This should be set to 'Yes' for subcontracted purchase order.
2. Add <Subcontract item> to order line dto. This is mapped to the 'item' on purchase line.
3. <Description> mapping should be changed to item description on service specification line for which the subcontracted order got created.
4. All other values should be mapped from service specification same as when PO is created from service specification.

6.3.12.4 Sending order updates to NAV

There could be changes in order- like changes in configuration or delivery dates, after the subcontracted order has been placed from production ticket. This is similar to the order update we send to NAV on SOTAS from sales order.

At present we are sending a 'create' message via wbi to NAV on purchase order confirmation which happens on start of a subcontracted operation. System should send 'update' wbi message to NAV on start of subcontracted operation if we already have a confirmation journal for the subcontracted purchase order. The format of update message and validation of IC state update allowed from production ticket should be same as for sales order update message. The mapping of fields should be same as mentioned in [Changes to WBI](#)

6.3.12.5 Changes in Biztalk

We need few modifications in biztalk as well. NAV has For purchase order of type subcontract, following changes need to be made in biztalk before sending file to NAV

1. In biztalk we need to create a mapping between < Subcontract item> and NAV subcontract item.
2. < NAV subcontract item > should be in the options section of the file to NAV.
3. Derive HI number from configuration and pass on to NAV as line item.

6.3.13 Auto-assigning worker to the operation

A new parameter "Auto-assign worker" should be added in the Production control parameters (Path: Production control/Setup/Production control parameters/Production ticket defaults). If parameter is active then after selecting the operation for production ticket the field "Assign to" should be automatically filled with the worker name related to the currently logged user. In case when currently logged user is not linked with any worker no value should be auto-added to the "Assign to" field.

The screenshot displays the 'Production control parameters' window with the 'Production Ticket Default' tab selected. The 'Auto-assign worker' checkbox is checked and highlighted with a red rectangle. The window is divided into three panes, each showing a set of parameters for different production scenarios. The left pane is the 'Production Ticket Default' pane, the middle pane is the 'Picking list journal' pane, and the right pane is the 'Report as finished journal' pane. The 'Auto-assign worker' checkbox is located in the 'Operations' section of the left pane. The status bar at the bottom indicates 'Specify system action if mandatory operation is not reported'.

The same functionality should also be created for the State scanning production ticket form and a new parameter "Auto-assign worker in State scanning production ticket" should control this functionality (Path: Sales and

marketing/Setup/Workflow/State machine parameters). If parameter is active then while opening State scanning production ticket form the field "Assign to" should be automatically filled with the worker name related to the currently logged user. In case when currently logged user is not linked with any worker no value should be auto-added to the "Assign to" field.

State machine parameters (1 - it50)

File

General settings

Default state at order creation: OE

Default user id in scan: ☒

Default marking in State scanning sales order: ☒

Default marking in State scanning service order: ☒

Default marking in State scanning production ticket: ☐

Auto-assign worker in State scanning production ticket: ☒

Batch group:

Transition note type (header): Internal

E-mail:

Transition note type (line): Internal

Next state for service order creation wizard: REVIEW

Warehouse for internal logistics: INTLOG

Use assign to: ☒

Default state at the time of order creation

Close

State scanning production ticket (1 - it50)

File

User ID: esie Elżbieta Sienkiewicz

Scan:

Operation:

Assign to:

Start

Finish

Enter

Clear

Order number	Production Ticket	Production	Operation	BC room	Box coordinate
SOR000114203	PRT000048844	PRO00006...	SLI	1.4	PRODUCTION
SOR000114203	PRT000048844	PRO00006...	SLI	1.4	PRODUCTION

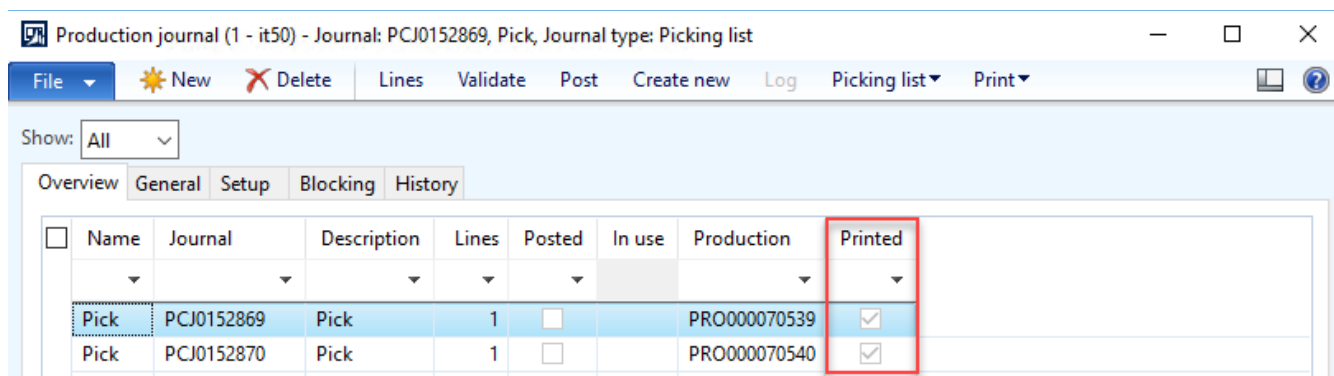
Log:

Close

6.3.14 Changing the BOM on the production ticket

Whenever a product configuration is edited and the changes are saved on the production ticket system should call EPDS BOM service in order to retrieve the new production BOM. The same criteria should be used as it is already when the BOM service is called for the sales order. It means that system should check if there is possible to create a BOM automatically based on the high runner/low runner kit and if these criteria are not fulfilled then system should show the form with all available BOMs for the product configuration. Once new BOM is selected it should be updated both on the sales order lines and on the related production order.

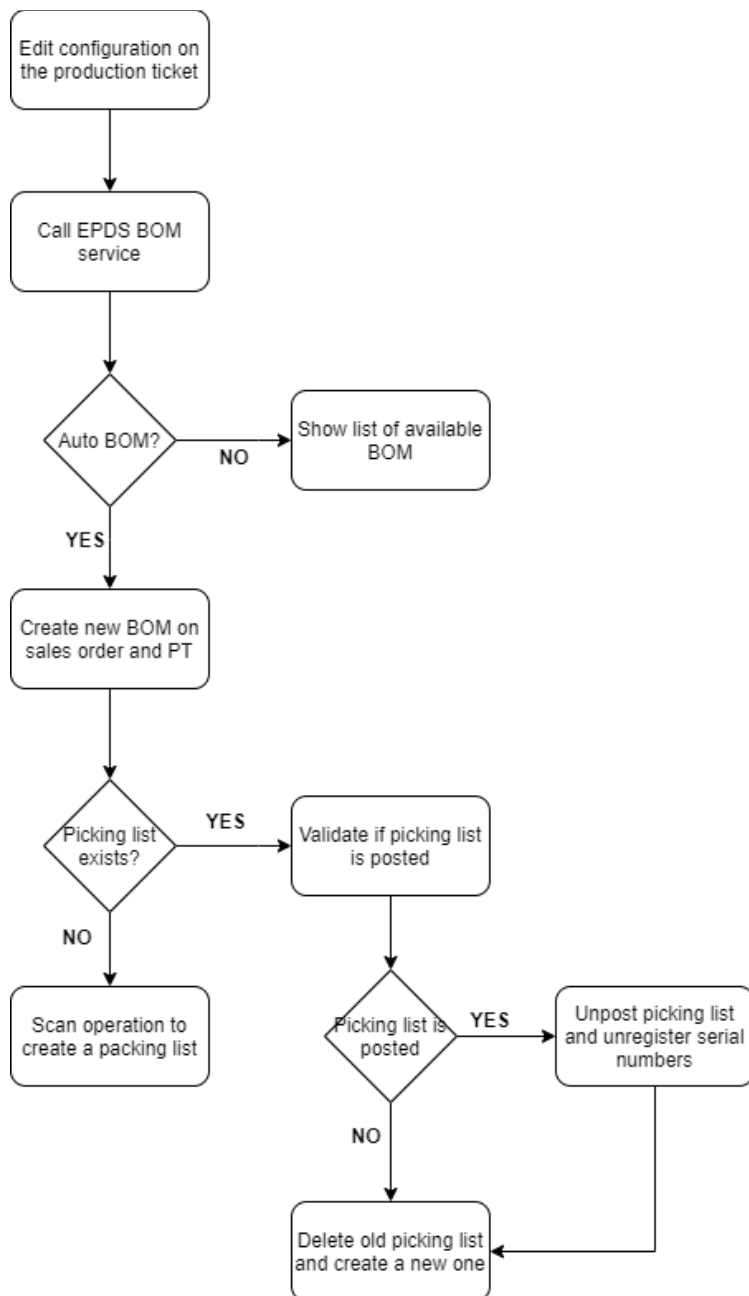
System should also validate if there is a kit picking list already created and posted for the production ticket. If picking list is not posted then system should delete it and create a new one based on the selected kit item. If picking list was already printed then Infolog should be shown: "Printed picking list has been deleted. Please print the new picking list."



	Name	Journal	Description	Lines	Posted	In use	Production	Printed
Pick	PCJ0152869	Pick	1	☐			PRO000070539	☒
Pick	PCJ0152870	Pick	1	☐			PRO000070540	☒

In case when picking list has already been posted system should show an Infolog with the message: "Picking list is already posted. Please unpost it before proceeding with the new picking list" and create a new picking list based on the selected kit item.

Graphical presentation of this process is visible on the flow chart below.



6.3.15 Site check (for USA)

-----wip-----

6.4 Quality order

Majority of the required design for the quality order is a part of Standard AX, but a few customizations are needed.

Each production order should have a quality order linked to it. In case a production ticket consists of two production orders that should generate two quality orders.

The form for quality orders is illustrated below. Three changes need to be incorporated to the form:

- 1) Ear type
- 2) Result of the tests and
- 3) Validation of the quality order.

6.4.1 Ear type

In the quality order provided by Standard AX (below form), it is not stated which ear type the quality order has. As one production ticket can trigger two quality orders – one with ear type right and one with ear type left, the ear type should figure in the overview on the quality order form.

Quality orders (1 - nl30) - Quality order: QOR12345679, Oticon Ria2 Pro ITC

File New Delete Validate Functions Inquiries Inventory

Overview General References Financial dimension Inventory dimensions

Quality order	Item number	Test group	Quantity	Status	Reference type	Ear type
QOR12345678	156807	Cosmetics	1.00	Open	Purchase	Right
QOR12345679	156807	Cosmetics	1.00	Open	Purchase	Left

Add Remove Results

Overview General Test

Sequence number	Test	Test result	Results entered	Include results	Documents	Outcome
10	CE		☐	☒		
20	CL		☐	☒		
30	CV		☐	☒		

Used symbols

Fail Pass

Quality order identification

EUR nl30 Atlas_Test_EMCA Close

6.4.2 Results of the test

The results of the quality orders in Standard AX are entered in the form that is retrieved by pressing 'results'. However, to reduce processing time, we would like to have the results of the tests handled directly from below form.

Quality orders (1 - nl30) - Quality order: QOR12345679, Oticon Ria2 Pro ITC

File New Delete Validate Functions Inquiries Inventory

Overview General References Financial dimension Inventory dimensions

Quality order	Item number	Test group	Quantity	Status	Reference type	Ear type
QOR12345678	156807	Cosmetics	1.00	Open	Purchase	Right
QOR12345679	156807	Cosmetics	1.00	Open	Purchase	Left

Add Remove Results

Overview General Test

Sequence number	Test	Test result	Results entered	Include results	Documents	Outcome
10	CE		☐	☒		
20	CL		☐	☒		
30	CV		☐	☒		

Used symbols

Fail Pass

Quality order identification (1) EUR nl30 Atlas_Test_EMEA Close

In below example, quality orders QOR12345678 (right ear) and QOR12345679 (left ear) have to go through a cosmetic check. The cosmetic check consists of three different tests that need to be performed for each quality order: 1) check engraving 2) check color and 3) check vent. Once the operator has finished each test, he/she should be able to insert the outcome from the dropdown on the right. From here the user should select value Y (Yes) or N (No). Y (Yes) equals a passed test and N (No) a failed test. In the first two tests, the outcomes are Y (Yes), which should populate the fields 'test result' with 'passed'. Oppositely, had the outcome been N (No) the field 'test result' should be populated with 'failed'.

The user should be able to select the outcome via the dropdown, but also by typing either 'Y' or 'N' and pressing enter.

The screenshot shows a software application window titled 'Quality orders (1 - n130) - Quality order: QOR12345679, Oticon Ria2 Pro ITC'. The window has a menu bar with 'File', 'New', 'Delete', 'Validate', 'Functions', 'Inquiries', and 'Inventory'. Below the menu bar are tabs for 'Overview', 'General', 'References', 'Financial dimension', and 'Inventory dimensions'. The 'Overview' tab is active, displaying a table with columns: Quality order, Item number, Test group, Quantity, Status, Reference type, and Ear type. The table contains two rows of data for quality order QOR12345679.

Quality order	Item number	Test group	Quantity	Status	Reference type	Ear type
QOR12345678	156807	Cosmetics	1.00	Open	Purchase	Right
QOR12345679	156807	Cosmetics	1.00	Open	Purchase	Left

Below the table are buttons for 'Add', 'Remove', and 'Results'. The 'Test' tab is active, showing a table with columns: Sequence number, Test, Test result, Results entered, Include results, Documents, and Outcome. The table contains three rows of data.

Sequence number	Test	Test result	Results entered	Include results	Documents	Outcome
10	CE	✓	☐	☒		Y (Yes)
20	CL	✓	☐	☒		Y (Yes)
30	CV		☐	☒		Y (Yes)

A dropdown menu for 'Outcome' is open, showing options: Y (Yes), Y (Yes), Y (Yes), Y (Yes), and N (No). The status bar at the bottom shows 'Quality order identification', 'EUR', 'n130', 'Atlas_Test_EMEA', and a 'Close' button.

6.4.3 Validation of quality order

With Standard AX the user is able to validate the quality order by pressing 'validate' (see below picture). However, as the user might forget pressing 'validate' when processing a quality order, validation should also be possible when the quality order is closed. When a user closes the quality order, the system should verify if the quality order has been validated already or not. If it has not, the system should generate a message stating 'Do you want to validate QORxxxxxx?' and include the options 'yes' and 'no'. By pressing 'yes', the system should validate the order and by pressing 'no' the order should not get validated. In both cases the quality order form should close.

Quality orders (1 - n130) - Quality order: QOR12345679, Oticon Ria2 Pro ITC

File New Delete Validate Functions Inquiries Inventory

Overview General References Financial dimension Inventory dimensions

Quality order	Item number	Test group	Quantity	Status	Reference type	Ear type
QOR12345678	156807	Cosmetics	1.00	Open	Purchase	Right
QOR12345679	156807	Cosmetics	1.00	Open	Purchase	Left

+ Add - Remove Results

Overview General Test

Sequence number	Test	Test result	Results entered	Include results	Documents	Outcome
10	CE	✓	☐	☒		Y (Yes) ▼
20	CL	✓	☐	☒		Y (Yes) ▼
30	CV	✓	☐	☒		Y (Yes) ▼

Used symbols

✗ Fail ✓ Pass

Quality order identification (1) EUR n130 Atlas_Test_EMEA Close

6.4.4 Quality Associations

A few changes need to be done to the form for quality associations. The form can be retrieved via the path: Inventory management > Setup > Quality control > Quality associations.

The test groups that need to be performed in quality orders vary, why different test groups can be linked to different types of orders with various flows. In a similar fashion, the operations at which a test group should be performed vary. It should therefore be possible to set up in which route and at which operation a test group should be executed. The three fields that need to be added to the form meet this purpose. All fields should be enum. They are described below.

- 1) **Route code:** From route code two different options should be available: 'All' and 'Table'. In case a user selects 'All', the quality order should appear every time the chosen operation appears in a route. In case a user select 'Table', the quality order should only appear when the chosen operation appears for the specific route chosen.
- 2) **Route:** Route should contain all the routes that are set up. The field should be locked until the field 'route code' has been populated.
- 3) **Operation:** Should contain those operations that belong to the route that is chosen, i.e. once the user has selected a route, he/she should only be able to see those operations that belong to the route number he/she has selected. The field should be locked until the field 'route' has been populated.

The screenshot shows a software window titled "Quality associations (1 - de30)". It features a menu bar with "File", "New", and "Delete". Below the menu is a table with columns: Site, Reference type, Item code, Item, Account code, Account selection, Resource code, and Res. The table contains one row: SNL01, Route operation, All. To the right of the table are several input fields: Site (dropdown), Reference type (Inventory), Item code (Table), Item (dropdown), Account code (Table), Account selection (dropdown), Resource code (Table), and Resource (dropdown). A red box highlights three fields: Route code (dropdown), Route (dropdown), and Operation (dropdown). Below these fields are sections for "Timing" (Event type, Execution, Show info checkbox), "Quality order process", and "Specifications". A "Close" button is at the bottom right.

6.4.5 Quality order usability

In case a site wants increased automatization of its quality management, it shall be possible to enable such automatization.

1) 'Use automatized quality management' eases the flow for quality management by reducing the amount of clicks a user has to execute in the system. 2) 'Use automatized approval' will approve non-conformances and corrections automatically. 3) 'Show quality acceptance' (Inventory management/Setup/Quality control/Quality associations/Timing) will show quality orders when created. If this check box is ticked then system should bring up the quality acceptance form when a quality order is created automatically as per setup in quality association (at present this functionality is controlled by Use quality management parameter, and this need to be changed to new parameter Show quality acceptance).

The functionalities should be enabled in Inventory and warehouse management > Setup > Inventory and warehouse management parameters.

Inventory and warehouse management parameters (1 - nl30)

File

General
Journals
Bills of materials
Locations
Transport
AIF
Quality management
RFID
Inventory dimensions
Print management
Picking and packing
Number sequences

Select to use quality management and set up an hourly rate

Quality management Report setup

Use quality management: ☐
Hourly rate:

Use automatized quality management ☒
Use automatized approval ☒

Should quality management functionality be used in this company Close

When the fields are unticked, quality management should work as it does with standard AX, but by enabling both parameters, below process should be applied for quality management.

Quality orders (1 - nl30) - Quality order: 12345, Oticon Ria2 Pro ITC

File New Delete Validate Functions Inquiries Inventory

Overview General References Financial dimension Inventory dimensions

Quality order	Item number	Test group	Quantity	Status	Reference type	Ear
12345	156807	Test 1	1.00	Failed	Production	Left

+ Add - Remove

Overview General Test

Sequence number	Test	Test result	Results entered	Include results	Documents	Outcome
10	Che...	✗	☐	☒	☐	N (No)
20	Che...	✗	☐	☒	☐	N (No)

Used symbols

✗ Fail ✓ Pass

The sequential order the test will be executed. (0) EUR nl30 Atlas_Test_EMEA Close

The quality order form should popup automatically only if the former quality order at this particular step was failed or is still open. Passed quality orders should not open again.

In cases where user want to send the item back to an operation when an item fails a quality order. In such scenario when user scan an operation which is previously completed, then only the failed item should be ticked off by the system on the production lines. The item that passed the quality order should be unticked, as this will not be included for the next operation(s) (the next operation (s) is intended to fix the item that failed the quality check).

Once the user validates a quality order that contains one or more failed steps, a non-conformance should be generated automatically. Today the user has to go to 'inquiries' and select 'non-conformances', however, by enabling automatization,

the form for non-conformance should appear once the user presses either 'validate' or selects 'yes' when system asks if quality order should be validated (see below picture).

Quality orders (1 - nl30) - Quality order: 12345, Oticon Ria2 Pro ITC

File New Delete Validate Functions Inquiries Inventory

Overview General References Financial dimension Inventory dimensions

Quality order	Item number	Test group	Quantity	Status	Reference type	Ear
12345	156807	Test 1	1.00	Failed	Production	Left

Microsoft Dynamics

Do you want to validate 12345

Yes No

Add Remove

Overview General Test

Sequence number	Test	Test result	Results entered	Include results	Documents	Outcome
10	Che...	✗	☐	☒		
20	Che...	✗	☐	☒		

Used symbols

✗ Fail ✓ Pass

Navigation icons | (0) | EUR | nl30 | Atlas_Test_EMEA | Close

Once the user selects either 'validate' or 'yes' to the info message, the non-conformance form should appear with a new line (see below picture). In case the user selects 'no', the quality order form should close and the non-conformance form should not appear.

, New, ☐, 000009. The 'Approval' field is highlighted in blue."/>

Non conformances (1 - nl30) - Non conformance number: NL30-00014, Production

File New Delete Related operations Corrections Functions Inquiries Print Inventory

Overview General Quality information Financial dimension Inventory dimensions

Non conformance number	Non conformance type	Problem type	Date	Reported by	Worker responsible	Item number	Treatment not required	Approval	Clos...	Approved by
NL30-00014	Production	3	07-01-20...	000009	Martin Errebo Møller	154709	☐	New	☐	000009

Navigation icons | Non conformance identification | (108356) | EUR | nl30 | Atlas_DEV13 | Close

As mentioned, a non-conformance should already be generated for the failed quality order. The user no longer has to select 'new' to generate it. Below is an explanation of the fields that appear in the form:

- Non-conformance number: Running number that is generated automatically once a new non-conformance is registered.
- Non-conformance type: Type of non-conformance. Value should be the same as the reference type in quality order. Field should be populated automatically.
- Problem type: Mandatory field which the user has to enter via a dropdown with list of values that are defined in the setup for problem types. The user should be able to enter the problem type by typing in the field, e.g. in above case the user should be able to type '3' as this is a value in the dropdown. Similarly, the user should also be able to

type a part of a value in the dropdown from where the system should recognize the full value. To emphasize that the field is mandatory, underline 'problem type' with red.

- Reported by: The worker who performed and failed the quality order. Field should be populated automatically.
- Worker responsible: The worker who caused the issue. Has to be entered by the user via a dropdown, however, not a mandatory field.
- Approval: Always set approval to 'new' until non-conformance has been approved, refused or closed. This is standard AX.
- Approved by: Same value as 'reported by'. Field should be populated automatically.

How the non-conformance gets approved depends on whether the tick mark for 'Use automatized approval' is enabled or not. If it is not enabled, the user has to approve the non-conformance by selecting 'approve' from the top bar.

Non conformance number	Non conformance type	Problem type	Date	Reported by	Worker responsible	Item number	Treatment not required	Approval	Dis...	Approved by
NL30-00014	Production	3	07-01-20...	000009	Martin Embo Møller	154709	☐	Approved	☐	000009

In case the tick mark for 'Use automatized approval' is ticked off, the approval of the non-conformance should happen automatically once the user is closing the non-conformance form. **NB:** The logic behind approving a non-conformance is already existing. With the current solution, the user will have to go to 'Functions' and select approve, which will mark the non-conformance as approved. This action should be possible to do via the header, thus, an 'approve' functionality has been included here.

After the approval of the non-conformance comes the corrections. The worker that has to perform the correction will retrieve the corrections form either via the non-conformance form or via the menu item for corrections. If corrections are retrieved from the non-conformance form, the system will automatically populate the non-conformance form with the non-conformance number (see below picture).

Diagnostic	Resolution	Worker	Correction priority	Created date and time	Requested date	Planned date	Completion date and time	Completed
			Low	07-01-2019 13:48			1	☐

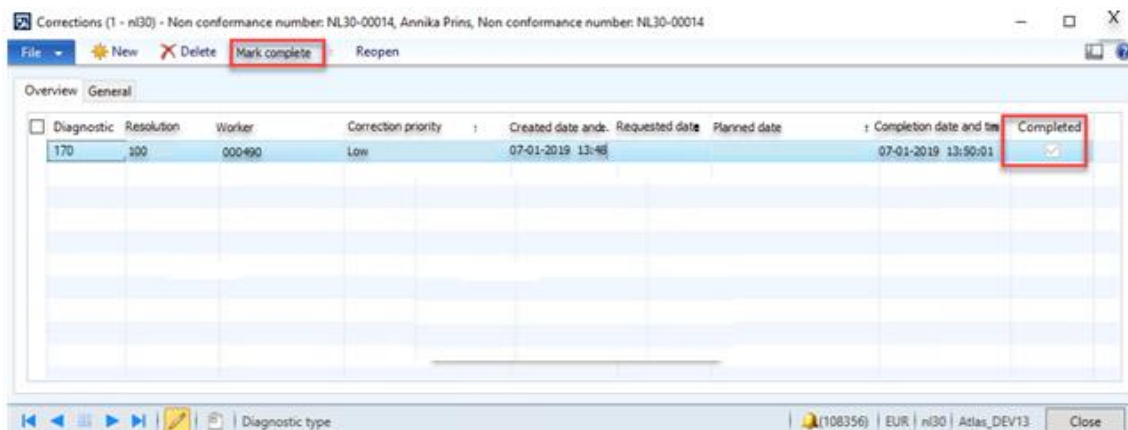
In case the user retrieves the corrections from the menu, the user must select the non-conformance number in the dropdown highlighted below.

Below is an explanation of the fields that figure in the corrections form:

- **Diagnostic:** Mandatory field which the user has to enter via a dropdown with list of values that are defined in the setup for diagnostic types. The user should also be able to enter the diagnostic by typing in the field. Similarly, the user should also be able to type a part of a value in the dropdown from where the system should recognize the full value.
- **Resolution:** Mandatory field which the user has to enter via a dropdown with list of values that are defined in the setup for resolution types or by typing a part of the value from where the system should recognize the full value.
- **Worker:** The worker who performed the correction. The field is mandatory and the user has to enter the information via a dropdown with a list of workers.

Just like problem type and diagnostic type have a menu item, resolution must also have a corresponding menu item 'resolution types'. The item should be placed in the same group as problem type and diagnostic type: Path is Inventory management > Setup > Quality management (see below picture).

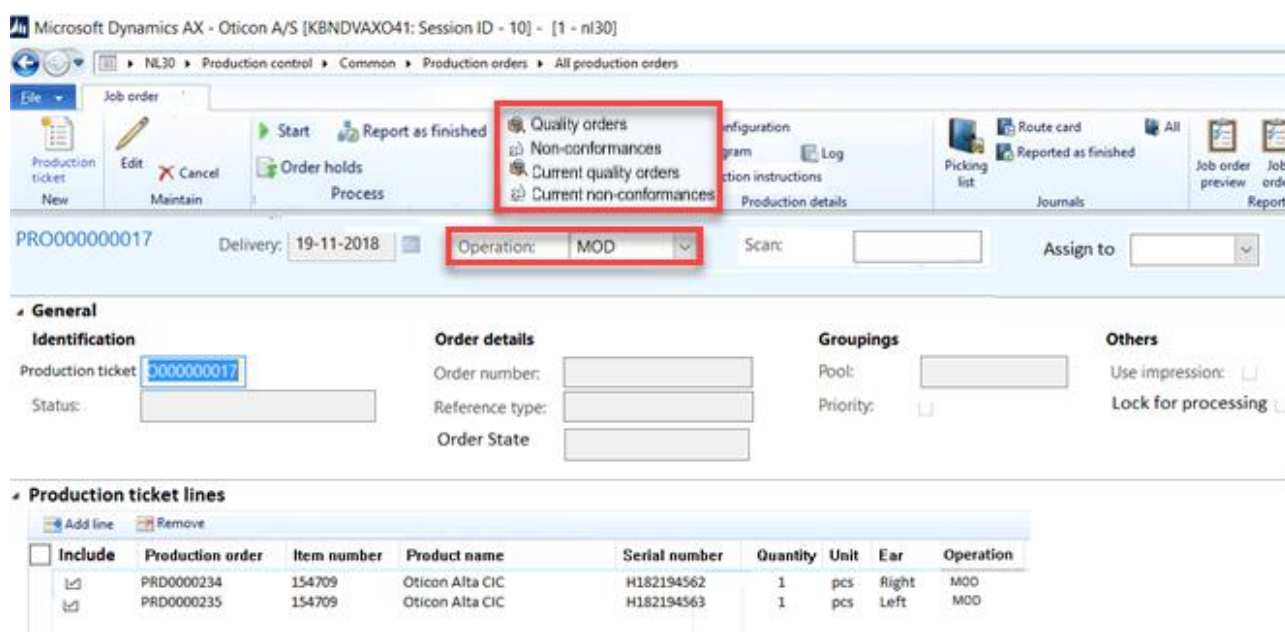
Once the worker has handled the correction, the correction has to be marked as completed. If the tick mark for 'Use automatized approval' is disabled, the worker should press 'mark complete' which should mark the corrections as completed.



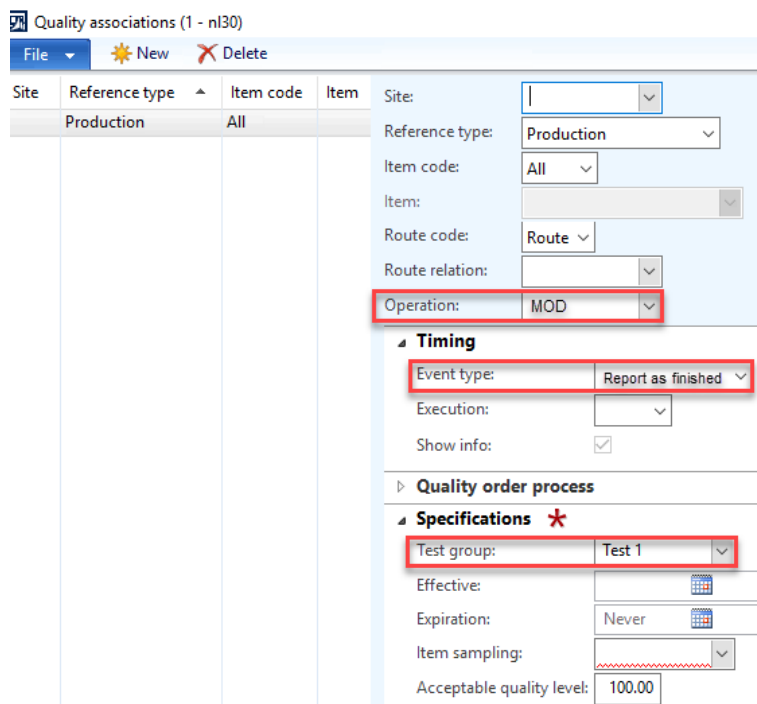
In case 'Use automatized approval' is enabled, the completion should happen automatically. In order for that to happen, an action "markCompleteActionForCorrection" must be linked to each state where corrections are likely to be performed. The action should complete any open corrections either when the user scans into or out of a particular operation (depending on which setup is selected) and it should also close related nonconformances. Finally, the last operation for a given route must hold a guard to ensure that all corrections have been marked as completed before an order can be reported as finished.

Access to quality order from production ticket header

On the current production ticket form, quality orders and non-conformances are retrieved from the header. If 'quality orders' is selected, all quality orders linked to the production ticket will appear. Similarly, if all 'non-conformances' is selected, all non-conformances linked to the production ticket will appear. The first two lines in the highlighted section will have the same purpose. Below, however, we should include 'current quality orders' and 'current non-conformances'. If the user selects 'Current quality orders', the quality orders linked to the current operation should open. Similarly, if the user selects 'Current non-conformances', the non-conformances linked to the current operation should open. In below example, the current operation is MOD (modeling). By selecting 'Current quality orders', the quality orders linked to operation 'modeling' should appear – both in case quality orders are linked to 'start' and 'report as finished'.



To identify which quality orders must appear once a user selects 'current quality orders' for the operation that the production ticket is scanned into, verify with 'quality associations'. Path: Inventory management > Setup > Quality control > Quality associations. In below example a quality order with test group 'Test 1' is linked to operation MOD. The quality order will appear once the user scans 'report as finished' for MOD. Knowing this, the system should show either one or two quality orders (depending on mo/binaural order) once the user selects 'current quality orders' and the operation is MOD. In case a quality order had been linked to 'start' as well, the system should show either two or four quality orders (depending on mo/binaural order) once 'current quality orders' is selected and operation is MOD. Register



Quality associations (1 - nl30)

File New Delete

Site	Reference type	Item code	Item
	Production	All	

Site:

Reference type:

Item code:

Item:

Route code:

Route relation:

Operation:

Timing

Event type:

Execution:

Show info: ☒

Quality order process

Specifications *

Test group:

Effective:

Expiration:

Item sampling:

Acceptable quality level:

When the user selects 'current non-conformances' and operation is 'MOD', any non-conformances linked to the quality orders that are linked to operation MOD should appear.

Link between problem type > diagnostic type and diagnostic type > resolution type.

The form for problem type should hold a link to the form for diagnostic type and similarly should the diagnostic type form hold a link to the resolution type form. The linking means that only the chosen diagnostic types for the given problem type should be available once that problem type is selected during a non-conformance. Similarly, only the chosen resolution types for the given diagnostic type should be available once that diagnostic type is selected during a correction.

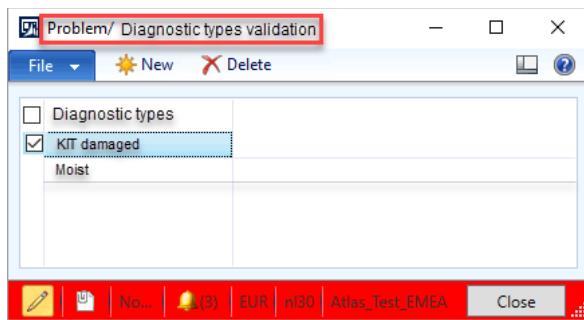
Problem, diagnostic and resolutions types are set up in Inventory management > Setup > Quality management. Resolution types are not a part of standard AX, thus, requires a new form. The form should be identical with the forms for 'problem types' and 'diagnostic types'. Example is shown below.

Problem type	Description
KIT replaced	
Moist removed	

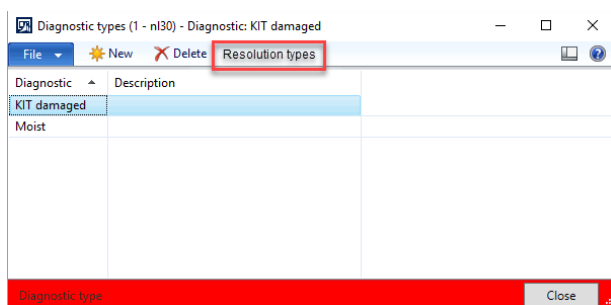
To link respectively problem type/diagnostic type and diagnostic type/resolution type, the user should go Inventory management > Setup > Quality management and select problem type to link those with diagnostic types and select diagnostic types to link those with resolution types.

In below example the problem type 'battery time low' is linked to one or more diagnostic types by selecting 'diagnostic types' and from here, a new form should open. The form should include all those diagnostic types that have been set up in the menu item 'diagnostic types'. In below example, only one diagnostic type 'KIT damaged' is linked to problem type 'battery time low'. That means that once the user selects problem type during a non-conformance, the user who performs the corrections will only be able to select 'KIT damaged' as diagnostic type.

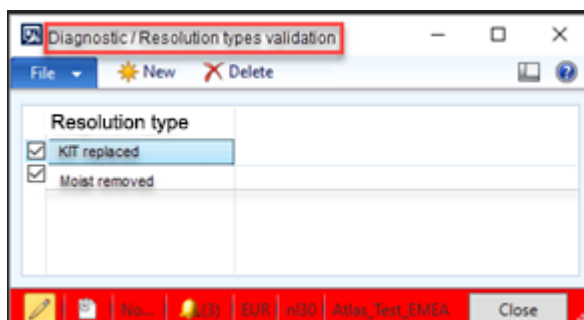
Problem type	Description
Battery time low	
Noise	



In a similar fashion, resolution types can also be linked to each diagnostic type. Here resolution types 'KIT replaced' and 'moist removed' are both linked to diagnostic type 'KIT damaged'. The worker performing the corrections will therefore be able to select either one of the resolution types.



Below form, 'Diagnostic/Resolution types validation' should contain all those resolution types that are created in the menu item for resolution types.



6.4.6 Corrections for failed quality orders

Once an operator has performed corrections for the failed quality order, the order will move forward to the operation at which it failed the quality check. Currently that triggers a new quality order. In case of a binaural order in which only one of the instruments failed the quality check, it triggers quality orders for both instruments. This requires an update, i.e.: the system should only trigger a quality order for the instrument that failed the quality check. If one instrument failed in a binaural order, then one quality order is generated. If both instruments in a binaural order failed, then two quality orders are generated.

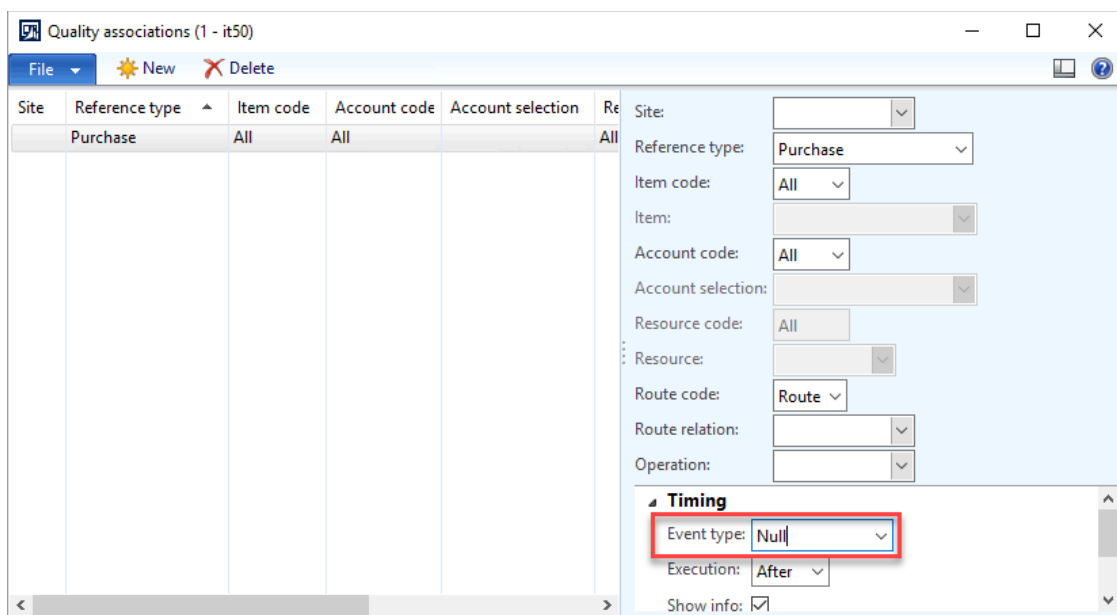
6.4.7 Quality order for service

Two new fields “Item number - service” and “Serial number - service” should be added the the quality order overview tab to register item number and serial number of the item related to the production order. Those fields should be updated only if quality order is created for the production ticket related to service order.

6.4.8 Quality order for purchase order

On receipt of items from supplier (say from Poland) wholesale company performs random quality testing. To record test result users have to create quality order manually. Creation of quality order is standard, we have to enhance the form with Ear information so that user can understand for which ear the test results are captured. The ear on quality order record should be fetched from sales line related to thePO line.

As the quality orders for purchase orders are performed randomly, the quality order for a purchase order should only be created when the user selects ‘new’ from the quality acceptance form. The needed information for the quality order should automatically be populated when selecting ‘new’ to avoid having the user to select the information manually. The needed information can be retrieved from the purchase order. The test group that should appear when selecting ‘new’ depends on the setup in ‘Quality associations’. To make this work, the form for Quality associations needs an adjustment. In field ‘Event type’ we need the value ‘Null’, as the quality order should not be triggered before or after any specific event.



The screenshot shows the 'Quality associations' form with a table on the left and a configuration panel on the right. The table has columns: Site, Reference type, Item code, Account code, Account selection, and Resource. The right panel contains various dropdown menus for configuration. The 'Timing' section is expanded, showing 'Event type' set to 'Null' (highlighted with a red box), 'Execution' set to 'After', and 'Show info' checked.

Site	Reference type	Item code	Account code	Account selection	Resource
	Purchase	All	All		All

Configuration Panel:

- Site: [Dropdown]
- Reference type: Purchase [Dropdown]
- Item code: All [Dropdown]
- Item: [Dropdown]
- Account code: All [Dropdown]
- Account selection: [Dropdown]
- Resource code: All [Dropdown]
- Resource: [Dropdown]
- Route code: Route [Dropdown]
- Route relation: [Dropdown]
- Operation: [Dropdown]
- Timing**
 - Event type: Null [Dropdown]
 - Execution: After [Dropdown]
 - Show info: ☒

6.4.9 Quality acceptance form

In standard AX quality orders are created for each production/purchase order. Due to this user has to navigate through multiple quality orders to document test results. The intent of introducing a new user interface ‘Quality acceptance’ form is to enhance the usability of quality order by making it possible to document test results of multiple quality order in one form.

The ‘Quality acceptance’ form will be called from ‘Quality acceptance’ button on production ticket and purchase order.

Quality acceptance - nl30j -

File New Validate Non conformance Reopen quality order Pass selected

References

Order number: SOR000088848

Reference type: Sales order

Show Open

Overview

Quality order	Operation	Test	Outcome	Reference number	Ear	Service specification	Status	Test group
NL30-000611	MOD	T1	Y	PRO000000411	Left		Open	Tg1
NL30-000612		T1	N	PRO000000412				
NL30-000633		T1	N	PRO000000411				
NL30-000633		T2	N	PRO000000411				
NL30-000634		T1	N	PRO000000412				
NL30-000634		T2	N	PRO000000412				

Used symbols

Navigation icons: Back, Forward, Search, etc.

Type of reference for quality order (81) EUR nl30 Atlas_DEV13 Close

Buttons on 'Quality acceptance' form:

New: Pressing 'new' should generate a quality order that contains the relevant information depending on the setup in quality associations.

Validate: This should open up Validate form (same as if it is called from quality order form) and it should validate all selected quality orders in overview tab of quality acceptance.

Non conformance: This should open up Non conformance form (same as if it is called from quality order form/Inquiries). This button should be disabled if more than one record is selected in overview tab of quality acceptance.

Reopen quality order: This should reopen (same as if it is called from quality order form/Functions) all selected quality orders in overview tab of quality acceptance.

Pass selected: The pass selected button should set "Outcome" with value, which has Outcome status as 'Pass' in 'Test variable outcome master', based upon 'Test variable' set for a particular 'Test' in 'Test group'. If for a Test, there are more than one Outcome for which Outcome status is 'Pass' in 'Test variable outcome master' then skip those Test from the update even if they are selected. Show Infolog describing all the tests which could not be updated to status Pass because of more than one pass outcome. Infolog: Outcome of following Test could not be marked as pass as there are more than one outcome which can result in Pass:

<Test> <Quality order>

<Test> <Quality order>

>>

Overview tab should show quality orders as per criteria selected in 'Show' field. If Show field is Open then overview should contain all quality orders with status Open related to the production ticket, similarly for options Pass and Fail in Show field. If user selects 'All' in Show field then show all quality orders related to the production ticket irrespective of quality order Status.

On Close of Order acceptance form system should auto validate all the quality orders for which Status is still open.

Add new field 'Test pending' on Test variable outcomes. This flag will determine if the value of this outcome indicates a pending test. If any of the test in a quality order has an outcome which has Test pending as true, then the validate function for that quality order should be skipped i.e even after validation the quality order status should remain as open.

The screenshot shows a window titled "Test variable outcomes (1 - it50) - Variable: V1, Pending". The window has a menu bar with "File", "New", and "Delete". Below the menu bar are two tabs: "Overview" and "General". The "General" tab is active, showing the "Identification" section with fields for "Variable" (V1), "Outcome" (Pending), and "Description". Below these is a checkbox labeled "Test pending" which is checked. The "Status" section shows "Outcome status" set to "Fail". At the bottom of the window is a red toolbar with various icons and a "Close" button.

6.4.10 Quality acceptance automation

To improve user experience while working with quality orders a following changes should be done in the current solution that supports simplification of this process. New automation feature should be enabled with the parameter "Enable quality order automation" (Path: Production control/Setup/Production control parameters/General).

The screenshot shows a window titled "Production control parameters" with a sub-header "KBNXDAPP203". The window has a menu bar with "File". On the left is a sidebar with a tree view containing "General", "Journals", "Production Ticket Default", "Automatic update", "Standard update", "AIF", "Status", "Inventory dimensions", "Lean manufacturing", "Unit of measure", "Number sequences", and "Default Service Part". The "General" tab is selected. The main area contains several sections: "Production recalculation", "Parameter usage", "Production creation", "Job scheduling", "Order triage management", "Routes", and "Posting". The "Enable quality order automation" checkbox under the "Production creation" section is checked and highlighted with a red box. At the bottom of the window is a status bar with the text "The maximum number of production orders that can be processed at once by a single production recalculation task" and a "Close" button.

If parameter is disabled in the legal entity then quality orders should be validated using already existing logic, without any automation. If parameter is enabled in legal entity then quality order validation should be automated in a way described below.

When operation is finished on the production ticket system should check if there is any open quality order related to this operation and if such quality order exists then system should show following Infolog.

The screenshot displays the 'Production Ticket' form in a software application. The top menu bar includes options like 'File', 'Cancel', 'Start', 'Finish', 'Order holds', 'Change route/operation', 'Quality Acceptance', 'Non conformances', 'Quality orders', 'Current quality orders', 'Current non conformances', 'Picking list', 'Route card', 'Reported as finished', 'All', 'Production details', 'List', 'Workflow', and 'Attach'. The main form area shows details for production ticket 'PRT000087296', including delivery date '1/9/2023', operation 'MOD', and assignee 'Volodymyr Yakubiv'. An infolog message states: 'The Production order PRO000115634, PRO000115635 is started for MOD operation'. A 'General' section contains fields for 'Production Ticket', 'Production ticket status' (set to 'Started'), 'Box coordinate', and 'BC room'. A 'Production Ticket Lines' table lists two lines with production numbers PRO000115634 and PRO000115635, both for 'Oticon Get CIC' with serial numbers W21126148 and W21126149, quantity 1.00, and operation 'MOD'. A 'Line details' section is also visible. A 'Right configuration details' table on the right lists various attributes and their values. A 'Microsoft Dynamics' dialog box is overlaid on the form, asking: 'Do you want to validate the quality order for production ticket PRT000087296'. The dialog has 'Yes' and 'No' buttons.

Include	Production	Item number	Name	Serial number	Quantity	Ear	Operation
☒	PRO000115634	154716	Oticon Get CIC	W21126148	1.00	Right	MOD
☒	PRO000115635	154716	Oticon Get CIC	W21126149	1.00	Left	MOD

Attribute type	Value
Custom Shell...	Beige
Fitting Level	N/A
WirelessRF	N/A
Auto Telecoil	No
Vent	Collection Vent 1.4 m
Receiver Prot...	ProWax
Microphone ...	Wind Shield
Program Co...	No
Pull out String	Standard (8 mm)
Large Ball Re...	No
Pull Out Clot...	N/A
Raised Push ...	N/A

When Yes button is clicked then all test results should be automatically entered as "Passed" and quality order should be validated and closed. When No button is clicked then system should show quality acceptance form on the screen, so user can enter test results manually (for example enter some test results as "Failed" then create non-conformance and correction).

When outcome of quality order is marked as failed then system should automatically open non-conformance form directly from the quality acceptance form.

A new button "Pass selected" should be added to quality orders form, to make manual entering of test results more user-friendly. When this button is clicked then all marked tests should be updated with the outcome "Pass".

Quality orders (1 - it50) - Quality order: QO-00064037, Oticon Get CIC

File New Delete Validate Functions Inquiries Inventory

Overview General References Financial dimension Inventory dimensions

Quality order	Item number	Test group	Quantity	Status	Reference type	Ear
QO-00064036	154716	Prod. HI	1.00	Open	Route operation	Right
QO-00064037	154716	Prod. HI	1.00	Open	Route operation	Left

+ Add - Remove **Pass selected**

Overview General Test

Sequence number	Test	Test result	Results entered	Include results	Documents	Outcome
10	Impression	✗	☐	☒		Pass
20	Test cube dimensions	✗	☐	☒		Pass
30	Shell treatment	✗	☐	☒		Pass
40	Serial number	✗	☐	☒		Pass
50	Vent size correct	✗	☐	☒		Pass
60	No leak from vent	✗	☐	☒		Pass
70	Pull out string	✗	☐	☒		Pass
80	Microphone/receiver	✗	☐	☒		Pass

Used symbols
✗ Fail ✓ Pass

Quality order identification | (0) | EUR | usr | it50 | 1/2/2023 | Atlas_DevTest_EMEA | Close

7 Algorithms

[This section makes a description of eventual algorithms in the solution. An algorithm is a calculation, data updated after an event happened, how data is selected based on queries or other details from the solution]

7.1.1 Changes to Route assignment

Assignment of route to a production order should also consider Route category. If a production order is created with reference to a service specification or type Remake then system should consider Route category 'Remake' to identify the route. For production order created with reference to sales order or created without any reference then system should consider Route category 'None' to identify the route.

7.1.2 Barcode operation

If a barcode starts with character mentioned in operation identifier field in Inventory management/ Inventory and warehouse management parameters then system looks into the Barcode operation setup to identify which operation has to be performed.

1. WDHAssignOperation: Assign the scanned operation to the production ticket.
2. WDHStartOperation: This barcode operation will start the route operation for production order which has included checkbox ticked. This action will do same action as if the operation has been started from current operation form in standard AX. Consider default setup in 'Operation start/Production ticket default' in Production control parameters. Before starting an operation system should check if any prior operation which is mandatory has been reported as finished or not. If not, then throw Error/Warning/ or no action based upon setup in production control parameters/mandatory operation validation.
 - a. Error message- 'Operation <operation name> is still pending for report as finished. Please do report as finish to move forward.'
 - b. Warning message- 'Operation <operation name> is still pending for report as finished'.
 - c. None- no action needed, system should allow to move forward with start of operation.

3. WDHReportAsFinishOperation: This barcode operation will report finish the route operation for production order which has included checkbox ticked. This action will do same action as if the operation has reported finished from current operation form in standard AX. Consider default setup in 'Operation report as finished/Production ticket default' in Production control parameters .

7.1.3 Changes to Transition log

Field mapping-

1. Document type: Production
2. Document number: Production ticket number
3. Header/Line: For production ticket record it should be Header and for production order record it should be Line.
4. Line number: For production ticket record it should be blank and for production order record it should be production order number.
5. From workflow Id: Route ID on production ticket
6. From state and To state:

Header: Operation selected on production ticket

Line: If the line is included then Operation selected on production ticket. If the line is not included then Operation from the production order record in Production ticket lines tab. (it may happen that only one of the production order need to be send back to a operation. That's why 'include' has to be considered for line level update in transition log)

7. Action: Put value as 'Start' if the record in transition log is being created on start of a operation on production ticket. Put value as 'Finished' if the record in transition log is being created on report as finish of a operation on production ticket.
8. Operator name- This should be the worker linked to the user id in Created by field.

Operator name- If operation update is done from Scan operation then this should be the worker assigned to user id selected on Scan operation form. If operation update is one from production ticket then this should be the worker assigned to the user id doing the update.

9. Created by- User id who is doing the update in system.
10. Assigned to worker- Worker in the Assign to field on Production ticket or on Scan operation form.

7.1.4 Changes to Quality association

During Start and report as finish of an operation system looks into Quality association and run through the setup to check if any quality order has to be generated. To control creation of quality order based upon route and operation we have created new parameter 'Route' and 'Operation' which will also be considered as a criteria for quality order creation.

On start and report as finish of an operation, in addition to existing criteria for finding quality association system should also check if there is any quality association record where 'Route' and 'Operation' parameter matches with the Route and current operation on production order. If it matches then create a quality order. Quality order creation should happen as standard i.e. as per the configurations mentioned in field groups-Timing, Quality order process and Specifications of quality association.

8 Example

[This section contains examples of typical situations where and how the solution is used. The examples illustrate data used, steps in a process calculations and how the system is operated.]

9 Data Model Considerations

[In this section both corrections to existing data models and new data structures are listed. Since the data model in Dynamics AX is heavily normalized, corrections to existing data model are only specified on a conceptual level, by specifying in a form, where the data field must exist.]

9.1 Changes to existing forms (new fields in forms)

Form name	Tab(s)	Field name	Help text	Attributes ¹
Production control parameters	General	Enable production ticket	This will enable production ticket related fields and functionalities	Boolean [N]
Route detail	General	Route category	Specify if the route is for new production or for remake	Enum : Production, Remake (default Production)
Route		Mandatory	Specifies if the operation is mandatory	Boolean [N]
Route	General	Category	Specify the category for automatic route version	Lookup for product category
Order hold codes	Production ticket	Lock for processing	Specify if the onhold code should stop further processing of production ticket	Boolean [N]
Order hold codes	Production ticket	Mapped state	Specify mapped state on order	Dropdown from workflow state

¹ Data type (eventually new data type), specific values, Specific validation, access rights

Form name	Tab(s)	Field name	Help text	Attributes ¹
Sales order	Quick access/Production	Production state	Indicates current operation of production ticket.	String (20) Non editable
Sales order lines		Production state	Indicates current operation of linked production order.	String (20) Non editable
Sales order overview		Production state	Indicates current operation of production ticket.	String (20) Non editable
Transition log		Production action	Specifies the production action	String (20) Non editable
Transition log		Ear	Specifies ear side of the instrument	<fetch value from sales line>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) /Route card journal (fieldgroup)	Route card	<i>Same as 'Route card' field on standard production 'Start' form</i>	<i>Same as 'Route card' field on standard production 'Start' form</i>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) /Route card journal (fieldgroup)	Automatic route consumption	<i>Same as 'Automatic route consumption' field on standard production 'Start' form</i>	<i>Same as 'Automatic route consumption' field on standard production 'Start' form</i>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup)/Route card journal (fieldgroup)	Post route card now	<i>Same as 'Post route card now' field on standard production 'Start' form</i>	<i>Same as 'Post route card now' field on standard production 'Start' form</i>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) /Route card journal (fieldgroup)	End mark route	<i>Same as 'End mark route' field on standard production 'Start' form</i>	<i>Same as 'End mark route' field on standard production 'Start' form</i>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) /Picking list journal (fieldgroup)	Picking list	<i>Same as 'Picking list' field on standard production 'Start' form</i>	<i>Same as 'Picking list' field on standard production 'Start' form</i>

Form name	Tab(s)	Field name	Help text	Attributes ¹
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) /Picking list journal (fieldgroup)	Automatic BOM consumption	<i>Same as 'Automatic BOM consumption ' field on standard production 'Start' form</i>	<i>Same as 'Automatic BOM consumption ' field on standard production 'Start' form</i>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) /Picking list journal (fieldgroup)	Create journal per	<i>Same as 'Create journal per ' field on standard production 'Start' form</i>	<i>Same as 'Create journal per ' field on standard production 'Start' form</i>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) /Picking list journal (fieldgroup)	Post picking list now	<i>Same as 'Post picking list now ' field on standard production 'Start' form</i>	<i>Same as 'Post picking list now ' field on standard production 'Start' form</i>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) /Picking list journal (fieldgroup)	End mark picking list	<i>Same as 'End mark picking list ' field on standard production 'Start' form</i>	<i>Same as 'End mark picking list ' field on standard production 'Start' form</i>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) /Picking list journal (fieldgroup)	Physical reduction	<i>Same as 'Physical reduction ' field on standard production 'Start' form</i>	<i>Same as 'Physical reduction ' field on standard production 'Start' form</i>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) /Picking list journal (fieldgroup)	Pick negative	<i>Same as 'Pick negative ' field on standard production 'Start' form</i>	<i>Same as 'Pick negative ' field on standard production 'Start' form</i>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) /Picking list journal (fieldgroup)	Pick negative	<i>Same as 'Pick negative ' field on standard production 'Start' form</i>	<i>Same as 'Pick negative ' field on standard production 'Start' form</i>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) /Picking list journal (fieldgroup)	Complete picking list journal	<i>Same as 'Complete picking list journal ' field on standard production 'Start' form</i>	<i>Same as 'Complete picking list journal ' field on standard production 'Start' form</i>

Form name	Tab(s)	Field name	Help text	Attributes ¹
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) /Picking list journal (fieldgroup)	Print picking list	<i>Same as 'Print picking list 'field on standard production 'Start' form</i>	<i>Same as 'Print picking list 'field on standard production 'Start' form</i>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) / Journals (fieldgroup)	Show journal info	<i>Same as 'Show journal info 'field on standard production 'Start' form</i>	<i>Same as 'Show journal info 'field on standard production 'Start' form</i>
Production control parameters	Production ticket default (new tab)/Operation start (fieldgroup) / Operations (fieldgroup)	Validate mandatory	Specify system action if mandatory operation is not reported	<i>Enum: None, Warning, Error</i>
Production control parameters	Production ticket default (new tab)/Operation report as finished (fieldgroup) / Route card journal (fieldgroup)	Route card	<i>Same as 'Route card' field on standard production 'Report as finished' form</i>	<i>Same as 'Route card' field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/Operation report as finished (fieldgroup) / Route card journal (fieldgroup)	Automatic route consumption	<i>Same as 'Automatic route consumption 'field on standard production 'Report as finished' form</i>	<i>Same as 'Automatic route consumption 'field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/Operation report as finished (fieldgroup) / Route card journal (fieldgroup)	End mark route	<i>Same as 'End mark route 'field on standard production 'Report as finished' form</i>	<i>Same as 'End mark route 'field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/Operation report as finished (fieldgroup) / Picking list journal (fieldgroup)	Picking list	<i>Same as 'Picking list 'field on standard production 'Report as finished' form</i>	<i>Same as 'Picking list 'field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/Operation report as finished (fieldgroup)	Automatic BOM consumption	<i>Same as 'Automatic BOM consumption 'field on standard</i>	<i>Same as 'Automatic BOM consumption 'field on standard</i>

Form name	Tab(s)	Field name	Help text	Attributes ¹
	/ Picking list journal (fieldgroup)		<i>production 'Report as finished' form</i>	<i>production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/Operation report as finished (fieldgroup) / Picking list journal (fieldgroup)	End mark picking list	<i>Same as 'End mark picking list 'field on standard production 'Report as finished' form</i>	<i>Same as 'End mark picking list 'field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/Operation report as finished (fieldgroup) / Report as finished journal journal (fieldgroup)	Report as finished	<i>Same as 'Report as finished 'field on standard production 'Report as finished' form</i>	<i>Same as 'Report as finished 'field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/Operation report as finished (fieldgroup) / Report as finished journal journal (fieldgroup)	End job	<i>Same as 'End job 'field on standard production 'Report as finished' form</i>	<i>Same as 'End job 'field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/Operation report as finished (fieldgroup) / Report as finished journal journal (fieldgroup)	Accept error	<i>Same as 'Accept error 'field on standard production 'Report as finished' form</i>	<i>Same as 'Accept error 'field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/Operation report as finished (fieldgroup) / Journals (fieldgroup)	Show journal info	<i>Same as 'Show journal info 'field on standard production 'Report as finished' form</i>	<i>Same as 'Show journal info 'field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/Operation report as finished (fieldgroup) / Operations (fieldgroup)	Report production as finished	<i>Same as 'Report production as finished 'field on standard production 'Report as finished' form</i>	<i>Same as 'Report production as finished 'field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/Operation report	Validate mandatory	Specify system action if mandatory	Enum: None, Warning, Error

Form name	Tab(s)	Field name	Help text	Attributes ¹
	as finished (fieldgroup) / Operations (fieldgroup)		operation is not reported	
Production control parameters	Production ticket default (new tab)/Production report as finished (fieldgroup) / Route card journal (fieldgroup)	Route card	<i>Same as 'Route card' field on standard production 'Report as finished' form</i>	<i>Same as 'Route card' field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/ Production report as finished (fieldgroup) / Route card journal (fieldgroup)	Automatic route consumption	<i>Same as 'Automatic route consumption' field on standard production 'Report as finished' form</i>	<i>Same as 'Automatic route consumption' field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/ Production report as finished (fieldgroup) / Route card journal (fieldgroup)	End mark route	<i>Same as 'End mark route' field on standard production 'Report as finished' form</i>	<i>Same as 'End mark route' field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/ Production report as finished (fieldgroup) / Picking list journal (fieldgroup)	Picking list	<i>Same as 'Picking list' field on standard production 'Report as finished' form</i>	<i>Same as 'Picking list' field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/ Production report as finished (fieldgroup) / Picking list journal (fieldgroup)	Automatic BOM consumption	<i>Same as 'Automatic BOM consumption' field on standard production 'Report as finished' form</i>	<i>Same as 'Automatic BOM consumption' field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/ Production report as finished (fieldgroup) / Picking list journal (fieldgroup)	End mark picking list	<i>Same as 'End mark picking list' field on standard production 'Report as finished' form</i>	<i>Same as 'End mark picking list' field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/ Production report as finished (fieldgroup) /	Report as finished	<i>Same as 'Report as finished' field on standard production</i>	<i>Same as 'Report as finished' field on standard production</i>

Form name	Tab(s)	Field name	Help text	Attributes ¹
	Report as finished journal journal (fieldgroup)		<i>'Report as finished' form</i>	<i>'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/ Production report as finished (fieldgroup) / Report as finished journal journal (fieldgroup)	End job	<i>Same as 'End job ' field on standard production 'Report as finished' form</i>	<i>Same as 'End job ' field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/ Production report as finished (fieldgroup) / Report as finished journal journal (fieldgroup)	Accept error	<i>Same as 'Accept error ' field on standard production 'Report as finished' form</i>	<i>Same as 'Accept error ' field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/ Production report as finished (fieldgroup) / Journals (fieldgroup)	Show journal info	<i>Same as 'Show journal info ' field on standard production 'Report as finished' form</i>	<i>Same as 'Show journal info ' field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/ Production report as finished (fieldgroup) / Operations (fieldgroup)	Report production as finished	<i>Same as 'Report production as finished ' field on standard production 'Report as finished' form</i>	<i>Same as 'Report production as finished ' field on standard production 'Report as finished' form</i>
Production control parameters	Production ticket default (new tab)/ Production report as finished (fieldgroup) / Operations (fieldgroup)	Validate mandatory	Specify system action if mandatory operation is not reported	Enum: None, Warning, Error
Production control parameters	Production ticket default (new tab)/ Production end (fieldgroup) / End (fieldgroup)	Report as finished	<i>Same as 'Report as finished ' field on standard production 'End' form</i>	<i>Same as 'Report as finished ' field on standard production 'End' form</i>
Production control parameters	Production ticket default (new tab)/ Production end	Use time and attendance cost	<i>Same as 'Use time and attendance cost ' field on standard</i>	<i>Same as 'Use time and attendance cost ' field on standard production 'End' form</i>

Form name	Tab(s)	Field name	Help text	Attributes ¹
	(fieldgroup) / End (fieldgroup)		<i>production 'End' form</i>	
Production control parameters	Production ticket default (new tab)/ Production end (fieldgroup) / Scrap (fieldgroup)	Scrap method	<i>Same as 'Scrap method 'field on standard production 'End' form</i>	<i>Same as 'Scrap method 'field on standard production 'End' form</i>
Production control parameters	Production ticket default (new tab)/ Production end (fieldgroup) / Scrap (fieldgroup)	Scrap account	<i>Same as 'Scrap account 'field on standard production 'End' form</i>	<i>Same as 'Scrap account 'field on standard production 'End' form</i>
Production control parameters	Default service part	Service type	Type of service	Enum (Service type)
Production control parameters	Default service part	Category hierarchy	The name of the category hierarchy	Dropdown from category hierarchy master (same as on Product categories on released product)
Production control parameters	Default service part	Category	Category identifier	Dropdown from category based upon selected hierarchy (same as on Product categories on released product)
Production control parameters	Default service part	Item	Identify item	Dropdown for item master where Service stock part is checked
Production control parameters	Default service part	Default service item	Specify item that should be used if system don't find any service stock item for a combination.	Dropdown for item master where Service stock part is checked
All production order	< list page >	Production ticket	Specifies production ticket reference	<production ticket number to which the production order is linked to>, it should be possible to filter on this field

Form name	Tab(s)	Field name	Help text	Attributes ¹
All production order	< list page >	Production ticket status	Specifies production ticket status	<Status of production ticket to which the production order is linked to>, >, it should be possible to filter on this field
All production order	< list page >	Hold production	Specifies production ticket is hold for production	<Hold production field on production ticket to which the production order is linked to>
All production order	< list page >	Operation	Specifies current operation of production	<Operation field on production ticket lines for related production order>
All production order	< list page >	Category	Specifies category of product	<Category from item master>
All Production ticket	< list page >	Customer account	Customer account that generated the order	<customer account from sales order/Service order>, >, it should be possible to filter on this field
All Production ticket	< list page >	Name	Customer name	<customer name for the customer>
State scanning sales order	<line grid>	Box coordinate	Box coordinate id	Display from sales order /line
State scanning sales order	<line grid>	BC room	Display row and column in a matrix format	Display from sales order /line
Released product	General/Administration	Service stock part	Specifies if the product is used to represent a service item on production order	Boolean [N]
Configit integration Id to AX	Attribute integration Ids	Show on production ticket	Show attribute on production ticket	Boolean [N]
Inventory and warehouse management	Quality management	Use automatized quality management	Functionality that eases quality management when enabled	Check box

Form name	Tab(s)	Field name	Help text	Attributes ¹
Inventory and warehouse management	Quality management	Use automatized approval	Functionality that eases the approval of non-conformances and corrections when enabled	Check box
Quality associations	Timing	Show quality acceptance	Functionality when enabled shows the quality acceptance form when quality orders is created automatically.	Boolean [N]
Corrections		Resolution	Dropdown from which all specified resolution types should be available	Dropdown
Problem types		Diagnostic types	New form will open from which diagnostic types can be linked to problem types	
Diagnostic types		Resolution types	New form will open from which resolution types can be linked to diagnostic types	
Picking list	General	Kit picking list number	Identifies the printed kit picking list	Text, non-editable
Picking list	Overview	Printed	Indicates if the picking list has already been printed	Boolean, non-editable
Picking list	Overview	Order number	Links each picking list to the right sales order.	Non-editable
Sales and marketing parameters	Sales and marketin/Setup/Sales and marketing parameters/General	Default marking in State scanning production ticket	Determines if default marking in the 'State scanning production ticket' form should be possible	Tick mark

Form name	Tab(s)	Field name	Help text	Attributes
Quality orders		Outcome	This is the field from which the user should be able to mark a test as passed (= Y (Yes)) or failed (= N (No)).	Enum field with two values: Y (Yes) N (No)
Quality associations		Route code	This is the field from which the test groups can be applied to 'All' routes or 'Table' routes	Enum field with two values: All Table
Quality associations		Route	This is the field from which existing routes should be visible	Enum field
Quality associations		Operation	This is the field from which existing operations belonging to a specific route should be available	Enum field
Quality associations		Null	'Null' needs to be added to the dropdown for the existing field 'Event type'.	Enum field
Create lines		Reverse to warehouse	Specify warehouse to which the item should be reversed to	Dropdown from warehouse master. This field should be visible only if production ticket function is enabled.
Service order	Maintain	Order holds		Button
Quality orders	Overview	Item number – service	Item number from service specification	Text
Quality orders	Overview	Serial number – service	Serial number from service specification	Text
Production control parameters	Journals/Default journal names	Movement	Movement journal to move damaged components during production out of scrap warehouse	Dropdown from journal master showing only journal type movement

Production control parameters	Production ticket default	Scrap warehouse	Default scrap warehouse to be used for holding damaged components during production	Dropdown of warehouse master. Show only those warehouse which 'Consignment/loan ' and 'Physical warehouse' are not ticked.
All production orders	listpage	Sellable brand	Brand of the customer	From the linked sales/service order header. Should be possible yo filter.
All production orders	listpage	WDHBrand	WDHBrand	From the financial dimension on linked sales/service order line.
All sales order line	listpage	Sellable brand	Brand of the customer	From the linked sales r. Should be possible yo filter.
All sales order line	listpage	WDHBrand	WDHBrand	From the financial dimension on linked sales
All sales order	listpage	Sellable brand	Sellable brand	From the linked sales r.
All service order	listpage	Sellable brand	Brand of the customer	From the linked service order.
All service specifications	listpage	Sellable brand	Brand of the customer	From the linked sales/service order header. Should be possible yo filter.
All service specifications	listpage	WDHBrand	WDHBrand	From the financial dimension on linked sales/service order line.
All sales order line	listpage	Production state	Production state	From sales order line
All service specifications	listpage	Production state	Production state	From service specification line
All sales orders for order entry	List Page	Order holds	Same as order holds button on all sales order list page	Button

Sales order entry form	Production	Order holds	Same as order holds button on sales order form	Button
Add / Add additional		Stock consumption	If this checkmark is active then consumption journal will be created for this item	Boolean (Y)
Test variable outcomes	General tab	Test pending	Identify if a test outcome represents awaiting test result	Boolean [N]
Production control parameters	Default service part	Route number	Specify route to be used for chosen service type	Dropdown: Route master 'All routes'
Production control parameters	Default service part	Name	Route description	Display route name
Production ticket	General	Route number	Indicates route number in use	Display route number from first production order
Production ticket	General	Name	Route description	Display route name
All production tickets		Route number	Indicates route number in use	Display route number from first production order. Should be able to filter on this field.
All production tickets		Name	Route description	Display route name
Subcontracted work		Production ticket	Indicates production ticket	Display production ticket
Subcontracted work		Order number	Indicates production order number	Display production order
Subcontracted work		IC state	Indicates intercompany state of the subcontracted work	Display IC state. Should be able to filter on this field.
Route	General/Route	Vendor item	Indicates the item representing subcontracted operations where BOM does not exist for the finished item	Enable only when route type is vendor. Dropdown: Released product
Purchase order	Purchase order lines	Subcontract item	Item number for which operation has	Display field.

			been subcontracted for in the purchase order.	
Production control parameters	Default service part	Operation for picking list	Specify operation at which picking list should be created	List of Operations from route mentioned in that record
Production control parameters	Production ticket defaults	Auto-assign worker	Automatically add worker name in "Assign to" field on the production ticket	Boolean, default NO
State machine parameters	General	Auto-assign worker in State scanning production ticket	Automatically add worker name in "Assign to" field on the State scanning production ticket	Boolean, default NO
Production control parameters	Default service part	Default warehouse for service	Default warehouse for the service order related production ticket	Lookup to the Warehouse master data table.
Production ticket	Work specification	Request area	Request code area	Lookup to the request area
Production ticket	Work specification	Request area description	Request code area description	Lookup to the request area description
Production ticket	Work specification	Request code	Request code	Lookup to the request codes
Production ticket	Work specification	Request code description	Request code description	Lookup to the request code description
Production ticket	Work specification	Failure area	Failure area	Lookup to the EPDS for the failure area
Production ticket	Work specification	Failure area description	Failure area description	Lookup to the EPDS for the failure area description
Production ticket	Work specification	Failure code	Failure code	Lookup to the EPDS for the failure code
Production ticket	Work specification	Failure description	Failure description	Lookup to the EPDS for the failure description

Production ticket	Work specification	Resolution	Resolution	Lookup to the EPDS for the resolution code
Production ticket	Work specification	Resolution description	Resolution description	Lookup to the EPDS for the resolution description
Production control parameters	General	Enable quality order automation	Enable quality order automation	Boolean

9.2 New form and fields

[Here details about the fields listed in the forms above are described]

New form Name (Label)	Help text	Attributes
Production ticket form		
Field Name (Label)	Help Text	Attributes
Production ticket	Identify the production ticket	Dropdown from production ticket
Status	Production ticket status	Enum (Created, Started, Onhold, Cancelled, Finished). (non editable)
Order number	Number of the order referred to	Number of of reference order (non editable)
Reference type	Module and number referred to	Sales, Remake (non editable)
Order status	Reference order status	Work flow status of reference order (non editable)
Box coordinate	Specify box coordinate	View field. Data should be populated from box coordinate field on sales order.
Priority	Is this a priority order	Boolean. Value from priority field on order number (non editable). Should work for both sales and service order In case of a linked service order, the system needs to check the setup of the "Priority" code. Priorities can be marked as "Express" on the setup. If the priority code on the service order has the "Express" = true or

		"High" = true , the priority on the production ticket should also be true.
Operation	Specifies the operation	Dropdown (operation list from the route for production order)
Pool	Specifies the pool that the production is attached to	Dropdown from Pool master. Non mandatory
Delivery date	Requested ship date	Requested ship date from sales order header.View only..
Scan	Scan barcode	String (30) Editable. Cursor should be on this field by default on production ticket form.
Assign to	Specify the worker to whom the production ticket is assigned to	Dropdown : Worker master. Non mandatory
Others/Lock for processing	Specifies that the production ticket is blocked from further processing	Boolean. Non Editable. The value will be derived from onhold codes on sales order
Use impression	Specify if existing impression to be used	Non editable. Value will come from sales order
Order processing/ Include	Specify the production order to be included in production ticket processing	Boolean [Y]
Order processing/ Production order	Specify the production order to be included in production ticket.	Dropdown (Production order master). Mandatory
Order processing/ Item number	Specifies the item number on the production order	Non editable. Item number from production order.
Order processing/ Product name	Description of the item.	Non editable.
Order processing/ Serial number	Serial number if the item.	Non editable. Serial number from sales order line.
Order processing/ Quantity	Scheduled production quantity	Non editable. Quantity from production order.
Order processing/ Ear	Identifies the ear	Non editable. Ear from sales line associated with production order
Order processing/ Operation	Identification of the operation	Non editable.
Configuration	Specifies configuration for the order	Non editable. Show configuration from sales line corresponding to the production order.

Attachments/Order No		Non editable. Show from attachment on sales order.
Attachments/Order type		Non editable. Show from attachment on sales order.
Attachments/Document type		Non editable. Show from attachment on sales order.
Attachments/Created date and time		Non editable. Show from attachment on sales order.
Attachments/Created by		Non editable. Show from attachment on sales order.
Attachments/Description		Non editable. Show from attachment on sales order.
Serial number used		Text
Item number used		Text
Add		Button
Add additional		Button
New item	Mark in case when new item has been used for swap	Boolean (N)
Battery serial number		Text
Battery serial number used		Text

New form Name (Label)	Help text	Attributes
All Production ticket/ Open production tickets/ On hold production tickets/ Delayed production tickets		
Field Name (Label)	Help Text	Attributes
Production ticket	Identify the production ticket	String (20) (Non editable)
Order number	Number of the order referred to	String (20) (Non editable)
Customer account	Customer account that generated the order	Non editable. Value from corresponding sales order header
Name	Name of the customer	Non editable. Value from corresponding customer account
Reference type	Module and number referred to	String (20) (Non editable)

Priority	Is this a priority order	Boolean. Value from priority field on order number. Should work for both sales and service order For service order, the priority handling should be marked with Express checkbox in the setup table.
Box coordinate	Displays box coordinate	View field. Data should be populated from box coordinate field on sales order.
Assign to	Specify the worker to whom the production ticket is assigned to	Dropdown : Worker master. Non mandatory
Status	Production ticket status	Show Status of Production ticket, non editable
Operation	Specifies the operation	Dropdown (operation list from the route for production order)
Pool	Specifies the pool that the production is attached to	Dropdown from Pool master. Non mandatory
Delivery date	Requested date of delivery	Requested delivery date from sales order header. Editable.
Locked for processing	Specifies that the production ticket is blocked from further processing	Boolean [N]. Non editable
Sellable brand	Brand of the customer	From the linked sales/service order header. Should be possible to filter.

New form Name (Label)	Help text	Attributes
Add product to route		
Field Name (Label)	Help Text	Attributes
All categories		
Item number	(Same as 'add products' form on sales order lines/Add products button)	(Same as 'add products' form on sales order lines/Add products button)
Product name	(Same as 'add products' form on sales order lines/Add products button)	(Same as 'add products' form on sales order lines/Add products button)

Search name	(Same as 'add products' form on sales order lines/Add products button)	(Same as 'add products' form on sales order lines/Add products button)
Translated name	(Same as 'add products' form on sales order lines/Add products button)	(Same as 'add products' form on sales order lines/Add products button)
Category	(Same as 'add products' form on sales order lines/Add products button)	(Same as 'add products' form on sales order lines/Add products button)

New form Name (Label)	Help text	Attributes
Create production ticket		
Field Name (Label)	Help Text	Attributes
Add order type	Specify type of order for production ticket	Enum: Sales, Remake, None
Order number	Specify order number	Dropdown of open order lines based upon 'Add order type'- For Sales show sales order line, For Remake show service specification lines. Show order number, item number, serial number in the dropdown.
Item number	Specify item number to be produced	Dropdown. User has to select it in case 'Add ordertype' is None. For other cases this will be populated based upon selection in Order number
Product name	Item description	As per Item number selection
Serial number	Serial number dimension	User has to fill it in case 'Add order type' is None. For other cases this will be populated based upon selection in Order number
Quantity	Specify quantity to be produced	Integer
Site	Specify site dimension	Site dropdown. User has to select it in case 'Add ordertype' is None. For other cases this will be populated based upon selection in Order number
Warehouse	Specify warehouse dimension	Warehouse dropdown. User has to select it in case 'Add ordertype' is None. For other cases this will be

		populated based upon selection in Order number
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New form Name (Label)	Help text	Attributes
Operation state mapping form		
Field Name (Label)	Help Text	Attributes
Route	Identify the route	Dropdown from production route
Operation	Identify the operation	Dropdown of operations for selected production route
Mapped state	Identify the mapped state	Dropdown from workflow state master

New form Name (Label)	Help text	Attributes
Resolution types	The form from which the user can create resolution types for the quality order	
Field Name (Label)	Help Text	Attributes
Resolution type	Resolution type	Text 20
Description	Resolution type description	Text 60

New form Name (Label)	Help text	Attributes
Problem /Diagnostic types validation	The form from which the user can link any diagnostic types to a specific problem type	
Field Name (Label)	Help Text	Attributes
Diagnostics types	Diagnostics types available for this problem type	Look up to the Diagnostics types

New form Name (Label)	Help text	Attributes
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Diagnostic /Resolution types validation	The form from which the user can link any resolution types to a specific diagnostic type	
Field Name (Label)	Help Text	Attributes
Resolution types	Resolution types available for this diagnostic type	Look up to resolution types

New form Name (Label)	Help text	Attributes
Registration overview form		
Field Name (Label)	Help Text	Attributes
Order number	Number of the order referred to	Number of reference order (drop down of sales order number)
Barcode	Scan barcode	String (60)
Unregistered lines / Item number	Specifies the item number on the picking list	Non editable. Item number from picking list line.
Unregistered lines / Product name	Description of the item.	Non editable.
Unregistered lines / Serial number	Serial number of the item.	Editable.
Unregistered lines / Quantity	Quantity	Non editable. Quantity from production order.
Unregistered lines / Ear	Identifies the ear	Non editable. Ear from the production order for relevant sales line.
Unregistered lines / Storage location	Specifies storage location of the item	Non editable. Storage location logic same as on Inventory management/picking overview / Unpicked lines
Unregistered lines / BC coordinate	Specifies box coordinate of the production ticket	Non editable. From production ticket
Unregistered lines / BC room	Specifies box coordinate room	Non editable. From production ticket
Unregistered lines / Production ticket	Identify the production ticket	Non editable. Production ticket linked to production order
Unregistered lines / Site	Site	Non editable. From production order picking line

Unregistered lines / Order number	Number of the order referred to	Number of reference order on production ticket (non editable)
Unregistered lines / Reference type	Module and number referred to	Sales, Service (non editable)
Registered lines / Item number	Specifies the item number on the picking list	Non editable. Item number from picking list line.
Registered lines / Product name	Description of the item.	Non editable.
Registered lines / Serial number	Serial number of the item.	Editable.
Registered lines / Quantity	Quantity	Non editable. Quantity from production order.
Registered lines / Ear	Identifies the ear	Non editable. Ear from the production order for relevant sales line.
Registered lines / Storage location	Specifies storage location of the item	Non editable. Storage location logic same as on Inventory management/picking overview / Unpicked lines
Registered lines / BC coordinate	Specifies box coordinate of the production ticket	Non editable. From production ticket
Registered lines / BC room	Specifies box coordinate room	Non editable. From production ticket
Registered lines / Production ticket	Identify the production ticket	Non editable. Production ticket linked to production order
Registered lines / Site	Site	Non editable. From production order picking line
Registered lines / Order number	Number of the order referred to	Number of reference order on production ticket (non editable)
Registered lines / Reference type	Module and number referred to	Sales, Service (non editable)

New form Name (Label)	Help text	Attributes
State scanning production ticket		
Field Name (Label)	Help Text	Attributes
User id	User ID of the scanning user	look up to user

Scan	Scan sales order or production ticket	String (60)
Operation	Transition to this operation	Lookup from route operation.
Assign to	Specify worker to whom the operation has to be assigned	Lookup worker master
Include	Orders to be included for transition	Boolean, by default it should be (Y) when a new line is created in the grid
Order number	Sales order number	As selected in the scan field , if production ticket number is scanned then the order reference as on production ticket
Production ticket	Production ticket number	As selected in the scan field , if sales/service order number is scanned then the order reference as on production ticket
Production	Production order number	Production order number linked to the production ticket
Operation	Operation of the production order	Operation for the production order as on production ticket line.
Box coordinate	Specifies box coordinate for the production ticket	Box coordinate as on production ticket.
BC room	Specifies BC room	BC room as on production ticket.

New form Name (Label)	Help text	Attributes
Additional consumption lines		
Field Name (Label)	Help Text	Attributes
Production	Identify the production order	View only, populated from the production order from which the additional consumption is called.
Date	Posting date for physical item posting	Editable, date field
Quantity	Quantity to be picked	Editable

Proposal	Specify proposed action for the specified quantity	Enum: Quantity, Full reversal
Reverse to warehouse	Specify the warehouse to which the item should be reversed to.	Dropdown from warehouse master

New form Name (Label)	Help text	Attributes
Quality acceptance		
Field Name (Label)	Help Text	Attributes
Order number	Number of the order	View only, Number of the Sales, Service or Purchase order
Reference type	Order referred to	View only, populated the referred document Sales order, Service order, Purchase order
New	Generates a new quality order	
Show	Select which quality order to be shown	Enum: Open, Pass, Fail, All
Quality order	Quality order identification	View only, from quality order
Operation	Identification of the operation	View only, from quality order
Test	Test identification	View only, from quality order
Outcome	Test variable outcome	Field behavior should be same as on Quality order.
Ear	Ear identification	View only, from quality order
Reference number	Production order identification	View only, from quality order
Service specification	Service specification identification	View only, from quality order
Status	Quality order status	View only, from quality order
Test group	Test group identification	View only, from quality order

New form Name (Label)	Help text	Attributes
Service consumption		
Field Name (Label)	Help Text	Attributes
Item number	< refer service consumption section in service specification form >	< refer service consumption section in service specification form >

Product name	< refer service consumption section in service specification form >	< refer service consumption section in service specification form >
Quantity	< refer service consumption section in service specification form >	< refer service consumption section in service specification form >
Storage location	< refer service consumption section in service specification form >	< refer service consumption section in service specification form >
Journal	< refer service consumption section in service specification form >	< refer service consumption section in service specification form >
Warehouse	< refer service consumption section in service specification form >	< refer service consumption section in service specification form >
New item	< refer service consumption section in service specification form >	< refer service consumption section in service specification form >
Picked	< refer service consumption section in service specification form >	< refer service consumption section in service specification form >
Posted	< refer service consumption section in service specification form >	< refer service consumption section in service specification form >
Serial number used	< refer service consumption section in service specification form >	< refer service consumption section in service specification form >
Batch number used	< refer service consumption section in service specification form >	< refer service consumption section in service specification form >

New form Name (Label)	Help text	Attributes
Change route/operation		
Field Name (Label)	Help Text	Attributes
Route	Specify the copying method	Radio button
Route number	Route identification	Route number dropdown
Operation	Specify the copying method	Radio button
From Oper. No.	From operation number	Operation number dropdown
To Oper. No.	To operation number	Operation number dropdown
New Oper. No.	New operation number	Operation number dropdown
Include	Specify orders to be changed	Checkbox, default Yes
Production	Production order number	Display field from production ticket
Item number	Item number	Display field from production ticket

Name	Item description	Display field from production ticket
Serial number	Serial number	Display field from production ticket
Ear	Ear	Display field from production ticket
Service type	Type of service	Display field from production ticket

10 Security

[In this section you describe the security setup related to the modification]

Security in Dynamics AX2012 is specified using the following definitions:

Menu item. The menu item is the menu entry or button used in the solution, on which security is eventually connected. (In the illustration they are the “Securable objects”)

Privilege. A privilege, typically called xxxMaintain or xxxView or xxxGenerate is defined and named, and to a privilege one or more menu items are connected. More menu items are connected to a privilege, when the access to these can be seen as one combined functional area. When I doubt only one menu item is connected to the privilege. The access rights are defined on the Privilege.

- When the privilege is called Maintain, the access right is Delete,
- When the privilege is called View, then access right is Read
- When the privilege is called Generate, then access right is Read, Create or Delete

Duty. A duty is used to grant access to a part of a process. They are typically called xxxMaintain or xxxApprove or xxxInquire and to a duty one or more privileges are connected. A new privilege can be connected to an existing duty, but usually new privileges are causing new duties.

Roles. Duties are connected with Roles. Dynamics AX2012 comes with a set of roles and the use of these are defined in configuration design. New roles can be created, but creation of roles must be coordinated in general.

All duties are connected with a process cycle. In AX the following process cycles are few, and the only the “Revenue Cycle” is used in WDH.

10.1 Menu items

Menu Item Name	Path	Label	Help text
All production tickets	Production control/Common/Production tickets	All production tickets	Open list of production tickets

Menu Item Name	Path	Label	Help text
Open production tickets	Production control/Common/Production tickets	Open production tickets	Open list of open production tickets
On hold production tickets	Production control/Common/Production tickets	On hold production tickets	Open list of on hold production tickets
Delayed production tickets	Production control/Common/Production tickets	Delayed production tickets	Open list of delayed production tickets
Resolution types	Inventory management/Setup/Quality management	Resolution types	Opens form with resolution types
Register serial number	Production control/Production ticket	Register serial number	Open form for serial number registration
State scanning production ticket	Production control/Production ticket	State scanning production ticket	Open form for bulk scanning of production ticket
Add	Production control/Production ticket	Add	Add new work specification line
Remove	Production control/Production ticket	Remove	Remove a work specification line
Work Spec Wizard	Production control/Production ticket	Work Spec Wizard	Open a work specification wizard form
Reset service codes	Production control/Production ticket	Reset service codes	Reset selected failure and resolution codes
Cancel Production ticket	Sales order form	Cancel Production ticket	Cancel Production ticket
Cancel Production ticket	Sales order entry form	Cancel Production ticket	Cancel Production ticket
Pass selected	Quality orders	Pass selected	Pass selected

10.2 Security hierarchy

Role	Duty	Privilege	Menu item	Access Right

Role	Duty	Privilege	Menu item	Access Right
Service and repair – Service order – Clerk	-	-	Add / Remove / Work Spec wizard / Reset service codes	Full access
Production control – Clerk	-	-	Add / Remove / Work Spec wizard / Reset service codes	Full access
Production control – Clerk	-	-	Quality orders/Pass selected	Full access

Security in DM and POS client is defined differently, but when the modification concerns these modules, then security for that is defined.

11 Setup

[Describe the specific setup in order to make the solution work. Both setup in existing standard AX2012 tables and eventual new parameters are defined here.]

N/A

12 Test Requirements

[The testing of the modification is specified in this section. The testing is both a list of test cases for functional test, process test and integration test. When seen relevant, also performance test cases are described. The actual test case/ test script is defined in Test Manager]

Functional test is the test of the functionality in the design. The functionality is tested in fragments and with different constructed data to get around the functions.

Process test is to test the entire process within the scope of the FDS. This includes the examples listed. Data is constructed to get into all branches of the process and also interruptions in the process

Integration test is the test on live-like data and with the process as part of a larger process. Technical integrations is tested against true test systems, not only on simulated data

Performance test is test of actual performance and experienced performance. This is number of clicks and steps, waiting time between field clicks, updates and also startup time of processes. Performance test is made as close to the normal use as possible]

Area	Test case name	Highlights	Details
Functional test	ATS Integration	Send all the EPDS feature details and order details to BizTalk	All the EPDS feature details and order details need to be sent to BizTalk to perform ATS testing.

Area	Test case name	Highlights	Details
Functional test	Sales order information	Send production instructions and item description in wbi file	Test that production instructions from sales order header is copied in 'Comment' section and item description in 'Description' (similar to how it happens in wbi file on doing SOTAS)

13 Risk Assessment

[What are the risks in introducing the modification in a running solution, and what measures are taken to minimize risk?]

Risk	Consequence	Severity	Measure
N/A			

14 Organizational Change Management

[Keywords for required training or change in processes in the organization are listed here. The training material and communication is not included in this document]

Training material	Description	Target group

15 Deployment

[This section describes migration issues, either from older systems or tasks/jobs run in order to update data for the solution to work after a system upgrade. This section also list eventual changes in setup of security roles.]